

Report



SPDC-BFPC UPSTREAM GAS SUPPLY, GATHERING, EVACUATION & PROCESSING (UGS-GEP) PROJECT

Conceptual Study Memorandum

Prepared for:

Brass Fertilizer & Petrochemical Company Limited

Doc. No. EA/SPDC-BFPC UGS-GEP/CSM-RPT-001

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SUMMARY PAGE

Client: BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED	
Project Ref: SPDC-BFPC UGS-GEP	Title: Conceptual Study Memorandum
Facilities: Gas Gathering & Evacuation Pipelines, and Gas Plant	Doc No.: EA/SPDC-BFPC UGS-GEP/CSM-RPT-001

The administration of this document is in respect of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project.

The document, in consideration with relational design elements & concepts, presents the project's concept selection study as a reference & premise to the Basic Design, as well as demonstrates readiness to progress to the Basic Design, and the subsequent FEED (Front End and Engineering Design).

It is also (in conjunction with the Basic Design) a presentation to DPR as a statutory requirement in the process of securing regulators' approval to execute the FEED / Detail Design that will lead to the project's FID and implementation.

The report has relied and used data and reference documents provided by client, and if further document and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEY WORDS: BASIS FOR DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES	
Document No.	Description
TO-D-AA-00001	Basis for Design
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Gas Gathering & Surface Facilities Concept Screening Study
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELERPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option

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Abbreviations

The Following definitions and abbreviations have been used in this document.

Organisational

BFCL	Brass Fertilizer Company Limited
EA	Epic Atlantic Limited
SPDC	The Shell Petroleum Development Company of Nigeria Limited

Technical Terms

BBOU	Bubuwei Bou
NUNR	Non River
SEIB	Seibou
ELEP	ELEP

OPUG	Opughene
T/L	Trunkline
B/L	Bulkline

Units of Measurements

km	Kilometre
m	Metre
M	Thousand
MM	Million
MMscfd	Million Standard Cubic Feet per Day
MTPD	Metric Tonnes Per Day

Technical Terms Used for PIPESIM Simulation

BBOU	Stream coming from Bubuwei Bou manifold
ELEP	Stream coming from Elepa Satellite manifold
ELEP MF	Stream coming from Elepa Field manifold
NUNR	Stream coming from Nun River manifold
NUNR MF	Stream coming from Nun River Field manifold
SEIB	Stream coming from Siebou manifold
OPUG	Stream coming from Opughene manifold

1. Administration

This Conceptual Study Memorandum is in respect of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project. It stems from the Concept Select Studies, and presents the selected design concepts as basis for the Basic Design of the project.

It is also an input and accompaniment to the Basic Design Memoranda to be presented to DPR as statutory requirements in the process of securing regulators' approval to execute the FEED / Detail Design that will lead to the project's FID and implementation.

1.1. Validity & Change Management

The report has relied and used data and reference documents provided by client. Some of the data are preliminary, based on limited information, but to a large extent, however, are adequate for this phase of work: - conceptual study and basic design.

For example, analogue PVT data was used in the absence of actual PVT, and preliminary pipeline distances were estimated based on desktop route selection study. These design data, and the consequent simulation results will invariably need to be updated during FEED, by which time actual PVT data would have been obtained, and the FP survey done to establish actual distances and elevations of the pipeline routes.

Similarly, the project timelines as set out in the execution schedule was predicated on FID date of June 2018, and that will correspondingly change with actual FID date.

Also, the security situation in the region with its wide spate of vandalism may dictate that certain design provisions be changed during the FEED. The situation will continue to be monitored, and if still perceived to be a cog in the wheel of innovation, some of the ideal designs will be replaced with what will be practically suitable for the situation.

For example, in as much as it is feasible to lay cable from the IPP to the remote manifold platforms to transmit electricity to motorise the platform valves in the Manifold Platforms Power Supply Options, or install solar systems on the platforms, there will be no need to implement such options if the facilities (cables & solar systems) will eventually be vandalised in no time. Instead, it will be rather more appealing to limit the process option to manual valve operations, without the features of valve motorisation that will require electricity.

Epic Atlantic therefore reserves the right to update accordingly, the opinion set out in this report, as further information and / or refinement become available.

1.2. Background and Introduction

The SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project is constituted by Brass Fertilizer & Petrochemical Company Limited (BFPC), in partnership with The Shell Petroleum Development Company of Nigeria (SPDC).

The project scope includes Gas Gathering Pipelines, and a Gas Process Processing Plant that will provide processed gas as feedstock to a Petrochemical Plant and a Power Plant. The Gas Processing Plant will process natural gas gathered from five SPDC fields – Bubouwe Bou, Elepa, Nun River, Seibou, and Opugbene - in OMLs 33, 32, & 36.

The Gas Plant will comprise three trains, (each with a processing capacity of 150 MMscfd) and condensate storage capacity of 50,000 Tonnes. In this respect, BFPC have concluded the Gas Supply & Processing Agreement (GSPA) with SPDC for 270 MMscfd of the natural gas feedstock. The Plants (Gas Plant, Petrochemical Plant, and Power Plant) will all be located in the same complex at Akabel, Brass Island, Bayelsa state of Nigeria, near the shores of Atlantic Ocean.

The Petrochemical Plant will comprise a Methanol Plant of 1.75 MTPA capacity, and a Urea Plant of 2.6 MTPA total capacity, built in two phases of 1.3 MTPA each. It will also have storage capacity for 150,000 Metric Tonnes of Methanol and 100,000 Metric Tonnes of Urea. The facilities complex will also comprise a captive export jetty of 4 berths for up to 35,000 DWT vessels.

The current on-stream date for the project is **December 2020**, and to achieve that, FID is planned for **June, 2018**. In pursuit of these business objectives, BFPC have virtually concluded Project Management & EPC contracts with EIL, Saipem, and TCC, who are internationally renowned contractors in the implementation of the subject projects type.

The Gas Processing Plant, and the associated Gas Gathering System / Pipelines which will transport wellhead gas from the SPDC fields to the Gas Processing Plant, constitute the key upstream enabler facilities for the realisation of the objective supply of processed gas supply as feedstock to the Petrochemical Plant for the product manufacturing of methanol & urea, and to the Power Plant for power generation.

1.3. Scope of Document

The scope of the document covers the Conceptual Study and its evolution of results that will shape the basic design of the project along the following subjects:

-  Project Design Data & Basis for Design Outline
-  Project Concept and Site for the Project
-  Feedstock Supply & Product Evacuation Scheme
-  Pipeline Route Selection

- + Process Simulation, - HYSYS Steady State, OLGA Dynamic Simulation, and PIPESIM
- + Feedstock Flow Assurance
- + Manifold Platforms Power Supply Options
- + Manifolds Platform Concept & Integration with SPDC Field Manifolds
- + Pipeline Material Selection & Corrosion Analysis
- + Gas Processing Technology
- + Safety Provisions & Preliminary Environmental Impact Assessment Study
- + Infrastructural Support Strategy & Manning
- + Decommissioning Philosophy
- + Concept Select Drawings

2. Project Design Data & Basis for Design Outline

Ref: TO-D-AA-00001-R01_Basis for Design

The Project Design Data & Basis for Design Outline as summarised here form the technical basis for the conceptual study, and subsequent development of the Basic Design and FEED for the project's EPC, albeit updates that will arise following additional data acquisition or refinement.

2.1. Project Design Data

The key design data required & used are: Project Location Plan & Operational Conditions, **the Natural Gas Conditions, Compositions, & Flow Rates**, Production Forecast & Fluid Characteristics, and Environmental Conditions.

2.1.1. Project Location & Operational Conditions

The project location plan & operational conditions are summarized in the table below.

Table 2.1: Project Location Plan & Operational Conditions

1	Location	OML 33 / 32 / 36 - with the fields widely spread & on both sides of Nun River
2	Target Fields	BBOU, ELEP, NUNR, SEIB, & OPUG BBOU & ELEP in OML 33 and NUNR in OML 32, - East of Nun River SEIB in OML 32 and OPUG in OML 36, - West of Nun River
3	Desired Offtake Rate / GSPA Contractual Rate	300 MMscfd / 270 MMscfd, starting at 110 MMscfd in Year 1, ramping up to 300 MMscfd in Year 4
4	Gas Offtake Points	SPDC Field Manifold Headers at the target fields, in phases starting with BBOU, then ELEP, NUNR, SEIB, & OPUG
5	Feed Stream Inlet Pressure to BFPC P/L System	80 – 100 barg
5	Feed Stream Outlet Pressure from BFPC P/L System to Gas Plant	60 – 65 barg
5	Gas Plant Processed Gas Outlet Pressure	50 – 55 barg
6	Condensate	5 – 45 Mbpd
7	Water Production	Starting at 1.5 Mbpd in 2026, increasing to about 20 Mbpd in 2032

2.1.2. Production Forecast & Fluid Characteristics

The production forecast and fluid characteristics are summarized below.

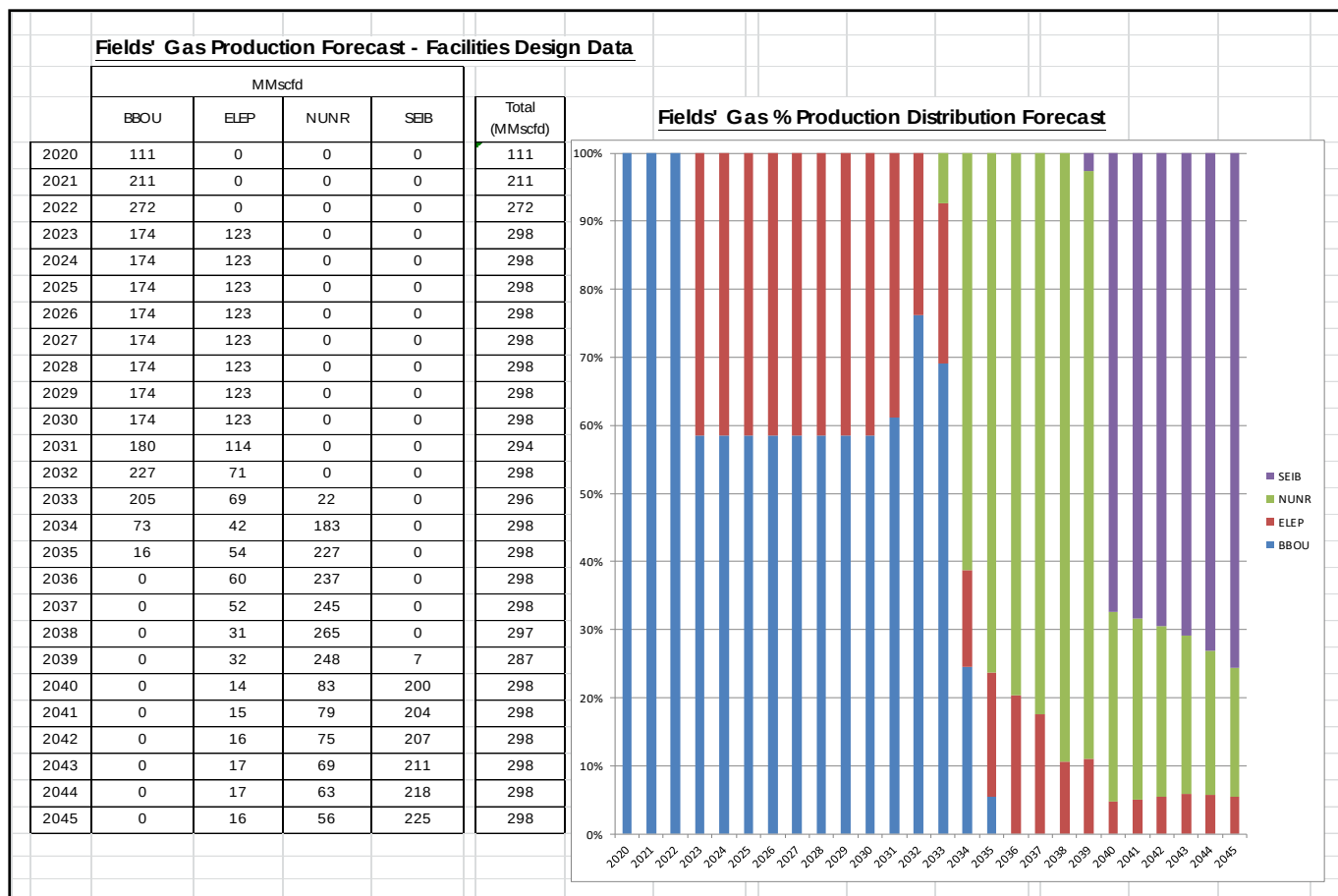


Figure 2.1: Fields' Gas Production Forecast – Facilities Design Data

Gas Composition

The gas compositions of the five fields are as summarised in the tables below:

Table 2.1: Fields' Reservoir Compositions & Conditions – as received.

Fields' Gas Compositions															
Component	BBO U			ELEP			NUNR			SEIB					
	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid
N2	0.07	0.00	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.07	0.07	0.00	0.07
CO2	0.59	0.00	0.59	2.30	0.00	2.22	0.96	0.00	0.89	1.63	0.00	1.55	1.63	0.00	1.55
H2S				0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1	93.26	0.00	92.92	84.63	0.00	81.66	80.42	0.00	74.63	83.24	0.00	78.62	83.24	0.00	78.62
C2	3.17	0.01	3.16	6.35	0.03	6.13	8.33	0.04	7.73	6.27	0.28	5.93	6.27	0.28	5.93
C3	1.01	0.03	1.01	2.67	0.27	2.59	4.76	0.97	4.49	3.49	1.88	3.40	3.49	1.88	3.40
iC4	0.61	0.07	0.61	0.76	0.37	0.75	1.37	1.23	1.36	1.20	1.55	1.22	1.20	1.55	1.22
nC4	0.44	0.07	0.44	0.82	0.54	0.81	1.61	2.35	1.66	1.31	2.36	1.38	1.31	2.36	1.38
iC5	0.25	0.16	0.25	0.43	0.87	0.45	0.73	2.43	0.85	0.73	2.34	0.83	0.73	2.34	0.83
nC5	0.19	0.16	0.19	0.32	0.68	0.33	0.52	1.90	0.62	0.50	1.96	0.58	0.50	1.96	0.58
C6	0.22	1.88	0.23	0.59	2.26	0.65	0.57	5.96	0.96	0.68	4.76	0.92	0.68	4.76	0.92
C7	0.17	6.21	0.19	0.69	6.75	0.90	0.49	10.05	1.18	0.62	9.71	1.13	0.62	9.71	1.13
C8	0.02	6.61	0.04	0.44	12.76	0.87	0.24	15.54	1.34	0.28	15.65	1.14	0.28	15.65	1.14
C9	0.00	7.13	0.03	0.00	10.87	0.38	0.00	10.75	0.77	0.00	11.71	0.65	0.00	11.71	0.65
C10	0.00	5.70	0.02	0.00	8.62	0.30	0.00	8.05	0.58	0.00	9.14	0.51	0.00	9.14	0.51
C11	0.00	7.83	0.03	0.00	6.64	0.23	0.00	5.96	0.43	0.00	5.68	0.30	0.00	5.68	0.30
C12+	0.00	64.14	0.22	0.00	49.34	1.73	0.00	34.76	2.51	0.00	33.00	1.82	0.00	33.00	1.82
Total	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.99	100.00	100.00	100.00	100.01	100.00	100.00	100.01
Molar ratio				0.9650	0.0350	1.0000	0.9280	0.0720	1.0000	0.0720	0.9280	1.0000	0.0720	0.9280	1.0000
Component	BBO U			ELEP			NUNR			SEIB					
	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid
Reservoir Pressure, psia			3,246			6,910			7,259			6,533			6,533
Reservoir Temp., °C			78			123			102			117			117
Mole %															
C7+	0.19	97.62	0.53	1.13	94.98	4.41	0.73	85.11	6.81	0.90	84.89	5.54	0.90	84.89	5.54
C12+	0.00	64.14	0.22	0.00	49.34	1.73	0.00	34.76	2.51	0.00	33.00	1.82	0.00	33.00	1.82
Mola mass	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol
C7+	100.89	167.69	143.17	105.67	186.72	166.70	104.82	176.38	169.26	104.48	174.90	164.20	104.48	174.90	164.20
C12+	-	191.60	191.60	-	191.60	191.60	-	191.60	191.60	-	191.60	191.60	-	191.60	191.60
fluid	17.97	165.61	18.47	20.84	181.08	26.44	21.82	160.91	31.84	21.38	158.85	29.00	21.38	158.85	29.00
Density	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³
C7+						832			830			829			837
C12+															
fluid		811			828			813			817			817	
Rel Density (air =1)	0.62			0.719		0.913	0.753		1.099	0.738			0.738		
CGR			stb/ MMscfd			stb/ MMscfd			stb/ MMscfd			stb/ MMscfd			stb/ MMscfd
fluid			4 - 36			62 - 158			62 - 158			62 - 158			62 - 158

Table 2.1: Fields' Reservoir Water-Saturated Compositions & Conditions – as constructed using HYSYS

Saturated Gas Compositions															
Component	BBO U			ELEP			NUNR			SEIB			OPUG		
	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid
N2	0.00	0.00	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.00	0.10
CO2	0.00	0.00	0.59	0.00	0.00	2.17	0.00	0.00	0.77	0.00	0.00	1.53	0.00	0.00	2.21
H2S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1	0.00	0.00	92.61	0.00	0.00	79.87	0.00	0.00	72.37	0.00	0.00	77.84	0.00	0.00	67.24
C2	0.00	0.00	3.15	0.00	0.00	6.00	0.00	0.00	7.63	0.00	0.00	5.87	0.00	0.00	7.88
C3	0.00	0.00	1.01	0.00	0.00	2.53	0.00	0.00	4.82	0.00	0.00	3.37	0.00	0.00	4.76
iC4	0.00	0.00	0.61	0.00	0.00	0.73	0.00	0.00	1.45	0.00	0.00	1.21	0.00	0.00	1.94
nC4	0.00	0.00	0.44	0.00	0.00	0.79	0.00	0.00	1.96	0.00	0.00	1.37	0.00	0.00	1.06
iC5	0.00	0.00	0.25	0.00	0.00	0.44	0.00	0.00	0.94	0.00	0.00	0.82	0.00	0.00	1.15
nC5	0.00	0.00	0.19	0.00	0.00	0.32	0.00	0.00	0.76	0.00	0.00	0.57	0.00	0.00	0.68
C6	0.00	0.00	0.23	0.00	0.00	0.64	0.00	0.00	1.02	0.00	0.00	0.91	0.00	0.00	1.06
C7+	0.00	0.00	0.53	0.00	0.00	4.31	0.00	0.00	7.65	0.00	0.00	5.48	0.00	0.00	10.86
H2O	0.00	0.00	0.34	0.00	0.00	2.19	0.00	0.00	0.63	0.00	0.00	0.95	0.00	0.00	1.08
Total	0.00	0.00	100.00	0.00	0.00	100.00	0.00	0.00	100.00	0.00	0.00	100.00	0.00	0.00	100.00

2.1.3. Environmental Data

The environment is mangrove swamp and brackish water of resistivity assumed 1000 - 5000 ohm – cm, based on regional data.

(i) Environmental (Outdoor) Conditions

Design Barometric Pressure	1.013 bara
Design Ambient Temperature - Maximum	42°C
Mean Maximum Ambient Temperature	31°C
Design Ambient Temperature - Minimum	18°C
Mean Minimum Ambient Temperature	23°C
Relative Humidity	100%
Thermal Insulation Design Criteria	23°C at 100% Humidity
Black Bulb Temperature	Up to 80°C
Ground Temperature	25 – 28 °C
Site Altitude	Sea Level
Lightning	Frequent in Rain
Average Annual Rainfall	3,800 in
Mean Maximum Hourly Rainfall	100 in
Design Wind Speed	36 m/s (130 km/hr)
Corrosion Environment	Brackish Swamp/Waters (50 -100 ohm-cm)

(ii) Indoor Conditions

The control equipment and HV/LV-boards will be installed in air-conditioned non-hazardous control rooms, with environmental conditions as follows:

Ambient Air Temperature	20 – 22 °C
Relative Humidity	50 – 75 %

Electronic Equipment (e.g. signal converters / barriers) shall also be capable of operating at a temperature of at least 75°C.

Equipment shall also be capable of operating at a room temperature of 55°C to cover shutdown of air conditioning.

The operator stations will be installed in an air-conditioned, non-hazardous control room with conditions as follows:

Ambient Air Temperature	21 ± 3 °C
Relative Humidity	50 – 75 %

Environment / Resistivities Vs Corrosion Current per m²

The assumption made for the current density for bare steel metal areas for this desk top calculations is 22 mA / m² at the assumed resistivity of 1000 – 5000 ohm-cm of the swampy, brackish water environment closed to the brass sea

Coat Efficiency

Coat efficiency, Mean Coating Breakdown and coating Breakdown at end of life time

2.2. Basis for Design Outline

The key BFD elements are summarised in the fishbone diagram below, and the matrix that follows.

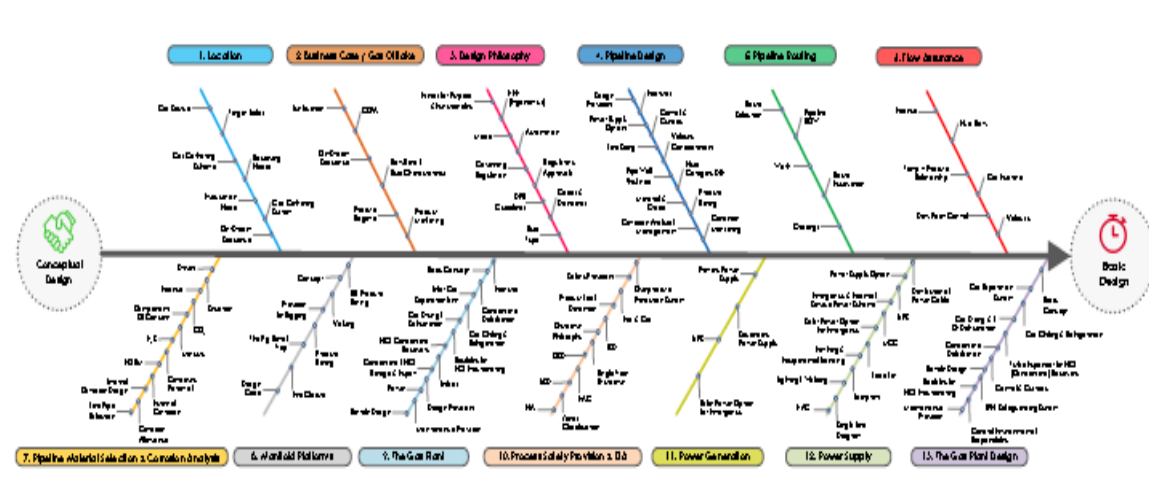


Figure 2.1: Fields' Gas Production Forecast – Facilities Design Data

Table 2.2: BFD Matrix

Situations & Systems	Design Basis	Remark / Reference
----------------------	--------------	--------------------

1	Location		
1.1	Gas Source		OML 33 / 32 / 36 - with Locations Widely Spread & on Both Sides of Nun River
	1.1.1	Target Fields	BBOU, ELEP, NUNR, SEIB, & OPUG
			BBOU & ELEP in OML 33 and NUNR in OML 32, - East of Nun River
			SEIB in OML 32 and OPUG in OML 36, - West of Nun River
1.2	Gas Gathering scheme		Nodal Raw Gas Gathering thru Pipeline Network to ELEP, and from there to Akabel
	1.2.1	Receiving Nodes	Three Nodes - ELEP, BBOU, & NURN Node - geographically convenient
			NUNR & SEIB Production to NUNR Node S-M/F - geographical convenience
			BBOU & OPUG Wells Production to BBOU Node S-M/F - geographical convenience
			ELEP Wells Production, BBOU Node Inventory, & NUNR Node Inventory to ELEP Node E-M/F - geographical convenience
	1.2.2	Evacuation Node	ELEP Manifold
			Inventory from ELEP Node E-M/F to AKAB PLEM, and from there to the Gas Plant

Situations & Systems	Design Basis	Remark / Reference
----------------------	--------------	--------------------

1	Location (Contd.)		
1.3	Gas Gathering System	A mix of Wellhead Flowlines, Field Manifolds, and Bulk Transport Lines, - optimally configured in terms of cost, layout, and choice of evacuation manifold (Bubouwe Bou or Elepa)	
		Wellhead Flowlines Gather Each Field Production into the Central Field Manifold (F-MF) within the Field, followed by: Transportation of the F-MF inventory through Bulkline to Satellite Manifold (S-MF) / Evacuation Manifold (E-MF). Transportation of the S-MF inventory through Bulklines to the E-MF Transportation of the E-MF inventory through a Trunkline to the Gas Processing Plant.	
1.4	The Gas Processing Plant	The Gas Plant will be located at Akabel, Brass Island, Bayelsa State of Nigeria, near the shores of Atlantic Ocean	
		The Plant Site is in the same complex with the Petrochemical Plant and the Power Plant that it will be supplying the processed gas	
2	Business Case		
2.1	Justification	To supply processed gas needed as feedstock to Petrochemical Plant and a Power Plant, - in the value chain for methanol & urea production by the petrochemical plant, and power generation by the power plant	
		GSPA with SPDC in place for the Supply of the Natural Gas to be Processed	

Situations & Systems	Design Basis	Remark / Reference
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2 Business Case (Contd.)

2.2	GSPA	Wellhead (Raw) Gas at Each of the Fields' Manifolds for offtake by BFPC through own Bulklines	
	2.2.1	Gas	270 MMscfd
	2.2.2	Supply Pressure	80 - 100 barg
	2.2.3	Condensate	5 - 45 Mppd
2.3	On-Stream Sequence	Phased Sequential (Production) Injection: BB ELEG + NUNR + SEIB + OPUG in that order, - with stop points driven / moderated by the GSPA	

3 Gas Offtake / Flow Conditions

3.1	Flow Rates / Fluid Characteristics		
	3.1.1	Gas Production	300 MMscfd Peak (Start 110 in Year 1, Ramping up to 300 in Year 4)
		Condensate Production	Up to 45,000 bpd
		Water Production	Up to 350 bpd from the water content of the Natural Gas
			Up to 20 Mbwpd reservoir production, starting later in the fields' life
		CGR	4 - 36 for BBOU, 62 - 158 for the rest
	3.1.2	Gas SG	0.6 - 0.74 (Air = 1)
		Condensate SG	0.74
		Water SG	1.00 - 1.02
	3.1.3	Corrosivity Potential	Envisaged, on account of CO ₂ in the gas stream
	3.1.4	Souring Potential	Not Envisaged, - since no H ₂ S in the gas stream
	3.1.5	Foaming Potential	Not Envisaged
	3.1.6	Scaling Tendencies	Not Envisaged
			5- 45Mbpd of Condensate Also water production will occur later, starting at 1.5 Mbpd in 2026 increasing to about 20 Mbpd in 2032 Fluid therefore to be considered two phase since no treatment will be done before injection into the Line, - prompting design of the Pipeline for two-phase Handling Two Phase Gas Transportation all the way to the Gas Processing Plant at Akabel Consequence is that Frequent Pigging will be needed to keep the Lines operating

Situations & Systems	Design Basis	Remark / Reference
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3 Gas Offtake / Flow Conditions (Contd.)

3.2	Pressure Regime			Provision to accommodate higher pressure (up to 110 barg) at the distant field manifolds of Seibou & Opugbene.
	3.2.1	Feed Stream Inlet Pressure	80 - 100 barg	
	3.2.2	Feed Stream Outlet Pressure	60 - 65 barg	
3.3	Product Metering		The NAG facility shall be provided with metering devices for gas and condensate flows. These meters shall be allocation type meters. All the instruments provided shall meet the Hazardous area requirements for field instruments	
	3.3.1	Raw Gas Metering	Conditioning Orifice Metering at the Custody Transfer Points at the Field Manifolds	
	3.3.2	Feed Stream Outlet Pressure	60 - 65 barg	

4 Design Philosophy

4.1	Fitness for Purpose & Functionality		Driven by an understanding of what constitutes a good (an excellent) design that will deliver technical & operational excellence; - requiring a good knowledge of how humans interact within the work system	
	4.1.1	Human Factors Engineering (Ergonomics)	Inclusion of User Requirement through: - Consideration to human capabilities, limitations and needs in the interaction between humans, technology and the working environment. - Equipment, Valves, Controls & Accessories Layout, degree of automation, and extent of protective systems.	Operability and Maintainability of the System to be given priority, considering the peculiarities of geographical locations of the Facilities
4.2	Mode		Not normally manned facility, with Automatic Shutdown Capability	Automatic Shutdown to be Fail Safe
	4.2.1	Automation	Remote Shutdown Capability from a PLC based Control Room.	Consideration to the use of Self Contained Fail Safe Shutdown System that Requires Human Intervention for Reopening
			Remote start-up capability with local permissives from the Field M/F & PLEM, Either as a provision, or intended startup process for the P/L	This will require the Installation of Patch Panels & Fieldbuses. For such a small installation, it will be more economical to keep automation as simple as possible
			Local Control Room, with HMI	
			PLC with SCADA capability	
Situations & Systems			Design Basis	Remark / Reference

4	Design Philosophy (Contd.)			
4.3	Governing Regulations		Oil Pipelines Act, Chapter 338, Laws of the Federation of Nigeria 1990 DPR Guidelines for Installation of Gas Plant	
	4.3.1	Regulatory Approvals	PTS (Permit To Survey) OPL (Oil Pipeline Licence) EIA (Environmental Impact Assessment Study) Staged Approvals contained in DPR Guidelines for Gas Plant Facility and Oil & Gas Pipeline Systems	
	4.3.2	Refence DPR Guidelines	Guidelines for the Establishment of Natural Gas Plant Facility in Nigeria DRP P-1P: Guidelines & Procedures for the Design, Construction, Operation and Maintenance of Oil & Gas Pipeline Systems in Nigeria - 2007	
4.4	Codes & Standards			
	4.4.1	Fluid Type (& Category)	Gas (Category D / E)	
			Category D: Non-toxic, Single-phase Natural Gas	
			Category E: Flammable, and/or toxic fluids which are gases at prevailing ambient temperature and atmospheric pressure conditions and are conveyed as gases and/or liquids. Typical examples are Hydrogen, Natural Gas (not otherwise covered in Category D), Ethane, Ethylene, LPG (such as Propane & Butane), Natural Gas Liquids, Ammonia and Chlorine.	
	4.4.2	Applicable Design Code	ANSI / ASME 31.8 - Pipeline & Field Manifolds ANSI / ASME 31.3 - PLEM	
	4.4.3	Specification Breaks	ANSI / ASME 31.3 - at the Gas Plant Fence, as PLEM is within the Gas Plant Fence; as well as at the Manifold Platforms	
			ISO 13623:2009 § BS EN 14161:200 - § 5.2 Shell DEP 31.40.00.10-Gen - § 2.1 FLUID IS TO BE CONSIDERED TWO PHASE SINCE NO TREATMENT WILL BE DONE BEFORE INJECTING INTO THE LINE Applicable Design Code for Category D and E fluids, ref: Shell DEP 31.40.00.10-Gen ISO 13623:2009 / BS EN 14161:200	

Situations & Systems	Design Basis	Remark / Reference
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5	Pipeline Design			
5.1	Design Provisions		Pipelines, Node M/Fs and PLEM	
	5.1.1	Features	Coated Linepipe / Field Joints Coating - for external corrosion protection	- 3 Layers PE (Poly Ethylene) Main Line Coating - GTS 65 Field Joints Coating
			Concrete Coating of River Crossing Sections to Combat Buoyancy	
			Pig Launcher / Receiver with Capability for Intelligent Pigging	
			Provision for Chemical Injection Facilities on the Manifold Platforms - for Corrosion Inhibition Administration to Mitigate Internal Corrosion	
			Pipeline End Manifolds on Manifold Platforms at Gas Gathering Nodes	
			Provision for Power Supply & Related Power Distribution Ancillaries at the Manifold Platforms that will enable powered equipment operations / automation, and data transmission for monitoring of the pipeline.	Concept selected among other options - Solar Power & Individual Power Generator options at Each Manifold
			Cathodic Protection System to Combat External Corrosion	Impressed Current
			Sectionalisation Valves at River Crossings / at 30 km Intervals	ASME Requirement
	5.1.2	Retrofit Design	Retrofit Design on Node Manifold Platforms in the light of sequential injection	
	5.1.3	Power Supply Options	Power Plant at Akabel Base to generate electricity for the base operations, and export to the Pipelines Manifold Platforms	
	5.1.4	Control & Custody	The automated block valves will be remotely controlled from BFPC Control Centre. Custody transfer metering will be installed at each of the SPDC M/Fs	
	5.1.5	Health & Safety	Design will provide for healthy and safe work environments for operation & maintenance.	
	5.1.6	General Environmental Responsibility	Protective devices and containment will be installed at the facilities; mainline valves will be remotely operable where practical	

Situations & Systems		Design Basis	Remark / Reference
5	Pipeline Design (Contd.)		
5.2	Line Sizing	To Meet Throughput, Pressure & Velocity Boundary Conditions	
	5.2.1 Inlet Pressure	80 - 100 barg (1,160 - 1,450 psi)	Delivery Pressure Available at CTP
	5.2.2 Outlet Pressure	60 - 65 barg (870 - 942 psi)	High enough to satisfy the requirements of the Gas Plant
	5.2.3 Pressure Drop	Applicable Equation: - Panhandle Equation (Long Distance, Large Diameter Pipelines)	
		Steady State Hydraulic Simulation done on HYSYS, & PIPESIM	
		Dynamic Hydraulic Simulation done on OLGA	
		Line Sizing Cosscheck with Panhandle Equation	
	5.2.4 Velocity Considerations	Minimum Velocity to prevent surging, and to keep line swept clear of entrained solids & liquids Maximum Velocity that will not cause Erosion, Excess Noise, or Water Hammer	
	5.2.5 Velocity	5 - 20 m/s: subject to Erosional Velocity Limit	API RP 14E
	5.2.6 Minimum Velocity	High enough to prevent surging, and to keep line swept clear of entrained solids & liquids	Too low a velocity leads to settlement of water, which may lead to bottom of pipe internal corrosion
	5.2.7 Maximum Velocity	Low enough to avoid erosion, excess noise, or water hammer	Too high a velocity can increase the overall corrosion rate and also destroy any protective scale or inhibitor films.

Situations & Systems		Design Basis		Remark / Reference
5	Pipeline Design			
5.3	Pipe Wall Thickness		To Meet Design Code, Pipe Material & Grade, Fluid's Pressure, Chemical Composition, Velocity, & Velocity Boundary Conditions	
	5.3.1	Fluid Category D / E	Category D: Non-toxic Singlephase Natural Gas Category E: Flammable fluids which are gases at prevailing ambient temperature and atmospheric pressure conditions and are conveyed as gases and/or liquids. Typical examples are Natural Gas (not otherwise covered in Category D), Ethane, Ethylene, LPG (such as Propane & Butane), Natural Gas Liquids.	ISO 13623:2009 Shell DEP 31.40.00.10-Gen
	5.3.2	Location Class	Class 1 / Class 2 Locations	5 - 30 residential buildings per km
	5.3.3	Design Factor	Geographical 0.72, Crossings without Casings 0.6	
	5.3.4	Material & Grade	API X65	
5.4	Pressure Rating		To Meet ANSI Standard B16.5 / API Spec 6A	
	5.4.1	Pressure Class	ANSI 900 (Class 900)	
5.5	Maintenance Provisions			
	5.5.1	Corrosion Management	Provision will be made for corrosion inhibitor injection to mitigate corrosion of the pipeline	The Gas is potentially corrosive on account of its CO ₂ content above threshold for corrosion; and water, NGL, & particulates that are inherently present
			Provision will also be made for cleaning and in-line inspection tools, shop, and in-situ maintenance	
	5.5.2	Condition Monitoring	UT Scanning	
			Intelligent Pigging	
			Leak Detection will be installed for pipelines and facilities	

Situations & Systems		Design Basis	Remark / Reference
6	Pipeline Routing		
	6.1	Route Selection	Desk-top Exercise, pending PTS / detailed survey that will enable Final Route Selection
		Route Optimised in the Light of Environmental Impact / Impact to Settlements	
		Consideration given to Economic Distances / Workable Routes	
		Current Concept for BBOU / ELEP is: 16" x 9km from BBOU - ELEP, and 24" x 28.5 km from ELEP - AKAB, running through Southern Ijaw, Nembe, & Brass LGAs	
		Alternative of 42 km predominatly offshore route, aimed at minimising communities traverse was considered, but dropped, as it did not achieve that objective, in addition to the intelligence report that it will not be acceptable to NIWA.	Further the alternative offshore route will mean higher installation, and risk of hydrate formation for the lower ambient temperature
	6.2	Pipeline ROW	Desk-top Exercise, pending PTS / detailed survey that will enable Final Route Selection
	6.2.1	Width	30 m - (15 m for P/L, including FOC, and 15m provision for Construction equipment & activities
	6.2.2	Route Excavation	Propose to excavate route as 1 - 1.5 m deep trench that will be backfilled
			The alternaive of open ditch (3m width X 4m depth), was considered in the light than it will be permanently flooded and will therefore pose as deterrent against vandalisation, - but was dropped as that objective is uncertain to be achieved.
			Further the alternative open ditch canal option will mean higher installation cost that will include additional equipment and weight (concrete) coating of linepipes
	6.2.2	Crossings	Crossings of Existing pipelines, Cables, Power Lines, Roads, and Waterways to be maintained at an angle between 60 & 90 degrees
7	Flow Assurance		
	7.1	Essence	Process conditions should be such that upstream pressure is sufficient to deliver fluid to destination, as well as avoid precipitation & deposition of solids

	Situations & Systems		Design Basis		Remark / Reference
7	Flow Assurance (Contd.)				
	7.2	Fluid Flow		Hydraulic Flow, on own Pressure	Hydraulic Analysis to be performed to establish range of operational parameters of the Pipeline
		7.2.1	Temp - Pressure Relationship	T & P to be maintained below equilibrium line for hydrate formation, otherwise chemical (antifreeze) injection	
		7.2.2	Gas Hydrate	Need for the Fluid Traverse to be maintained outside Hydrate Formation Envelop	
		7.2.3	Dew Point Control	Dew Pointing & Hydrate Handling in respect of the Gas Processing will be addresses at the Gas Plant FEED	
8	Pipeline Materials Selection & Corrosion Analysis				
	8.1	Drivers		Internal & External Corrosion Predictions and Design Life, Fitness & Compatibility	
	8.2	Essence		Specification of Materials that can withstand the physical conditions and application, as well as work together	
	8.3	Situations		High Pressure Transportation of Category D/E Fluid (flammable fluid), in the pressure region of 100 barg, and temperature region of 60 °C	
	8.4	Components of Concern		Obnoxious Components of Concern are: CO ₂ , H ₂ S, Mercury, and Naturally Occurring Radioactive Materials (NORM)	
		8.4.1	CO ₂	CO ₂ content is up to 2%, which for the empirically established 0.1 bar partial pressure treshhold for significant CO ₂ corrosion, implies corrosion of unprotected mild CS at pressures above 5 barg	DeWaard-Milliams CO ₂ Corrosion Nomogram
		8.4.2	H ₂ S	Nil in the gas streams, - hence no sour gas corrosion to contend with	
		8.4.3	Mercury	No Mercury Present , - to cause mercury amalgamation and embrittlenes of alumunium and other metals	
		8.4.4	NORM	No NORM (Natrurrally Occuring Radioactive Materials), - Radon, Radium, - which can accumulate within P/L & Process Eqpt to render them radioactive over time	

Situations & Systems		Design Basis	Remark / Reference
8	Pipeline Materials Selection & Corrosion Analysis (Contd.)		
8.5	Issues & Consideration		The CO ₂ content of up to 2%, renders the gas corrosive at pressures higher than 5 barg (corresponding to the empirically established 0.1 bar partial pressure threshold for significant CO ₂ corrosion), - prompting the need for CRA materials or matching corrosion management design
	8.5.1	Corrosivity Potential of the Transported Fluid (Gas)	Internal Corrosion Issue
		Corrosivity Potential of the Ambient Environment	Assessments Based on: - HydroCor Software and External Resistivity Analysis - DeWaard-Milliams CO ₂ Corrosion Nomogram
	8.5.2	Internal Corrosion Design	Humidity, Acidity, and Oxygen in Air, and known potentially corrosive nature of the soil in the area jointly contribute to render the ambient environment corrosive
	8.5.3	External Corrosion	Materials Selection - NACE Codes, Consideration for MS (WT), MS (WT)+CA, MS+Inhibitor, MS with CRA
			Coating Selection and Painting - NACE Code
			CP Design NACE Code IP
			T/R, Junct Boxes, Cabling, (IAC Code), Groundbed design, Electrical Separation
8.6	Line Pipe Selection		Main Line Coating with 3 Layers PE (polyethylene) Field Joint Coating with GTS 65
			Duplex Stainless Steel for the Flowlines & Field Manifold Headers on the Manifold Platforms
			High Yield Strength (a favourable trade-off with required wall thickness) that can accommodate high corrosion allowance that will be needed
8.7	Corrosion Allowance		Up to 10 mm, and will be supported with Corrosion Inhibition

Situations & Systems		Design Basis	Remark / Reference
9	Manifold Platforms		
	9.1	Concept	Integrated Field Manifold Platform for both SPDC Field Manifold Header & BFPC Bulklines
	9.2	The Manifold Platform	Comprises Field Manifold Header, Incoming & Outgoing Bulklines and Scraper Barrels, Chemical Injection System, Electrical Transformer, & Electric Room
	9.2.1	B/L Pressure Rating	ANSI 900
	9.2.2	Provision for Pigging	The Pipeline Manifold equipped with Pigging Facilities (Pig Launcher / Receiver) for pigging / pipe cleaning that will be done during precommissioning, commissioning & decommissioning of pipeline, and routine cleaning and condition monitoring of the pipeline during its operations
	9.2.3	Valving	Valving arranged and labelled so that they can be accessed safely and efficiently; the required valve can be visually identified and distinguished without error; the information required for decision making is provided; and the physical and mental workload required remains within reasonable limits. Task demands should not exceed operator capabilities
			IVA (Identification Valve Analysis) to ensure that all valve configurations are well designed from an operational task performance perspective, and to avoid design misfits in category 1 valves (critical). A key objective is to ensure that valves are properly categorised and prioritised at an early stage in design.
			Valve Interlock System
			Unrestricted Access for Valves used for emergency operations (i.e. Category 1 valves)
	9.2.4	Manifold Platform Property Boundary	Will be Fenced / Caged to Ward off Intrusion
	9.2.5	The Pig Barrel Trap	Horizontal Pig Trap, with capability for intelligent pigging
	9.2.6	Pressure Rating	ANSI 900
	9.2.7	Design Code	ASME 31.8 for the Infield M/F Pig Traps, and ASME Section VIII for Pig Trap EnclosureE ASME 31.3 for the PLEM Piping

Situations & Systems		Design Basis		Remark / Reference
9	Manifold Platforms (Contd.)			
9.2	The Manifold Platform (Contd.)			
9.2.8	Corrosion Allowance	Yes, - as raw gas contains CO ₂ at threshold concentration for Corrosion		
9.2.9	Double Block & Bleed	Two separate valves in series with a bleed (vent) point in between to allow diversion to safe location of any fluid leaking through either valve		
9.2.10	End Closure	End Closure pointing away from personnel areas and critical items of equipment, in the event of pig being ejected from the pig trap	Consideration Given TO ADDITIONAL SAFETY MEASURES LIKE USING INTERLOCKING KEY SYSTEM TO OPERATE THE PIG TRAP SYSTEM.	
9.2.11	Valving	Valving arranged and labelled so that they can be accessed safely and efficiently; the required valve can be visually identified and distinguished without error; the information required for decision making is provided; and the physical and mental workload required remains within reasonable limits. Task demands should not exceed operator capabilities	PIPELINE VALVES TO BE DESIGNED AS PER ASME B16.34 AND ASME B16.10	
		IVA (Identification Valve Analysis) to ensure that all valve configurations are well designed from an operational task performance perspective, and to avoid design misfits in category 1 valves (critical). A key objective is to ensure that valves are properly categorised and prioritised at an early stage in design.		
		Valve Interlock System		
		Unrestricted Access for Valves used for emergency operations (i.e. Category 1 valves)		

Situations & Systems		Design Basis	Remark / Reference
10	The Gas Plant Design		
10.1	Basic Concept		Full Spectrum Gas Plant with Capability for NGL (Condensate) Extraction
	10.1.1	Features	Inlet Gas Separation Unit
			Condensate Stabilisation
			Gas Drying and / or Dehydration
			Gas Chilling & Refrigeration
			Turbo Expander for NGL (Condensate) Recovery
			Flexibility for NGL Fractionating
			Condensate / NGL Storage & Export Facilities
			Utilities
10.2	Design Provisions		
	10.2.1	Features	Provision for Power Supply & Related Power Distribution Ancillaries that will enable powered equipment operations / automation, and data transmission for monitoring of the Plant performance.
			Earthing Ring Main
	10.2.2	Retrofit Design	Retrofit Design for Booster Compression
	10.2.3	Power	Power Plant at Akabel Base to generate electricity for the Gas Plant operations
	10.2.4	Control & Custody	The automated block valves will be remotely controlled from BFPC Control Centre. Custody transfer metering will be installed at each of the SPDC M/Fs
	10.2.5		The automated block valves will be remotely controlled from BFPC Control Centre. Custody transfer metering will be installed at each of the SPDC M/Fs

	Situations & Systems		Design Basis		Remark / Reference
10	The Gas Plant Design (Contd.)				
	10.2	Design Provisions (Contd.)			
	10.2.6	Maintenance Provision	Provision will be made for corrosion inhibitor injection to mitigate corrosion of the pipeline	The Gas is potentially corrosive on account of its CO ₂ content above threshold for corrosion; and water, NGL, & particulates that are inherently present	
	10.2.7		Provision will also be made for cleaning and in-line inspection tools, shop, and in-situ maintenance		
	10.2.8	Health & Safety	Design will provide for healthy and safe work environments for operation & maintenance.		
	10.2.9	General Environmental Responsibility	Protective devices and containment will be installed at the facilities; mainline valves will be remotely operable where practical		
11	Process Safety Provision & Preliminary EIA Study				
	11.1	Process Safety Provision	Design Provision ranges from Overpressure Protection, Fire & Gas, Product Leak Detection, and Shutdown	Provision with Safeguarding (ESD) requirement in accordance to Shell standards.	
	11.1.1	Pipeline Overpressure Protection System	A High Integrity Pressure Protection System (HIPPS) shall be provided for the protection of the Bulklines & Manifold Platforms		
			PSVs will be installed on the pig traps to provide additional safety especially during pigging where it pups up and closes to maintain pressure balance		
			A complete blowdown / evacuation of the Pipeline will be required, and can only be achieved through Akabel where it will be possible to install such facility		
			On the other hand, for maintenance purposes, the Spec requires sectionalisation valves to be installed at pre-determined spacings on the Pipeline in order to reduce the volume of inventory to be evacuated in case damage to any section of the Pipeline. This will require the installation of small venting systems around the Sectionalisation Valve Stations which include of course the PLEM to vent the sections of the Pipeline requiring maintenance		

Situations & Systems		Design Basis		Remark / Reference
11	Process Safety Provision & Preliminary EIA Study (Contd.)			
11.1	Process Safety Provision (Contd.)			
11.1.2	Fire & Gas	Fire and Gas system signals will be implemented into the Pipelines & Gas Plant safeguarding system		
		A fire & gas mimic panel will be installed within the control room indicating main F&G detectors/initiators / F & G loops / push button, etc.		The Mimic Panel will be designed in accordance with Shell DEP and SPDC guidelines
		Mimic panel will be operated from the SIS system with dedicated SIS input & outputs to final indication lamps. Horns acknowledge button, lamp test button, ESD button and reset button for activated lamps will also be included on the mimic panel.		
11.1.3	Product Leak Detection	ROW Inspection by Line Patrols		
		Hydrocarbon Sensing (of External Leaks) via Fibre Optic / Dielectric Cable Cable		
11.1.3	Shutdown	To be initiated as a last resort of the Process Safeguarding System		
11.2	Shutdown Philosophy		Four types of shut downs, viz: - ESD, OSD, Single Train Shutdown, and Unit Shutdown, - will be designed to be initiated depending on process upsets and their hazardous levels.	The process operating, control and start-up / shutdown philosophy to be detailed at FEED
11.2.1	Emergency Shutdown System (ESD)	The ESD will be designed to be initiated by fire, manual push buttons or by loss of instrument air supply		
11.2.2	Operational Shutdown (OSD)	The Operational Shut Down (OSD) will cut off the fluid supply from the pipelines as well as block the gas supply to the Petrochemical & Power Plants, albeit that an ESD will always initiate the OSD		
11.2.3	Single Train Shutdown	The single train shutdown will cut off all fluid supply and discharges from a single equipment train allowing the parallel equipment train(s) to continue operation		

Situations & Systems		Design Basis		Remark / Reference
11	Process Safety Provision & Preliminary EIA Study (Contd.)			
11.2	Shutdown Philosophy (Contd.)			
11.2.4	Unit Shutdown (USD)	The Unit Shutdown will stop a dedicated unit of the plant		
		On the other hand, for maintenance purposes, the Spec requires sectionalisation valves to be installed at pre-determined spacings on the Pipeline in order to reduce the volume of inventory to be evacuated in case damage to any section of the Pipeline. This will require the installation of small venting systems around the Sectionalisation Valve Stations which include of course the PLEM to vent the sections of the Pipeline requiring maintenance		
11.3	Hazardous Area Classification (HAC)		Will be in compliance with Shell DEP 80.00.10.10-Gen in cross reference with API RP 505 and IP 15	Shell DEP 80.00.10.10-Gen
11.3.1	Zones Classification		Zone 1 - where a flammable atmosphere is continuously present for long periods Zone 2 - where a flammable atmosphere is likely to occur in normal operation Zone 3 - where a flammable atmosphere is not likely to occur in normal operation and if it occurs it will exist only for short period	
11.3.2	Extent of Zone		The standard 6m Radius will be used as general guide in this Basic Design scope, while specifics & details will be covered at FEED	
11.3.3	Location of Equipment		Will be done with the consideration to avoid ignition of flammable hydrocarbons by minimising the probability of coincidence of a flammable atmosphere and a source of ignition.	
			In general will be more than 3m away from the launcher and receiver barrels which are the major points of release on the manifold.	

Situations & Systems		Design Basis	Remark / Reference
11	Process Safety Provision & Preliminary EIA Study (Contd.)		
11.4	Preliminary EIA Study	An environmental impact assessment is currently in progress for the gas plant site, and will also be carried out for the Pipeline Route and associated Manifold Platforms	
		Will cover the entire Plant Complex (Gas Plant, Power Plant, and the Petrochemical), the Pipeline Route, and the associated Manifold Platforms	
		Only the phase-1 Pipeline Route (BBOU - ELEP - AKAB) will be covered now. The other routes will be covered later, close to their on-stream dates, otherwise we will end up with obsolete EIA when their times come	
12	Power Generation	IPP at the Gas Plant to provide the power needs of the Gas Plant, the Pipelines & Manifold Platforms System, and the Petrochemical Plant Complex	Full Specification to be detailed by the IPP Scope
12.1	Primary Power System	Gas Turbines	N+1 philosophy, and to be synchronised
		Power Generation at 11 kV, and stepped down to 6.6 kV and 415 V for high voltage and low voltage supply to the Gas Plant and Manifold Platforms as required	Emergency → essential to safe start up / shutdown
12.2	Secondary Power System	Diesel Gen Set	Emergency → essential to safe start up / shutdown
		Power requirement and classification - normal or emergency depending on equipment specification.	Reference with Load List Vital & Essential Loads to be defined & agreed
12.2.1	Uninterruptible Power System	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	The Vital & Essential Loads to be defined & agreed
12.2.2	Solar Power Option for Emergency	Option for Emergency Back-up with Solar Power will be considered during FEED	

Situations & Systems		Design Basis	Remark / Reference
13	Power Supply		
13.1	Power Supply to the Remote Manifold	Underground Power Cable to the Manifold Platforms	
13.1.1	Power Supply Options	Six main options evaluated, from which underground power cable was selected	
		1. Underground Power Cable at all manifolds - Selected Option	Buried underground cables
		2. Overhead Power Cable at all manifolds	Power cables run through transmission towers
		3. Composite Power Cable (with FOC) at all manifolds	Composite cable with power and data transmission capability.
		4. Solar power at all manifolds	Photo-voltaic cells and power bank arrays with inverters
		5. Combination of Cable and Solar Power	Part underground and part photo-voltaic cell array
		6. IPP in all manifolds	Gas engines, thermo-electric generators and micro-turbine generators
13.1.2	Distribution of the Cable Power	Underground power cable buried along bulk pipeline From the Power Plant at Akabel to the Remote Manifold Platforms at ELEP, BBOU, NUNR, SEIB, and OPUG	Cable sizing with reference to Load requirement, ampacity, voltage drop and short-circuit current.
		Underground power cable buried along bulk pipeline From the Power Plant at Akabel to the AKAB PLEM	
		Atex (Ex) / NEMA Junction boxes provided along the MF for cable hook-up.	
13.1.3	Emergency & Essential Services Power Systems	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	vital load —> needed to run / be available during emergency s/d for safety of personnel
		Power requirement and classification - normal or emergency depending on equipment specification.	Reference with Load List Vital & Essential Loads to be defined & agreed
13.1.4	Uninterruptible Power System	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	The Vital & Essential Loads to be defined & agreed

Situations & Systems		Design Basis		Remark / Reference	
13	Power Supply (Contd.)				
13.1	Power Supply to the Remote Manifold (Contd.)				
	13.1.5	Solar Power Option for Emergency	Option for Emergency Back-up with Solar Power will be considered during FEED		
	13.1.6	MCC	Cubicle provision for supply of field MFs dependent on load requirement.	Cubicle provision and sizing dependent on load requirement	
	13.1.7	Earthing & Equipotential bonding	Earthing of MFs to desired earth resistance $\leq 4\Omega$	SHELL DEP 33.64.10.33-GEN	
	13.1.8	Lighting, small power and welding receptacle	Minimum illumination provision (10-20 lux) photocell controlled on MFs with Atex (EX) / NEMA lighting fixture. Small power provision for potable hand tools and welding socket provision	Will be addresses during FEED. Welding during shutdown TAM only	
			Perimeter and area lighting for facility location and manifold operation area. Small power welding receptacles to be provided with the hazardous area classification as a consideration.		
	13.1.9	Single Line Diagrams	Overall and individual single line diagrams to show flow of power supply from the power generation equipment to the required loads.		
	13.1.10	Load List	Detailed load list of each M/F based on equipment that require power supply, area and fence lighting.	To determine the load requirement for each M/F for adequate sizing of generation equipment.	
	13.1.11	Hazardous Area Classification	Will be in compliance with Shell DEP 80.00.10.10-Gen in cross reference with API RP 505 and IP 15	Shell DEP 80.00.10.10-Gen	
	13.1.12	Foot prints	Detail the extents of the selected option so as to have an estimated Manifold Platform size		
	13.2	Power Supply to the Gas Plant			
		13.2.1	Power Supply Options	Step Down of IPP Voltage from 11 kV to 6.6 kV & 415 V for Power Supply to the Gas Plant	
		13.2.2	Distribution of the Cable Power	Underground power cable from the Power Plant to the Gas Plant	Cable sizing with reference to Load requirement, ampacity, voltage drop and short-circuit current.
			Atex (Ex) / NEMA Junction boxes provided along the MF for cable hook-up.		

Situations & Systems		Design Basis	Remark / Reference
13	Power Supply (Contd.)		
13.2	Power Supply to the Gas Plant (Contd.)		
13.2.3	Emergency & Essential Services Power Systems	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	vital load → needed to run / be available during emergency s/d for safety of personnel
		Power requirement and classification - normal or emergency depending on equipment specification.	Reference with Load List Vital & Essential Loads to be defined & agreed
13.2.4	Uninterruptible Power System	UPS supply via dedicated panel connected to emergency bus (switchboard) to supply vital & essential load	The Vital & Essential Loads to be defined & agreed
13.2.5	MCC	Gas Plant Scope / Co-Design with MCC Vendor	
13.2.6	Earthing & Equipotential bonding	Earthing of Plant Skids to desired earth resistance $\leq 4\Omega$	Shell DEP 33.64.10.33-Gen
13.2.7	Lighting, small power and welding receptacle	Minimum illumination provision (10-20 lux) photocell controlled on Plant Units with Atex (EX) / NEMA lighting fixture. Small power provision for portable hand tools and welding socket provision	Will be addressed during FEED.
		Perimeter and area lighting for facility operation area. Small power welding receptacles to be provided with the hazardous area classification as a consideration.	Welding during shutdown TAM only
13.2.8	Load List	Will be addressed at FEED	
13.2.9	Hazardous Area Classification	Will be addresses at FEED, and in accordance with Shell DEP 80.00.10.10-Gen in cross reference with API RP 505 and IP 15	
		The location of equipment to be more than 3m away from the launcher and receiver barrels of the Plant's PLEM, and Process Vessels which are the major points of release of Hydrocarbon.	Intent is to minimise the probability of coincidence of a flammable atmosphere and a source of ignition.

3. Project Concept and Site for the Project

Ref: EA SPDC-BFPC UGS-GEP CSM RPT-001-A01_Concept Select Study – Project Concept & Proposed Site for the Project

The project's concept is to gather raw wellhead gas, up to the tune of 300 MMscfd, and transport same to a Gas Processing Plant for processing as feedstock to Petrochemical Plant, and Power Plant.

The gas sources cut across five fields, - Bubouwe Bou, Elepa, Nun River, Seibou, and Opugbene, - in OMLs 33, 32, & 36, – with locations widely spread, and on both sides of Nun River.

The gas gathering & evacuation concept constitutes a mix of wellhead flowlines, field manifolds, bulklines, remote satellite manifolds, evacuation manifolds, and trunkline, as shown in the schematics below, - optimally configured in terms of cost, as driven by manifolds layout and choice of evacuation manifold (Bubouwe Bou or Elepa)

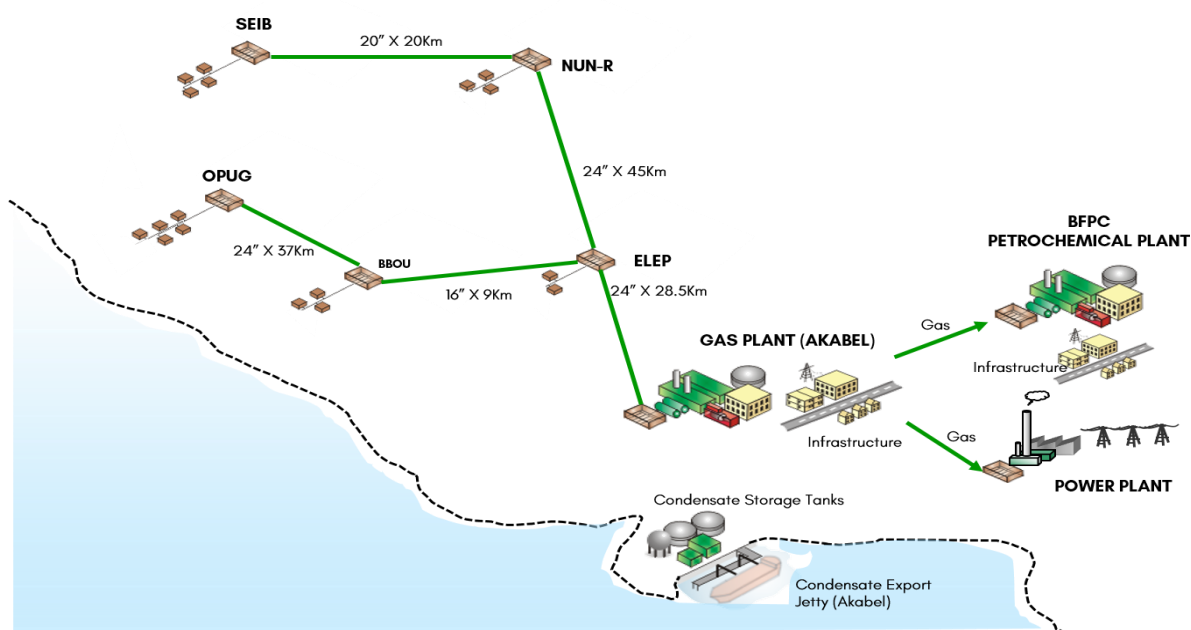


Figure 3.1: Regional Location Plan and the Gas Gathering & Evacuation Concept - Schematics

The Gas Plant will comprise three trains, (each with a processing capacity of 150 MMscfd, - corresponding to 300 MMscfd capacity at 3 x 50%); and condensate storage system capacity of 50,000 Tonnes.

It will be located in Akabel, Brass Island, in Brass LGA of Bayelsa State of Nigeria, near the shores of Atlantic Ocean, - in the same complex as the Petrochemical Plant & Power Plant it will be supplying the processed gas.

3.1. The Gas Plant Site

The aerial photograph showing the plant site is shown below, holding all the three plants, - the Gas Plant, the Petrochemical Plant, and the Power Plant.

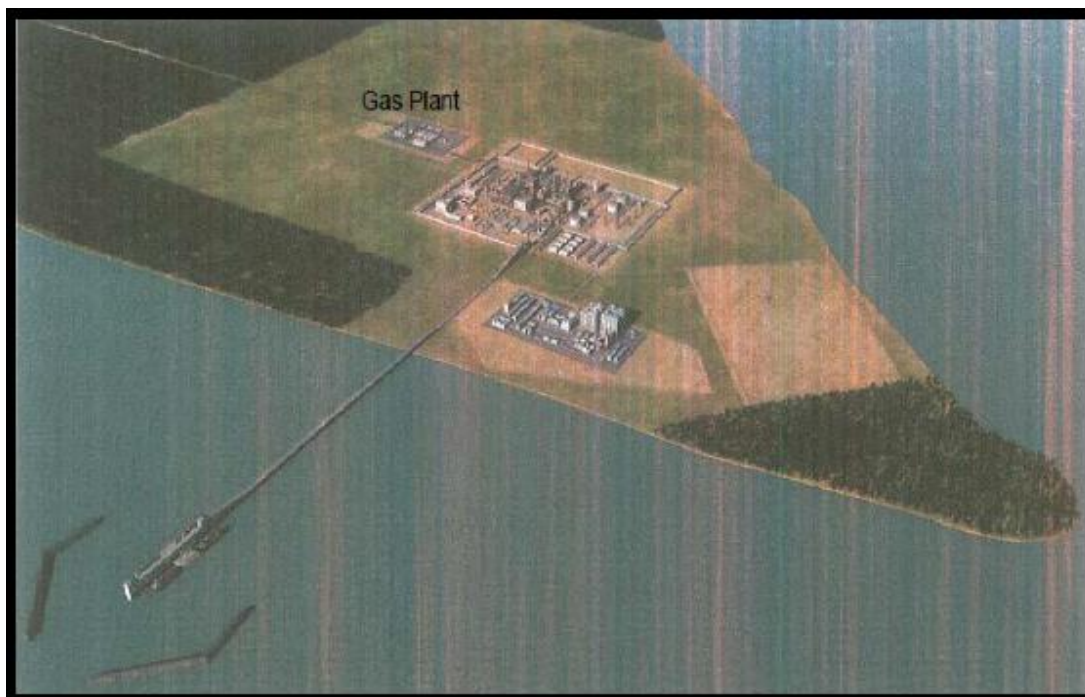


Figure 3.2: Aerial Photograph of Plants Site

3.2. The Gas Gathering & Evacuation Concept

The gas gathering concept & the gas transportation to the Gas Plant will be as follows:

1. Gathering of each field well production through its wellhead flowline to the Field's Central Manifold Header (FC-MH) / Field's Central Manifold (F-MF).
2. Transportation of the F-MF inventory through Bulkline to a remote Satellite Manifold (S-MF), or to the Evacuation Manifold (E-MF), as the case may be for fields (BBOU & ELEP) that are within the vicinity of the E-MF.
3. Transportation of the S-MF inventory through Bulklines to the E-MF.
4. Transportation of the E-MF inventory through a Trunkline to the Pipeline End Manifold (PLEM) at Akabel, and from there as feedstock inlet stream to the Gas Processing Plant.

The selected wellhead gas gathering and evacuation concept is shown below, with on-stream schedule for the various phases. The selection resulted from CAPEX comparison of six options that made the final shortlist in the concept select exercise.

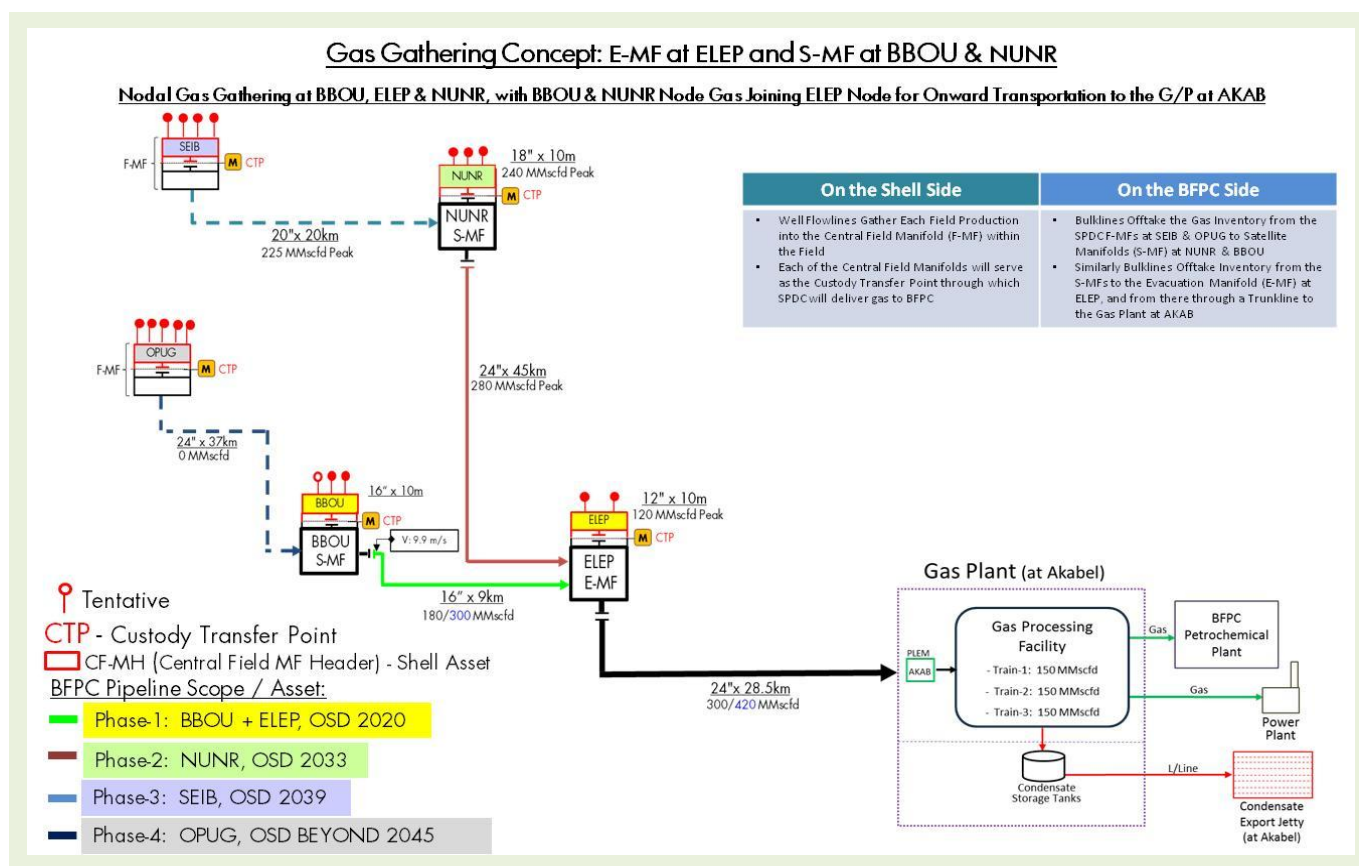


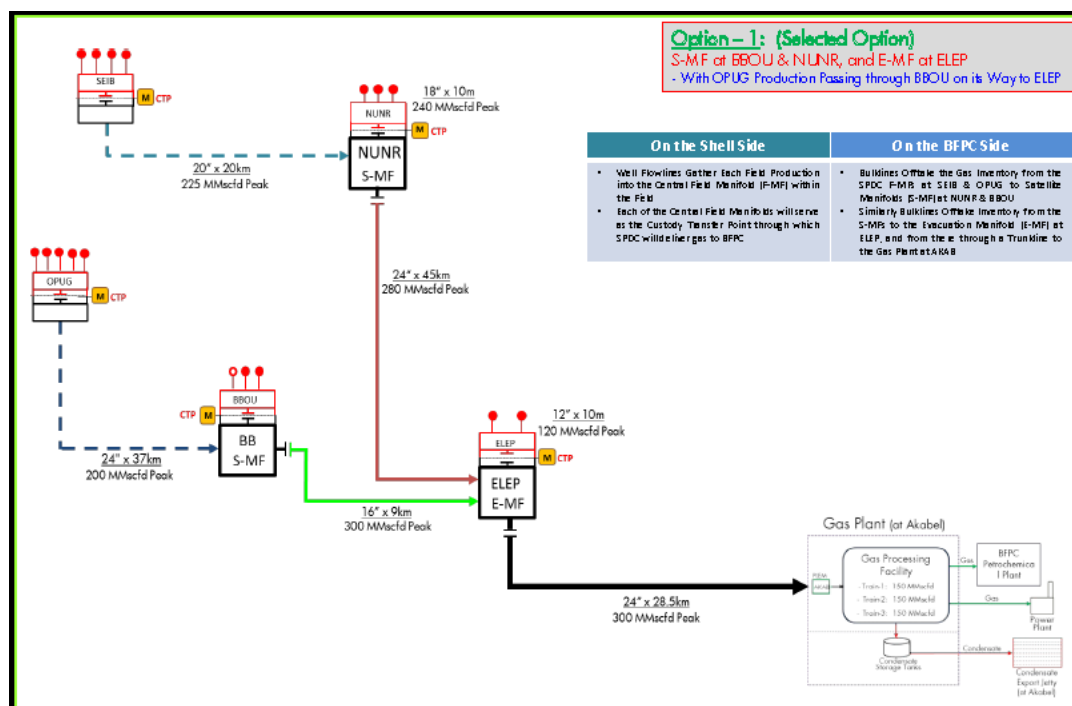
Figure 3.1: Wellhead Gas Gathering & Flow Schematics to Gas Processing Plant

The gas will be delivered raw to the gas processing plant for NGL extraction and conditioning of the lean gas to specifications of the petrochemical / power plants.

3.3. The Gas Gathering & Evacuation Options

The options considered and their installation cost estimates are shown as follows:

Option – 1: S-MF at BBOU & NUNR, and E-MF at ELEP - With OPUG Production Passing through BBOU to ELEP
Capex Estimate **\$469.9 mln** - Eliminates need for BBOU – ELEP 9km section for the 24" OPUG B/L.



Option – 2: S-MF at NUNR & E-MF at ELEP - With OPUG Production Direct to ELEP
Capex Estimate **\$501 mln**

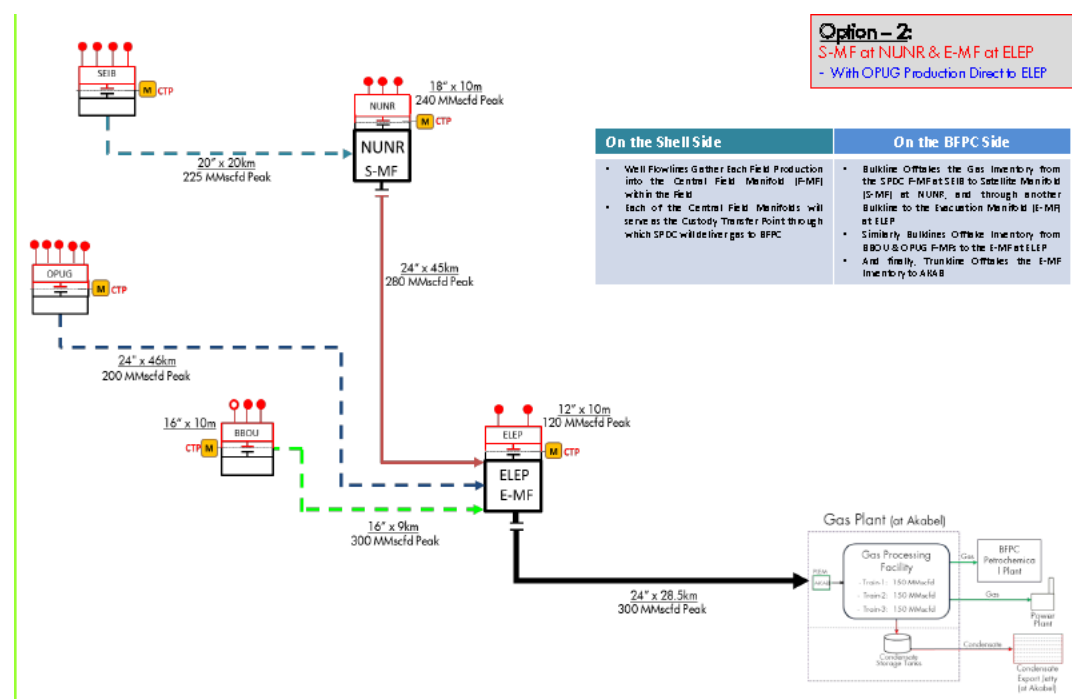
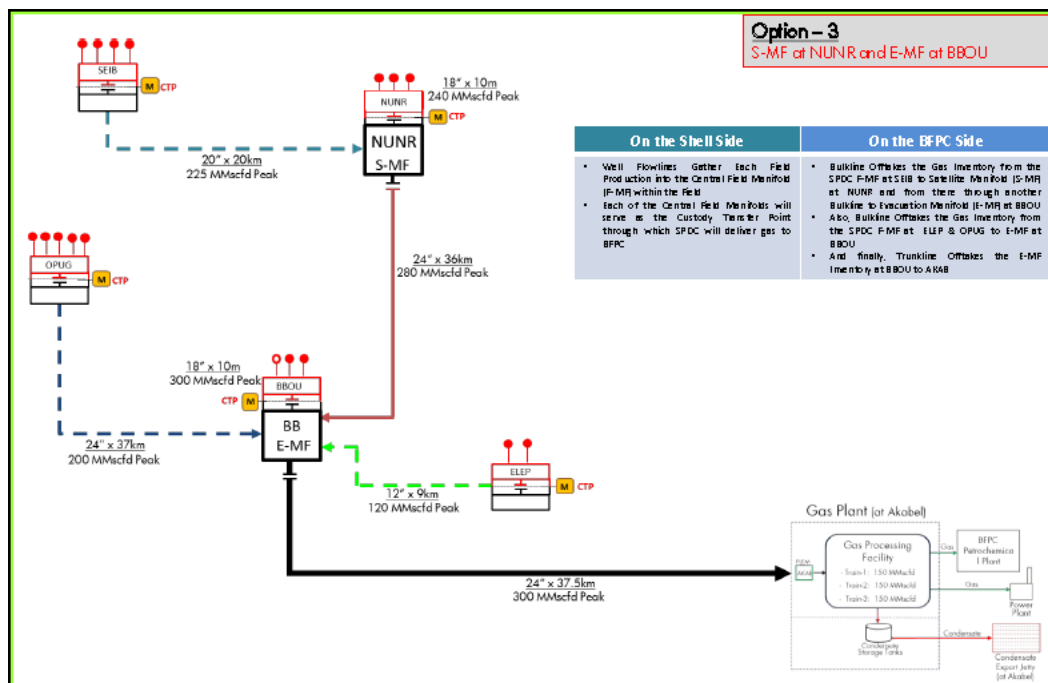


Figure 3.2: Wellhead Gas Gathering & Flow Schematics – Options 1 & 2

Option – 3: S-MF at NUNR, and E-MF at BBOU
Capex Estimate **\$497.9 mln**



Option – 4: S-MF at DBUC & E-MF at ELEP

Capex Estimate **\$449.8 mln** – Cost attractive but with S-MF at an external location at DBUC, - a situation not acceptable to BFPC owing to the additional ER risk it will pose

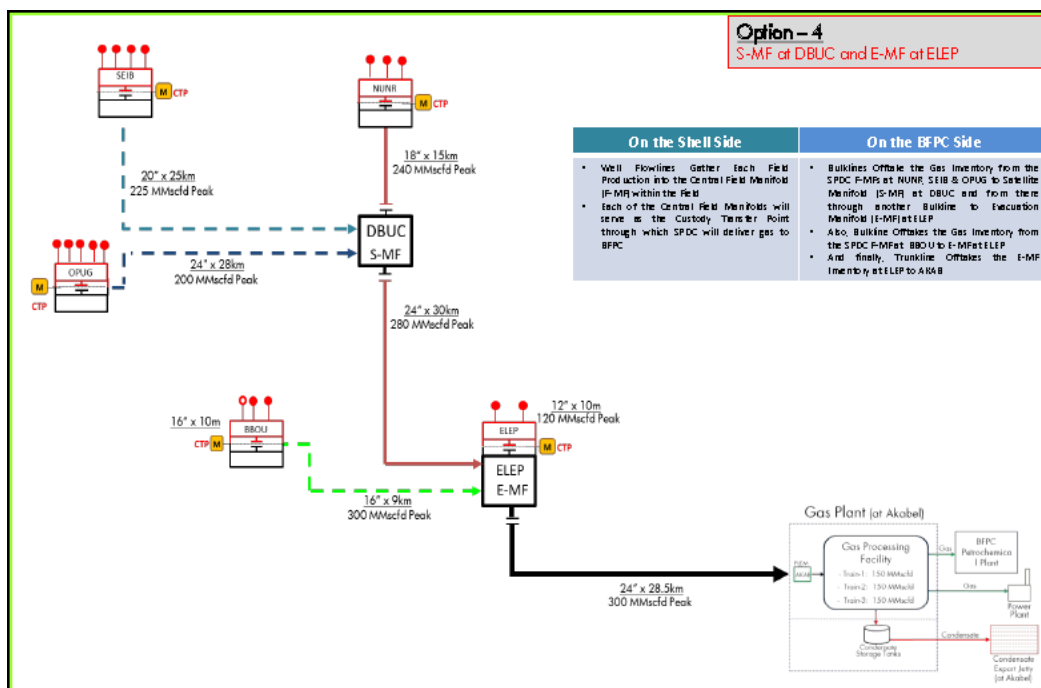
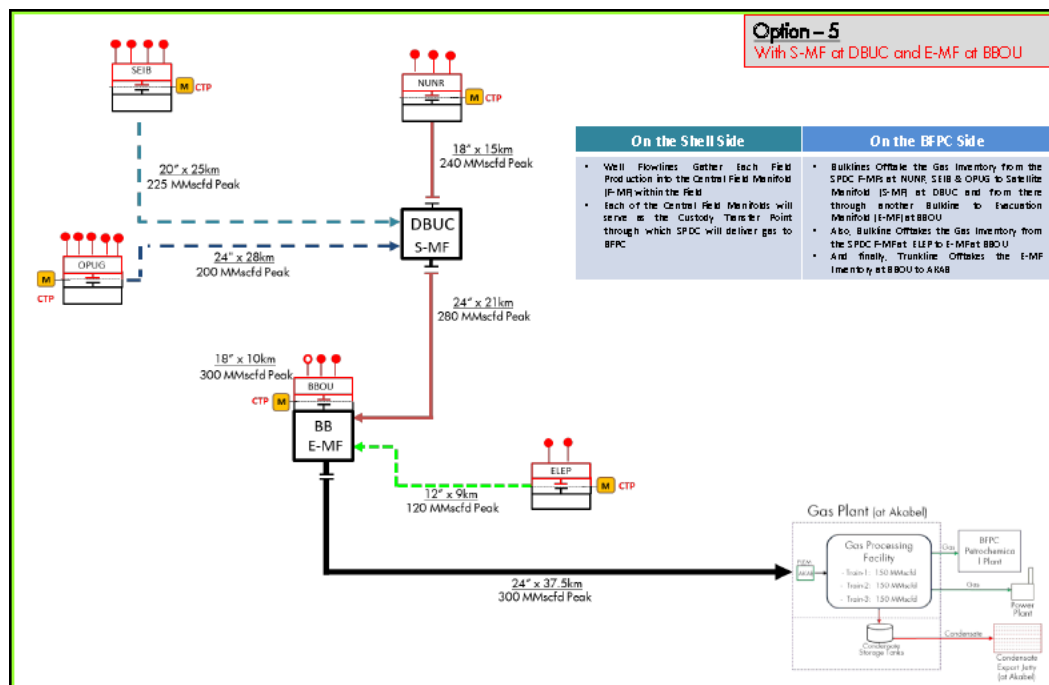


Figure 3.3: Wellhead Gas Gathering & Flow Schematics – Options 3 & 4

Option – 5: S-MF at DBUC, and E-MF at BBOU

Capex Estimate **\$446.3 mln** – **Cost Attractive / Lowest** but with S-MF at an external location at DBUC, - a situation not acceptable to BFPC owing to the additional ER risk it will pose



Option – 6: Bulkline From Each Field Direct to AKAB

Capex Estimate **\$926.3 mln** - **Demonstrates the high cost of individual manifold transport, compared with satellite concept**

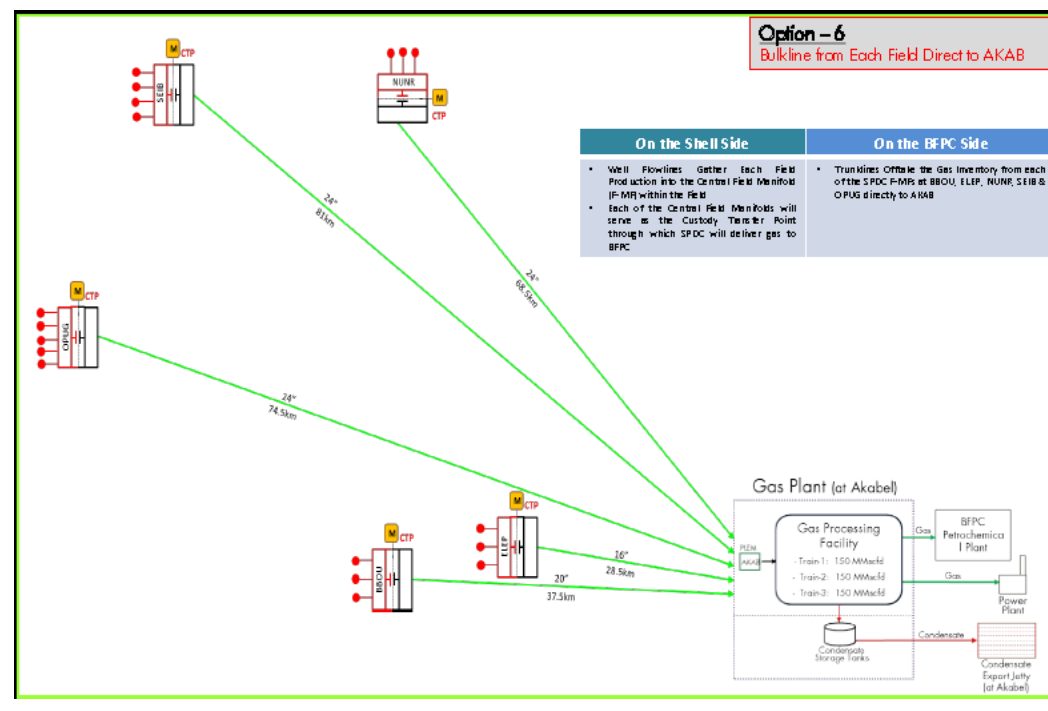


Figure 3.4: Wellhead Gas Gathering & Flow Schematics – Options 5 & 6

3.4. The Manifolds Concept

Three categories of manifolds were conceptualised, - Field Manifold (F-MF), Satellite Manifold (S-MF), and Evacuation Manifold (E-MF).

1. Field Central Manifold (F-MF) that receives the field's production gathered through its wellheads flowlines to the Field's Central Manifold Header (FC-MH).
2. Satellite Manifolds (S-MF) which receives inventory from the Fields' Manifolds (F-MF) through their bulklines. There are two satellite manifolds, - at BBOU and NUNR.
3. Evacuation Manifold (E-MF) which receives inventory from the Satellite Manifolds (S-MF) through their bulklines. The evacuation manifold serves as the last gathering point of all fields production, and delivers same to the gas processing plant.

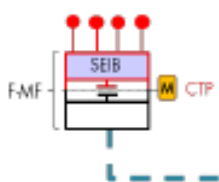
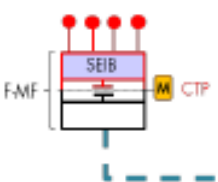
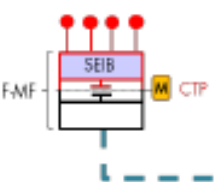
Field Manifold (F-MF) At SIEB & OPUG	Satellite Manifold (S-MF) At BBOU & NUNR	Evacuation Manifold (E-MF) At ELEP
		
At SIEB & OPUG	At BBOU & NUNR	At ELEP

Figure 3.5: Manifold Concept & Categories

A desirable situation will be to power the manifolds to enable automation, - motorisation of valves, etc, and in that case underground armoured cables will be installed to transport electricity from the IPP at Akabel to the various remote manifolds. However, in as much as that will be ideal, the concept will be reviewed during FEED in the light of cost savings, and risk assessment with particular reference to widespread vandalism of such facilities in the region, - to re-confirm that the concept will be worth doing. Otherwise, the platform will be designed without the concept of automation, with provision only for solar power to provide basic warning lighting at night.

3.5. The Gas Processing

The proposed gas plant will be full spectrum gas processing plant with capability for NGL extraction, condensate stabilisation & storage, and conditioning of the lean gas to specifications of the petrochemical & power plants, in the following main steps:

1. Gas Separation
2. Gas Drying
3. NGL Recovery
4. Fractionation
5. Condensate Stabilisation

The plant will be located in Akabel, in the same complex as the Petrochemical Plant & Power Plant it will be supplying the processed gas, but 28.5 km downstream of the evacuation manifold at Elepa from where it will be receiving its feedstock of wet gas.

It will comprise three trains, (each with a processing capacity of 150 MMscfd) and condensate storage capacity of 50,000 Tonnes. Two trains will be installed initially, in strategic fit with the current GSPA for 270 MMscfd ACQ, while the 3rd train will be installed later, as will be dictated by additional GSPA.

Processing of gas will start with phase-1 production, - initially from BBOU, then joined by ELEP a couple of years later, and in both cases with peak receipt of 300 MMscfd.

The process flow scheme of the process systems of the gas plant is shown in Fig 3.6 below, including the expected products yields for 300 MMscfd wet gas feed from the BBOU / ELEP phase-1 production, with indications for 150 MMscfd and 450 MMscfd.

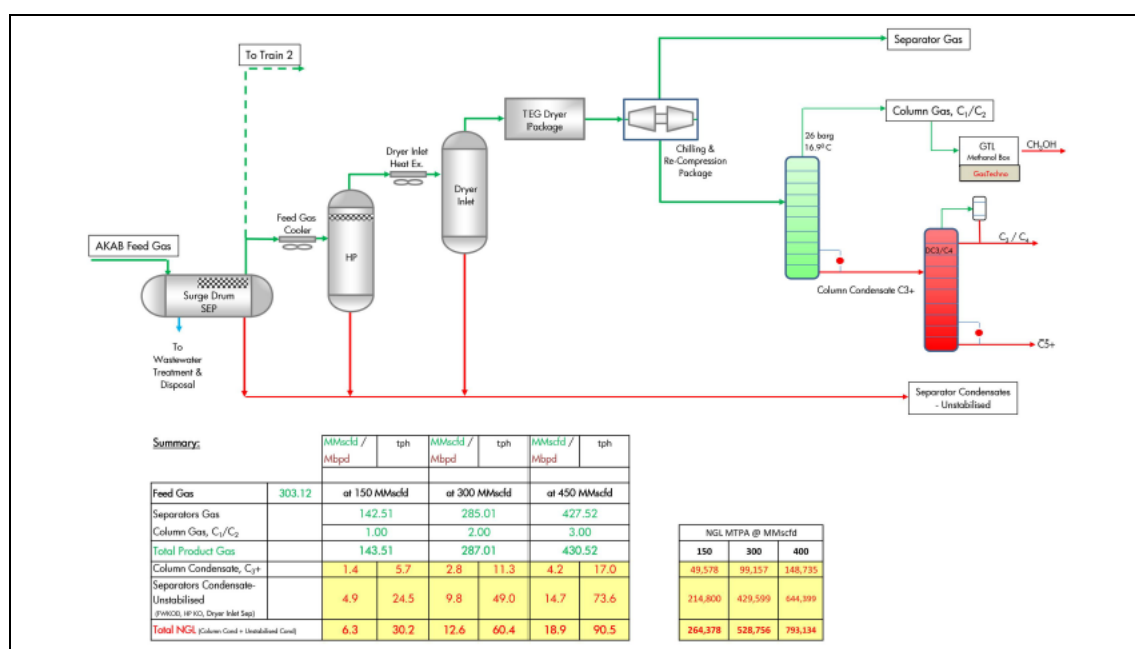


Figure 3.6: Process Flow Scheme – AKAB Gas Processing Plant

options considered and their installation cost estimates are shown in table 3.1:

Table 3.1: Wellhead Gas Gathering & Flow Schematics – Options 1, 2, & 3.

	Option Description	Cost Estimate \$mln	Schematics	Remarks
1	S-MF at NUNR, and E-MF at ELEP With OPUG Production Passing through BBOU on its Way to ELEP	469.9	Option – 1: (Selected Option) S-MF at BBOU & NUNR, and E-MF at ELEP - With OPUG Production Passing through BBOU on its Way to ELEP On the Shell Side - Wellhead Gas Gathering Field Production into the Central Field Manifold (CFM) within the field. - Each of the Central Field Manifolds will serve as the primary transfer point through which SPDC will deliver gas to BFPC. On the BFPC Side - Building Offshore Gas Inventory from the SPDC into a S-MF & OPUG on Shell Side Manifold (S-MF) at NUNR. - Building Offshore Gas Inventory from the Shell Side Manifold (S-MF) at NUNR into a S-MF & OPUG on Shell Side Manifold (S-MF) at BBOU. - Building Offshore Gas Inventory from the Shell Side Manifold (S-MF) at BBOU into a S-MF & OPUG on Shell Side Manifold (S-MF) at ELEP. - Building Offshore Gas Inventory from the Shell Side Manifold (S-MF) at ELEP into a S-MF & OPUG on Shell Side Manifold (S-MF) at Akabab.	Savings on 9km installation cost for OPUG 24" B/L. Though constrained to reduced 16" conduit to ELEP, but that is manageable
2	S-MF at NUNR, and E-MF at ELEP With OPUG Production Direct to ELEP	501	Option – 2: S-MF at NUNR & E-MF at ELEP - With OPUG Production Direct to ELEP On the Shell Side - Wellhead Gas Gathering Field Production into the Central Field Manifold (CFM) within the field. - Each of the Central Field Manifolds will serve as the primary transfer point through which SPDC will deliver gas to BFPC. On the BFPC Side - Building Offshore Gas Inventory from the SPDC into a S-MF & OPUG on Shell Side Manifold (S-MF) at NUNR. - Building Offshore Gas Inventory from the Shell Side Manifold (S-MF) at NUNR into a S-MF & OPUG on Shell Side Manifold (S-MF) at BBOU. - Building Offshore Gas Inventory from the Shell Side Manifold (S-MF) at BBOU into a S-MF & OPUG on Shell Side Manifold (S-MF) at ELEP. - Building Offshore Gas Inventory from the Shell Side Manifold (S-MF) at ELEP into a S-MF & OPUG on Shell Side Manifold (S-MF) at Akabab.	

3	S-MF at NUNR, and E-MF at BBOU	497.9	Option - 3 S-MF at NUNR and E-MF at BBOU On the Shell Side - Well Headlines Gather each field production into the Central Field Manifold (CFM) within the field. - Each of the Central Field Manifolds will serve as the custody transfer point through which SPDC will deliver gas to BFPC. On the BFPC Side - Building Offtake the Gas Inventory from the SPDC Unit at 318 to Shell Side Manifold (S-MF) at NUNR and from there through another Building to Intermediate Manifold (I-MF) at BBOU. - Also Building Offtake the Gas Inventory from the SPDC Unit at 318 to OPUG on Shell at BBOU. - And finally, Building Offtake the Gas Inventory at BBOU to AKAB.	Savings on 9km installation cost for OPUG 24" B/L. Though constrained to reduced 16" conduit to ELP, but that is manageable
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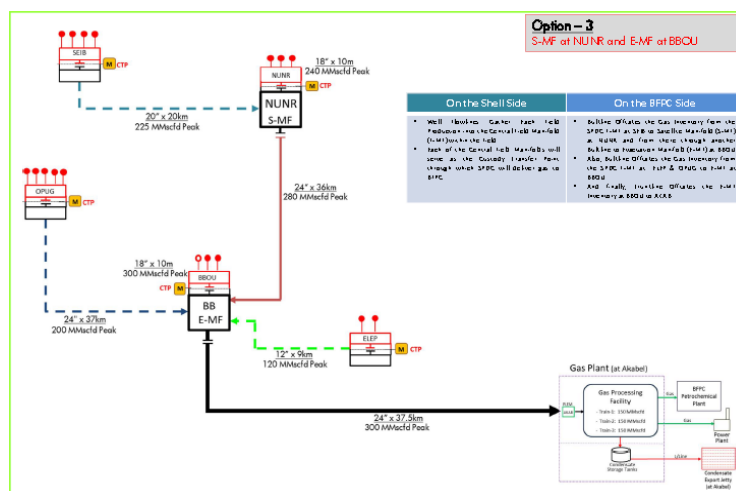
Table 3.1 (contd.): Wellhead Gas Gathering & Flow Schematics – Options 4, 5, & 6.

	Option Description	Cost Estimate \$mln	Schematics	Remarks
4	S-MF at NUNR, and E-MF at ELP With OPUG Production Passing through BBOU on its Way to ELP	469.9	Option - 1: (Selected Option) S-MF at BBOU & NUNR, and E-MF at ELP - With OPUG Production Passing through BBOU on its Way to ELP On the Shell Side - Well Headlines Gather each field production into the Central Field Manifold (CFM) within the field. - Each of the Central Field Manifolds will serve as the custody transfer point through which SPDC will deliver gas to BFPC. On the BFPC Side - Building Offtake the Gas Inventory from the SPDC Unit at 318 to Shell Side Manifold (S-MF) at NUNR and from there through another Building to Intermediate Manifold (I-MF) at BBOU. - Also Building Offtake the Gas Inventory from the SPDC Unit at 318 to OPUG on Shell at BBOU. - And finally, Building Offtake the Gas Inventory at BBOU to AKAB.	Savings on 9km installation cost for OPUG 24" B/L. Though constrained to reduced 16" conduit to ELP, but that is manageable
4	S-MF at NUNR, and E-MF at ELP With OPUG Production Passing through BBOU on its Way to ELP	501	Option - 2 S-MF at NUNR & E-MF at ELP - With OPUG Production Direct to ELP On the Shell Side - Well Headlines Gather each field production into the Central Field Manifold (CFM) within the field. - Each of the Central Field Manifolds will serve as the custody transfer point through which SPDC will deliver gas to BFPC. On the BFPC Side - Building Offtake the Gas Inventory from the SPDC Unit at 318 to Shell Side Manifold (S-MF) at NUNR and from there through another Building to Intermediate Manifold (I-MF) at BBOU. - Also Building Offtake the Gas Inventory from the SPDC Unit at 318 to OPUG on Shell at BBOU. - And finally, Building Offtake the Gas Inventory at BBOU to AKAB.	Savings on 9km installation cost for OPUG 24" B/L. Though constrained to reduced 16" conduit to ELP, but that is manageable

6

S-MF at NUNR, and E-MF at ELEP
With OPUG Production Passing through BBOU on its Way to ELEP

497.9



Savings on 9km installation cost for OPUG 24" B/L.

Though constrained to reduced 16" conduit to ELEP, but that is manageable

4. Feedstock Supply & Product Evacuation Scheme

Ref: EA SPDC-BFPC UGS-GEP CSM RPT-001-A01_Concept Select Study – Project Concept & Proposed Site for the Project

The feedstock supply and product evacuation involves the gathering of the fields' raw wellhead gas, and transporting same through a network of bulklines and trunkline to the Gas Processing Plant for processing into lean gas and condensate.

The lean gas is supplied as feedstock to the Petrochemical Plant for methanol & urea production, and also to the Power Plant for power generation.

The condensate is exported via a delivery to the condensate export jetty from where offtakers take delivery. The condensate export jetty is within the BFPC plants' captive export jetty, from where all plants' products (condensate, methanol, and urea) are exported.

4.1. The Feedstock & Products Evacuation Schematics

The feedstock and product evacuation schematics is shown below, - involving gathering and transporting the fields' raw wellhead gas via a network of flowlines & pipelines to the Gas Processing Plant, and the delivery of the Gas Plant lean gas (product) to the the Petrochemical & Power Plants, and the condensate (product) to the condensate export jetty for export to buyers.

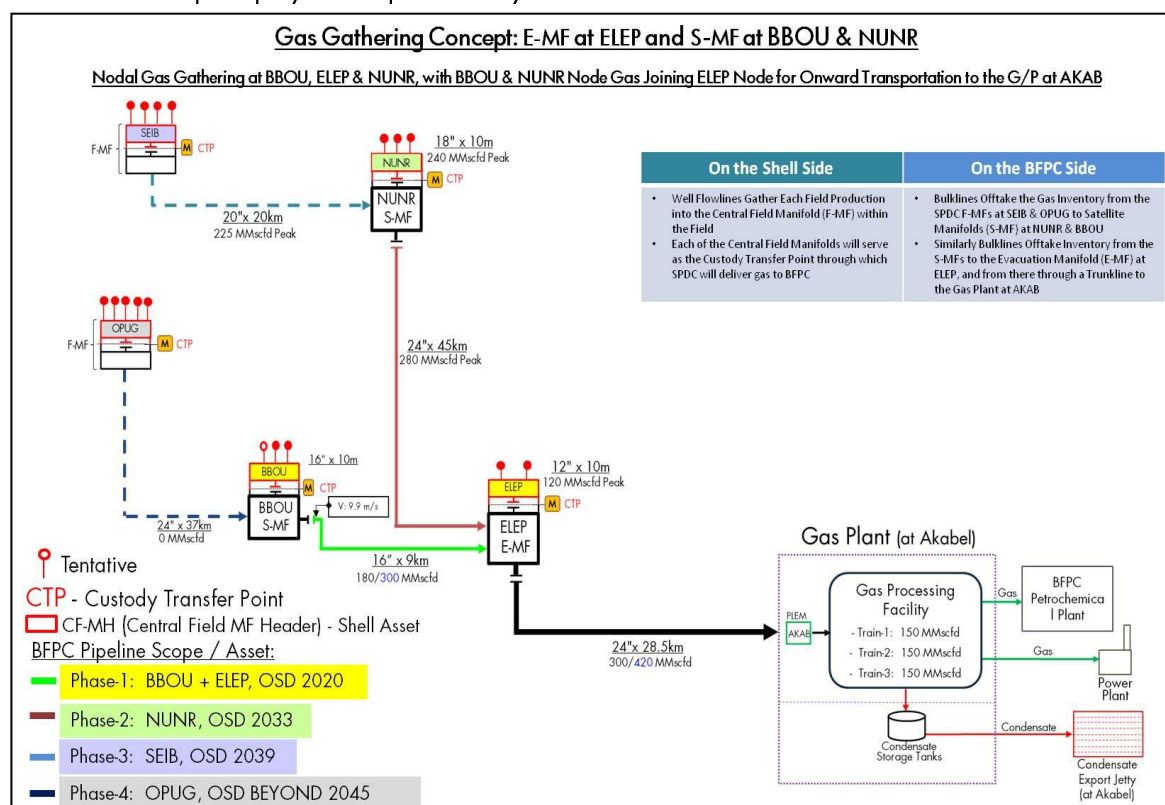


Figure 4.1: Feedstock & Products Evacuation Schematics

5. Pipeline Route Selection

Ref: EA SPDC-BFPC UGS-GEP SVY RPT-001-A01_Concept Select Study – Pipeline Route Selection Study

The Pipeline Route Selection focused on the requirement for phase-1 production, which involves Bubouwe & Elepa gas transportation to Akabel Gas Plant, and has prescribed an optimal route based on desktop studies. Indicative routes have also been earmarked for requirement beyond phase-1 production, but proper desktop studies will be done for each of the phases when the time draws near.

Ideal to follow the 'desk-top' selected route was the field investigation to evaluate its viability, involving consultations with relevant stake holders. That was planned but could not be carried out owing to security risk and excessive FTO demand at the time. It was therefore decided to carry out the investigation as part of the full preparatory (FP) survey by which time the security and FTO issues would have been fully addressed.

5.1. Pipeline Route for Phase-1 Production

The pipeline route for phase-1 is shown in Fig 5.1 below, - a composite route comprising two segments, viz: BBOU - ELEP, and ELEP – AKAB.



Figure 5.1: Composite Plan of the Selected Pipeline Route – for Phase-1 Production

5.2. Routes Options Considered

Three routes shown in Fig 5.2 below were in the final screening, with the 9 km BBOU – ELEP segment common to all of them, viz:

1. The Proposed Route, - Total Length 37.5 km, comprising: 9km BBOU – ELEP, and **28.5 km ELEP – AKAB**.

This route is the **shortest of the three**, with moderate river crossing lengths (± 1 km) that will not pose extra-ordinary construction challenges.

2. The Alternative Route 'A', Total Length 40 km, comprising: 9km BBOU – ELEP, and **31 km ELEP – AKAB**.

This route is 2.5 km longer than the proposed route, and with some 4 km wide river crossing length that will pose extra-ordinary construction challenges.

3. The Alternative Route 'B', Total Length 51 km, comprising: 9km BBOU – ELEP, and **42 km ELEP – AKAB** predominantly offshore.

This route is the longest, and with about 35 km of the ELEP – AKAB segment in Brass River / Offshore Coastal water, - rendering it the most expensive to install. The idea behind route was to minimise traverse through community settlements, but it turned out to be that it is equally within view and reach of those settlements it aimed to avoid. Furthermore from intelligence report gathered, the route is not what NIMASA will be disposed to support, - as it will compromise on dredging programme that may arise in respect of waterways navigation in the area.

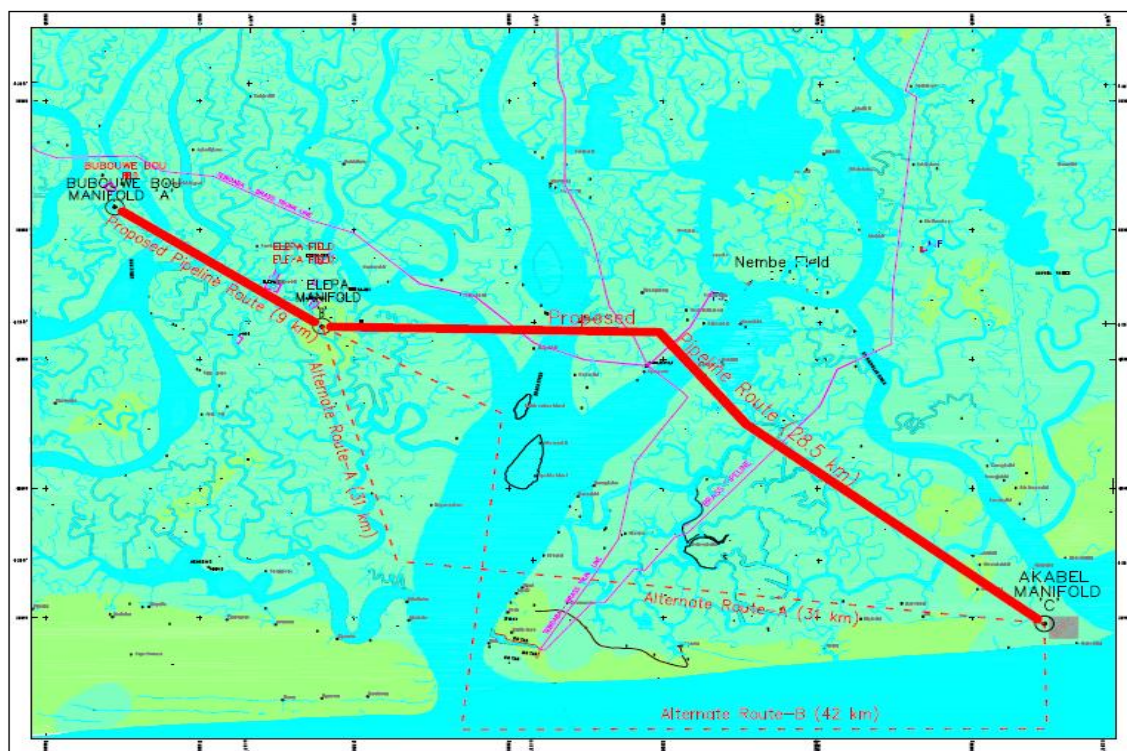


Figure 5.2: Alternative Pipeline Routes – for Phase-1 Production

The features of the three routes are summarised in table 5.1 below; based on which the proposed route is assessed the most acceptable of the three.

Table 5.1: Assessment Matrix of the Proposed Route & Alternative Routes.

s/n	Description	Features	Positive	Negatives
1	Proposed Route	1. Distance 1.1, BB-Elepa = 9km 1.2, Elepa – Akabel = 28.5 km. 2. Key Communities 2.1, Elepa (Nembe) 2.2, Golubokiri (Nembe) 2.3, Kpongbokiri (Nembe) 2.4, Fantuo (Fantuo) 2.5, Oyankia (Oodioama) 2.6, New Jerusalem (Oodioama) 2.7, Akabeleu-Ama (Oodioama) 3. LGAs 3.1, Southern Ijaw 3.2, Nembe 3.3, Brass 4. River Crossings 4.1, Major Crossings (3 x 1.2km). 4.2 Minor Crossings(13 nos)	1. Shortest in length. 2. Easily manageable river crossing lengths.	No significant challenges.
2	Alternative Route 'A'	1. Distance 1.1, BB-Elepa = 9km 1.2, Elepa – Akabel = 31km. 2. Key Communities 2.1, Elepa (Nembe) 2.2, Ekwema (Ekwema) 2.3, Twon-Brass (Twon-Brass) 2.4, Ewoama (Ewoama/Okpoama) 2.5, Okpoama (Okpoama) 2.6, Diema (Diema) 2.7, Oyankia (Oodioama) 2.8, Akabeleu-Ama (Oodioama) 3. LGAs 3.1, Southern Ijaw 3.2, Nembe 3.3, Brass 4. River Crossings 4.1, Major Crossings (1 x 4km) 4.2 Minor Crossings(13 nos)	Only one major river crossing	Key Installation challenge of 4km river crossing by HDD. Though feasible.

Table 5.1 (contd): Assessment Matrix of the Proposed Route & Alternative Routes

s/n	Description	Features	Positive	Negatives
3	Alternative Route 'B'	1. A section of Route runs near parallel to the sea front. This will pose enormous engineering, construction and logistics challenges, including the Brass estuarine crossing 2. Distance 2.1 BB-Elepa = 9km 2.2, Elepa – Akabel = 42km. 3. Key Communities 3.1, Elepa (Nembe) 3.2, Brass 4. LGAs 4.1, Southern Ijaw 4.2, Nembe 4.3, Brass 5. River Traverse 5.1, 20% on land 5.2, 30% in Brass River. 5.3, 50% Offshore.	No significant benefit.	1. Engineering and installation challenges and high cost associated with offshore terrain 2. Challenge of managing estuarine wave turbulence during installation works 3. Rapid sea bed dynamics that will be experienced in the region 4. Exposure to damages in case of possible future dredging works to improve Brass River navigability.

Note:-

Recommended major river crossing method for the pipeline, power and telecommunication cables shall be by HDD process through a single hole in one pull-pass. While the minor river/creek crossing shall be by open cut.

5.3. Field Investigation Works

The field investigation works that will be done along with the FP Survey will address the following:

1. The river and creek crossing points shall be identified and investigated for construction convenience and other potential hazards and constrains.
2. Assessment of the elevation difference at the overland area beyond the rivers/creeks banks, including height of adjacent banks. Abrupt changes of elevation (i.e. steep banks) to be avoided.
3. Due consultation shall be held with the local communities to identify if any, some sacred or forbidden bush, biodiversity, legacy, traditional concerns.
4. Due consultation shall be held with the relevant government agencies for information of planned future land use and any environment requirements.

5.3.1. Other Considerations

Other key considerations to the route selection are:

1. Safety of the public and personnel - the route must provide a safe and secured environment for the pipeline during construction and over its operational life and ideally be routed away from populated areas.
2. Minimum negative impact on the environment, nature, biodiversity and minimal land take.
3. Ability to meet the project's economic objectives, without compromising safety and environment, and minimizing local economy impact on communities that the pipeline passes through, and have the smallest footprint feasible (ideally the shortest distance between pipeline start and end points).
4. Land ownership related factors e.g. the number of landowners, anticipated ease and cost to obtain/purchase consents.
5. Contents of the pipeline and operating conditions, e.g. consideration of leakage of a high vapour pressure liquids.
6. Terrain and subterranean conditions, including geotechnical, hydrographical, and meteorological conditions should be acceptable. These include ground stability, and other land uses which may create instability (e.g. sand mining subsidence, land development).
7. Cultural heritage sites, should be strictly identified and avoided.
8. Existing and future land plan usage, determinable through research of public records and consultation with the local land owners & government planning agencies which should identify:-
 - Third-party activities
 - Agricultural practice
 - Facilities and services
 - Future development requirements
 - Permanent access; – the pipeline must ideally be accessible for inspection and maintenance all year round over its operational life
 - Security; - the pipeline system should be routed to minimise security concerns, particularly due to trespass and sabotage, during both construction and operation.
 - Throgh out the operational life span, the pipeline route limits shall as far as possible be made visible, either by regular devegetation or marking with above markers.
 - Other unforeseen hazards.

5.4. Right of Way (ROW) Acquisition

Construction Right of Way of 30 m width is proposed. The idea is to make sufficient provision for the various requirements as shown in the sketch below.

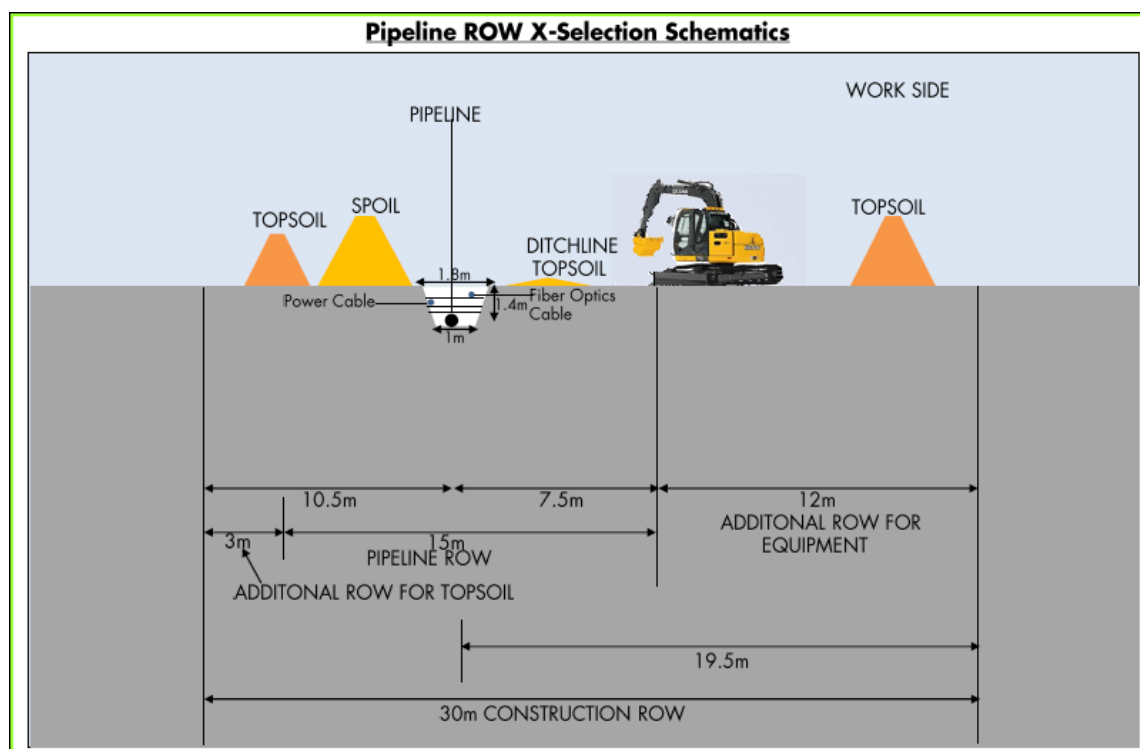


Figure 5.2: Alternative Pipeline Routes – for Phase-1 Production

5.5. PTS Application Plan

PTS Application plan followed DPR guide for processing as two composite gas pipelines: 16" x 9km Bubouwe – Elepa segment, and 24" x 28.5km Elepa - Akabel segment, as shown in the sketch below.

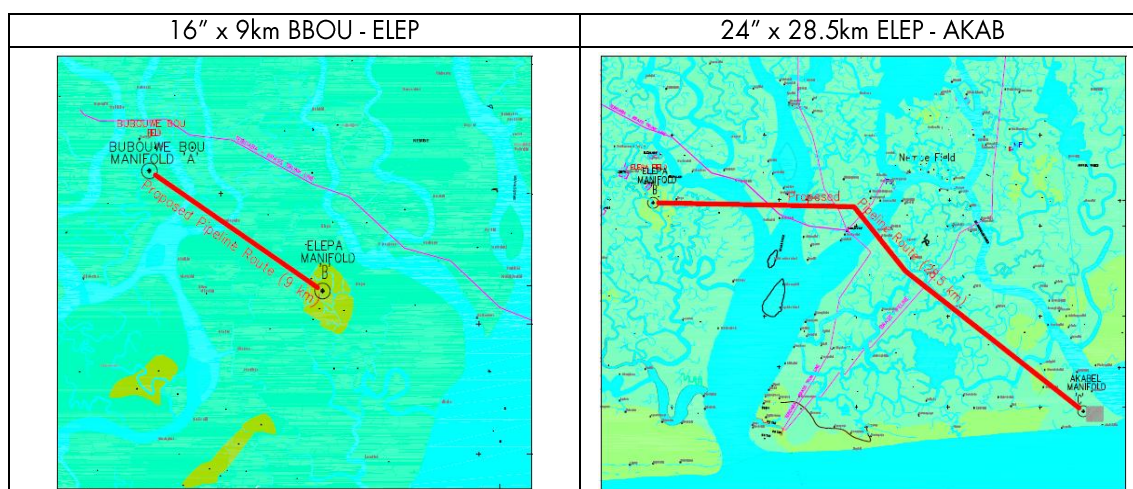


Figure 5.2: Alternative Pipeline Routes – for Phase-1 Production

The proposed pipeline route skirts major community settlements and traverses mangrove vegetation with multiple river and creek crossings. Four existing pipelines have been identified and will be duly crossed. It is currently perceived that the entire pipeline route traverses three Local Government Areas of Bayelsa State: - Southern Ijaw, Brass and Nembe Local Government Areas.

The start and end point co-ordinates in Geographic Co-ordinate System (referenced to Minna Datum) of the 16" x 9km Bubouwe Bou Manifold "A" to Elepa Manifold "B", pipeline with wall thickness of 0.875 inches are as contained in Table-1.

Table 5.2: Bubouwe Bou Manifold "A" to Elepa Manifold "B"

Point	Easting (m)	Northing (m)	Latitude	Longitude
Start 'A'	407,242.91	50,888.32	04° 27' 23.46"N	06° 07' 39.49"E
End 'B'	413,949.37	46,280.51	04° 24' 54.22"N	06° 11' 17.36"E

The start and end point co-ordinates in Geographic Co-ordinates System (referenced to Minna Datum) of 24" x 28.5km Elepa Manifold "B" to Akabel Manifold "C" pipeline with wall thickness of 1.25 inches are as contained in Table-2.

Table 5.3: Elepa Manifold "B" to Akabel Manifold "C"

Point	Easting(m)	Northing(m)	Latitude	Longitude
Start 'B'	413,949.37	46,280.51	04° 24' 54.22"N	06° 11' 17.36"E
End 'C'	437,351.70	34,771.29	04° 18' 41.92"N	06° 23' 57.06"E

5.6. The Full Preparatory Survey

The Full Preparatory (FP) Survey will be carried out following receipt of PTS approval, and all field investigation works will then be carried out along side / as part of the FP Survey.

6. Process Simulation

Ref: EA SPDC-BFPC UGS-GEP PRC RPT-001-A01_Concept Select Study – Process Simulation - HYSYS

Ref: EA SPDC-BFPC UGS-GEP PRC RPT-002-A01_Concept Select Study – Process Simulation - PIPESIM

Ref: EA SPDC-BFPC UGS-GEP PRC RPT-003-A01_Concept Select Study – Process Simulation - OLGA

Ref: EA SPDC-BFPC UGS-GEP PRC RPT-004-A01_Concept Select Study – Gas Plant Process Simulation - HYSYS

This chapter summarises the process simulation results of the above reference Process Simulation Studies which are as follows:

Enablers to the FA

Pressure drop is such that we can offtake at inlet and arrive destination at stipulated pressure band.

Velocity, ok – b/w min & max

Hydrate – nil (either as it is or under treatment.

Material that will corrode acceptably (minimally) or with CI treatment to stay within limit that will deliver over the life cycle.

Etc

last life Corrosion or treatment

Deliverin

1. Steady State Process Simulation of the Pipeline Network, using HYSYS
2. Steady State Process Simulation of the Pipeline Network, using PIPESIM
3. Transient State (Dynamic) Process Simulation of the Pipeline Network, using OLGA
4. Steady State Process Simulation of the Gas Plant Process, using HYSYS

The objective of the simulations is in two folds:

- ✚ To establish a gas gathering concept, with the appropriate line sizes that will guarantee flow assurance and gas deliverability to the Gas Plant at the required pressure; as well as establish operating conditions and treatment required in the pipeline to meet the deliverability
- ✚ To also establish gas processing technology and operating conditions to meet product mix specifications at the Gas Plant.

The process simulation delivers this objective through the following scopes:

- i. Generate mass and material balances for flow sources in-situ the pipelines and at the delivery outlets.
- ii. Establish appropriate line sizes suitable for the anticipated range of flow from the fields.
- iii. Establish flow assurance through the following parameter checks, - pressure, temperature, velocity and hydrate formation.
- iv. Develop a suitable operating flow conditions, especially in respect of maximum velocity to prevent erosion, and minimum velocity to prevent surging, and to keep line swept clear of entrained solids & liquids; and temperature.

The HYSYS, PIPESIM & OLGA Simulation results for the Pipeline Network are quite comparable, and established both deliverability and flow assurance of the feed gas to the Gas Processing Plant.

Also, the HYSYS Simulation results for the Gas Plant established both deliverability and flow assurance of the feed gas to the Gas Processing Plant.

This chapter summarises the process simulation results of the above reference Process Simulation Studies which are as follows:

5. Steady State Process Simulation of the Pipeline Network, using HYSYS
6. Steady State Process Simulation of the Pipeline Network, using PIPESIM
7. Transient State (Dynamic) Process Simulation of the Pipeline Network, using OLGA
8. Steady State Process Simulation of the Gas Plant Process, using HYSYS

The objective of the simulations is in two folds:

- ✚ To establish a gas gathering concept, with the appropriate line sizes that will guarantee flow assurance and gas deliverability to the Gas Plant at the required pressure; as well as establish operating conditions and treatment required in the pipeline to meet the deliverability
- ✚ To also establish gas processing technology and operating conditions to meet product mix specifications at the Gas Plant.

The process simulation delivers this objective through the following scopes:

- v. Generate mass and material balances for flow sources in-situ the pipelines and at the delivery outlets.
- vi. Establish appropriate line sizes suitable for the anticipated range of flow from the fields.
- vii. Establish flow assurance through the following parameter checks, - pressure, temperature, velocity and hydrate formation.
- viii. Develop a suitable operating flow conditions, especially in respect of maximum velocity to prevent erosion, and minimum velocity to prevent surging, and to keep line swept clear of entrained solids & liquids; and temperature.

The HYSYS, PIPESIM & OLGA Simulation results for the Pipeline Network are quite comparable, and established both deliverability and flow assurance of the feed gas to the Gas Processing Plant.

Also, the HYSYS Simulation results for the Gas Plant established both deliverability and flow assurance of the feed gas to the Gas Processing Plant.

6.1. The Gas Gathering & Process Flow Scheme

The overall gas gathering and processing scheme, with phased on-stream dates is as shown in Fig 6.1 below.

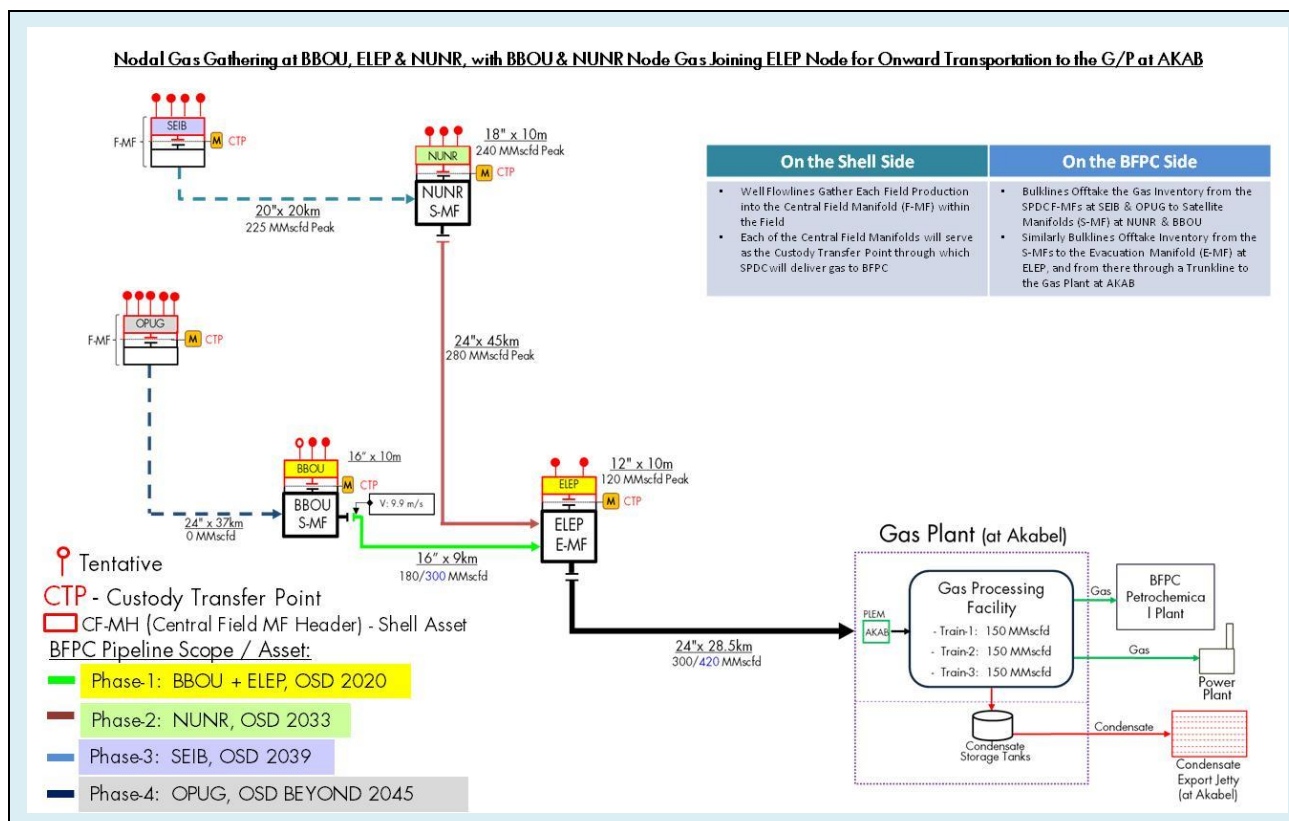


Figure 6.1: Overall Gas Gathering & Processing Schematics, with phased on-stream dates

Production starts in 2020 with BBOU + ELEP as phase-1, followed in sequence by production from other phases, from 2033 as reflected in the schematics. The peak production associated with each phase is also indicated.

The 420 MMscfd peak for ELEP – AKAB Trunkline represents a situation where the line handles both the BBOU & ELEP peak at the same time, - simulated to establish robustness of the trunkline up to that capacity level. That will guide decision in a situation where BFPC would still wish to pursue the aspiration to intake up to 400 MMscfd into the Gas Processing Plant.

6.2. Basis and Assumptions

The pipeline temperature ambient medium was taken as 25 °C.

Also, it is assumed for this conceptual study, that there are no elevation changes across the pipeline route.

6.2.1. Software & Simulation Property Package

The hydraulics process simulation study was undertaken using Aspen HYSYS v7.3, PIPESIM, and OLGA

Water saturation extension was explored to determine the saturated water in the gas key considerations to the route selection are:

The thermodynamic equation of state (EOS) used for the process simulation is; Peng-Robinson (PR) fluid property package.

PVT Compositions

For BBOU stream Analogue Composition (Bonny PVT) was used.

For ELEP stream, ELEP Upper U4 sand PVT was used.

For NUNR stream, NUNR H4 sand PVT was used.

For SEIB stream, the average of PVT compositions of SEIB G1.1M & F9.3M was used.

For OPUG, the average of OPUG F2000 & F2001 was used.

6.2.2. Flow Rates, Fluid Condition & Composition

Flow Rates

The forecast production profiles for BBOU, ELEP, NUNR, & SEIB are summarized below.

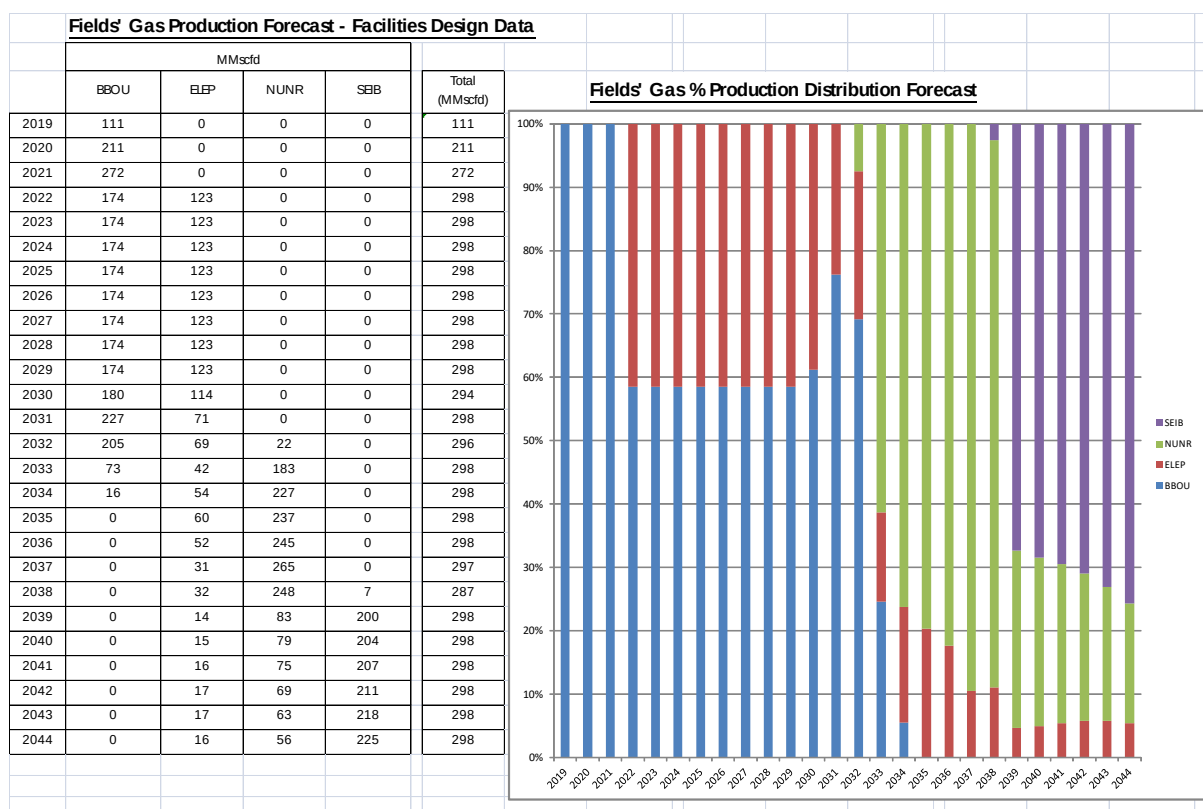


Figure 2.1: Fields' Gas Production Forecast – Facilities Design Data

Fluid Condition & Composition

The Reservoir Fluid Conditions and PVT Compositions for BBOU, ELEP, NUNR, SEIB, & OPUG are summarized in table 6.2 below

Table 6.2: BBOU, ELEP, NUNR, SEIB, and OPUG Reservoir Compositions & Conditions.

Field's Gas Compositions																			
Component	BBO U			ELEP			NUNR			SEIB			OPUG						
	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid				
N2	0.07	0.00	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.07	0.12	0.00	0.10				
CO2	0.59	0.00	0.59	2.30	0.00	2.22	0.00	0.00	0.77	1.63	0.00	1.55	2.53	0.00	2.23				
H2S				0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00				
C1	93.26	0.00	92.92	84.63	0.00	81.66	0.00	0.00	72.83	83.24	0.00	78.62	77.45	0.00	68.00				
C2	3.17	0.01	3.16	6.35	0.03	6.13	0.00	0.00	7.68	6.27	0.28	5.93	9.06	0.13	7.97				
C3	1.01	0.03	1.01	2.67	0.27	2.59	0.00	0.00	4.85	3.49	1.88	3.40	5.40	0.53	4.81				
iC4	0.61	0.07	0.61	0.76	0.37	0.75	0.00	0.00	1.46	1.20	1.55	1.22	2.13	0.76	1.96				
nC4	0.44	0.07	0.44	0.82	0.54	0.81	0.00	0.00	1.97	1.31	2.36	1.38	1.00	1.59	1.07				
iC5	0.25	0.16	0.25	0.43	0.87	0.45	0.00	0.00	0.95	0.73	2.34	0.83	1.07	1.79	1.16				
nC5	0.19	0.16	0.19	0.32	0.68	0.33	0.00	0.00	0.76	0.50	1.96	0.58	0.53	1.97	0.69				
C6	0.22	1.88	0.23	0.59	2.26	0.65	0.00	0.00	1.03	0.68	4.76	0.92	0.53	5.23	1.07				
C7	0.17	6.21	0.19	0.69	6.75	0.90	0.00	0.00	1.41	0.62	9.71	1.13	0.17	8.16	1.07				
C8	0.02	6.61	0.04	0.44	12.76	0.87	0.00	0.00	1.61	0.28	15.65	1.14	0.05	16.75	1.92				
C9	0.00	7.13	0.03	0.00	10.87	0.38	0.00	0.00	0.90	0.00	11.71	0.65	0.00	8.81	0.99				
C10	0.00	5.70	0.02	0.00	8.62	0.30	0.00	0.00	0.65	0.00	9.14	0.51	0.00	5.87	0.67				
C11	0.00	7.83	0.03	0.00	6.64	0.23	0.00	0.00	0.47	0.00	5.68	0.30	0.00	6.44	0.79				
C12+	0.00	64.14	0.22	0.00	49.34	1.73	0.00	0.00	2.66	0.00	33.00	1.82	0.00	42.00	5.55				
Total	100.00	100.00	100.00	100.00	100.00	100.00	0.00	0.00	100.00	100.00	100.00	100.01	100.00	100.01	100.00				
Molar ratio				0.9650	0.0350	1.0000				0.0720	0.9280	1.0000							

	BBOU			ELEP			NUNR			SEIB			OPUG AVG.		
	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid
Reservoir Pressure, psia			3,246			6,910			7,259			6,533			5,176
Reservoir Temp, °C			78			123			102			117			117
Mole %															
C7+	0.19	97.62	0.53	1.13	94.98	4.41	0.00	0.00	7.70	0.90	84.89	5.54	0.21	88.01	10.98
C12+	0.00	64.14	0.22	0.00	49.34	1.73	0.00	0.00	2.66	0.00	33.00	1.82	0.00	42.00	5.55
Mola mass	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol
C7+	100.89	167.69	143.17	105.67	186.72	166.70	104.82	176.38	169.26	104.48	174.90	164.20	98.81	159.87	158.76
C12+	-	191.60	191.60												
fluid	17.97	165.61	18.47	20.84	181.08	26.44	21.82	160.91	31.84	21.38	158.85	29.00	22.29	149.53	38.11
Density	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³
C7+						832			830			837			
C12+									829						
fluid		811			828			813			817			805	
Rel Density (air = 1)	0.62			0.719		0.913	0.753		1.099	0.738			0.770		
CGR			stb/ MMscfd			stb/ MMscfd			stb/ MMscfd			stb/ MMscfd			stb/ MMscfd
fluid			4 - 36			62 - 158			62 - 158			62 - 158			

Water Saturated Compositions

The Re-Constructed Reservoir Fluid PVT Compositions saturated with water is shown below in table 6.3.

Table 6.3: BBOU, ELEP, NUNR, SEIB, and OPUG Reservoir Compositions Saturated with Water

Saturated Gas Compositions															
Component	BBOU			ELEP			NUNR			SEIB			OPUG		
	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid
N2	0.00	0.00	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.00	0.10
CO2	0.00	0.00	0.59	0.00	0.00	2.17	0.00	0.00	0.77	0.00	0.00	1.53	0.00	0.00	2.21
H2S	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C1	0.00	0.00	92.61	0.00	0.00	79.87	0.00	0.00	72.37	0.00	0.00	77.84	0.00	0.00	67.24
C2	0.00	0.00	3.15	0.00	0.00	6.00	0.00	0.00	7.63	0.00	0.00	5.87	0.00	0.00	7.88
C3	0.00	0.00	1.01	0.00	0.00	2.53	0.00	0.00	4.82	0.00	0.00	3.37	0.00	0.00	4.76
iC4	0.00	0.00	0.61	0.00	0.00	0.73	0.00	0.00	1.45	0.00	0.00	1.21	0.00	0.00	1.94
nC4	0.00	0.00	0.44	0.00	0.00	0.79	0.00	0.00	1.96	0.00	0.00	1.37	0.00	0.00	1.06
iC5	0.00	0.00	0.25	0.00	0.00	0.44	0.00	0.00	0.94	0.00	0.00	0.82	0.00	0.00	1.15
nC5	0.00	0.00	0.19	0.00	0.00	0.32	0.00	0.00	0.76	0.00	0.00	0.57	0.00	0.00	0.68
C6	0.00	0.00	0.23	0.00	0.00	0.64	0.00	0.00	1.02	0.00	0.00	0.91	0.00	0.00	1.06
C7+	0.00	0.00	0.53	0.00	0.00	4.31	0.00	0.00	7.65	0.00	0.00	5.48	0.00	0.00	10.86
H2O	0.00	0.00	0.34	0.00	0.00	2.19	0.00	0.00	0.63	0.00	0.00	0.95	0.00	0.00	1.08
Total	0.00	0.00	100.00	0.00	0.00	100.00	0.00	0.00	100.00	0.00	0.00	100.00	0.00	0.00	100.00

6.3. Gas Deliverability & Flow Assurance

Gas deliverability flow assurance outlook is presented for the peak flow production for each of the four phases.

The deliverability being adjudged by the following:

1. Ability of the system to offtake the gas at 80 – 100 barg from the Custody Transfer Manifold and deliver to the Gas Processing Plant within the stipulated pressure band of 60 – 65 barg.
2. Fluid velocity below erosional velocity limit, - to avoid pipewall erosion, excess noise or water hammer; and above the minimum level required for efficient flow, - to keep line swept clear of entrained solids & liquids.
3. Fluid traverse outside hydrate formation region of the phase envelope.

6.3.1. Phase-1 Production (BBOU & ELEP)

There are three scenarios associated with this phase-1 production:

1. BBOU – ELEP Bulkline Flow, **with BBOU as the only production to AKAB**, flowing at 300 MMscfd; with its resultant flow into ELEP – AKAB Trunkline at the 300 MMscfd BBOU production.
2. BBOU – ELEP Bulkline Flow, **with production from both BBOU and ELEP**, with BBOU flowing at 180 MMscfd and ELEP at 120 MMscfd; with its resultant flow into ELEP – AKAB Trunkline at 300 MMscfd, being the combined BBOU & ELEP production.
3. Examines the robustness of the ELEP to AKAB Trunkline to handle up to 420 MMscfd, simulated by combining BBOU peak production of 300 MMscfd with the 120 MMscfd of ELEP, to arrive at the 420 MMscfd flow into the ELEP – AKAB Trunkline.

The ELEP Field – ELEM Manifold Flow is assumed to have the constant scenario of ELEP flowing at 120 MMscfd, and therefore not considered as generating a different scenario.

Scenario-1 (Phase-1 Production)

The flow parameters give positive indication of deliverability and flow assurance, giving the pressure, velocity, & fluid traverse profiles shown in figs 6.3 & 6.4 below.

Flow Parameters – Pressure & Velocity Profile: Scenario-1, Phase-1 Production

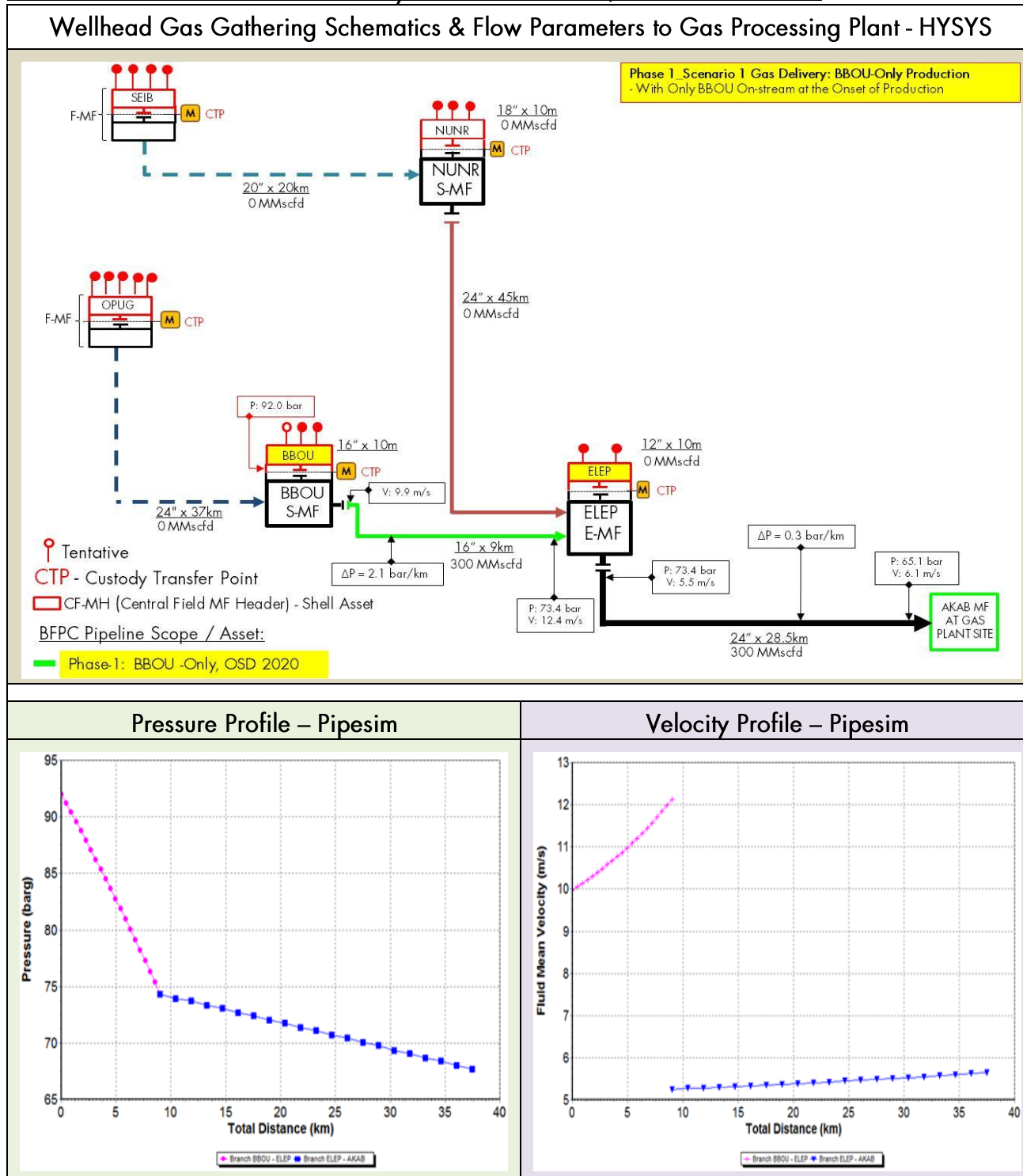
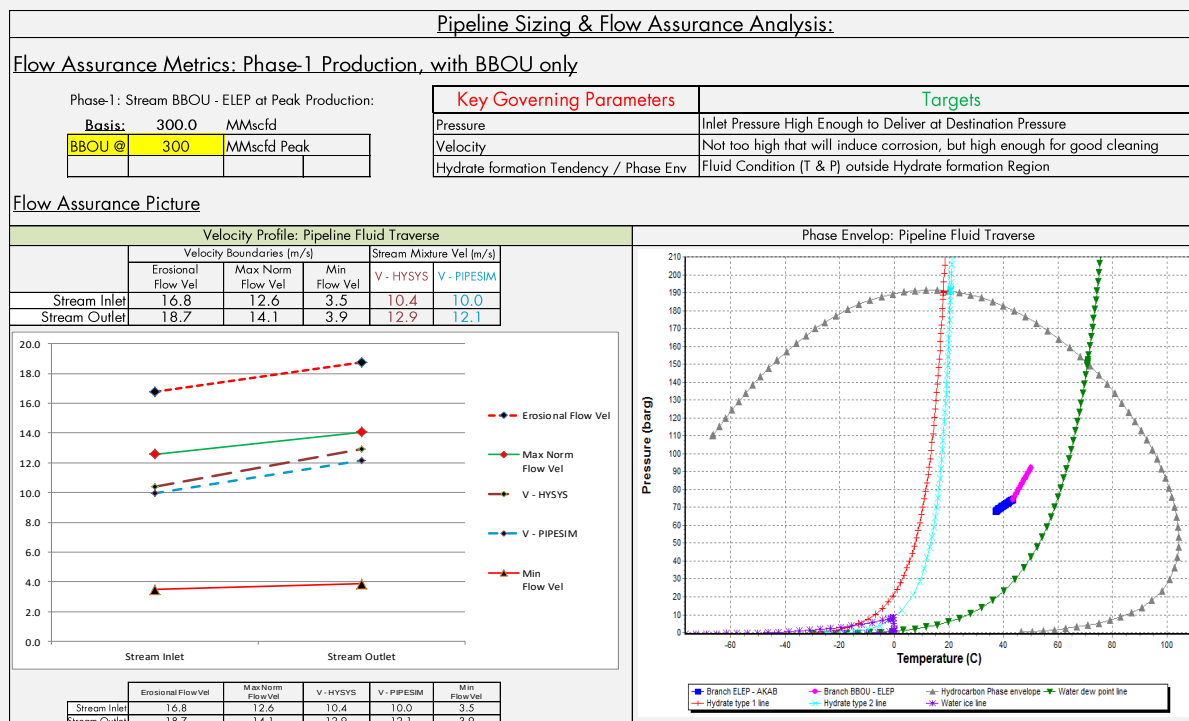


Figure 6.3: Flow Parameters and Pressure & Velocity Profile – Scenario-1, Phase-1 Production

Flow Parameters – Flow Assurance Analysis

BBOU – ELEP Flow Stream: Scenario-1, Phase-1 Production



ELEP – AKAB Flow Stream: Scenario-1, Phase-1 Production

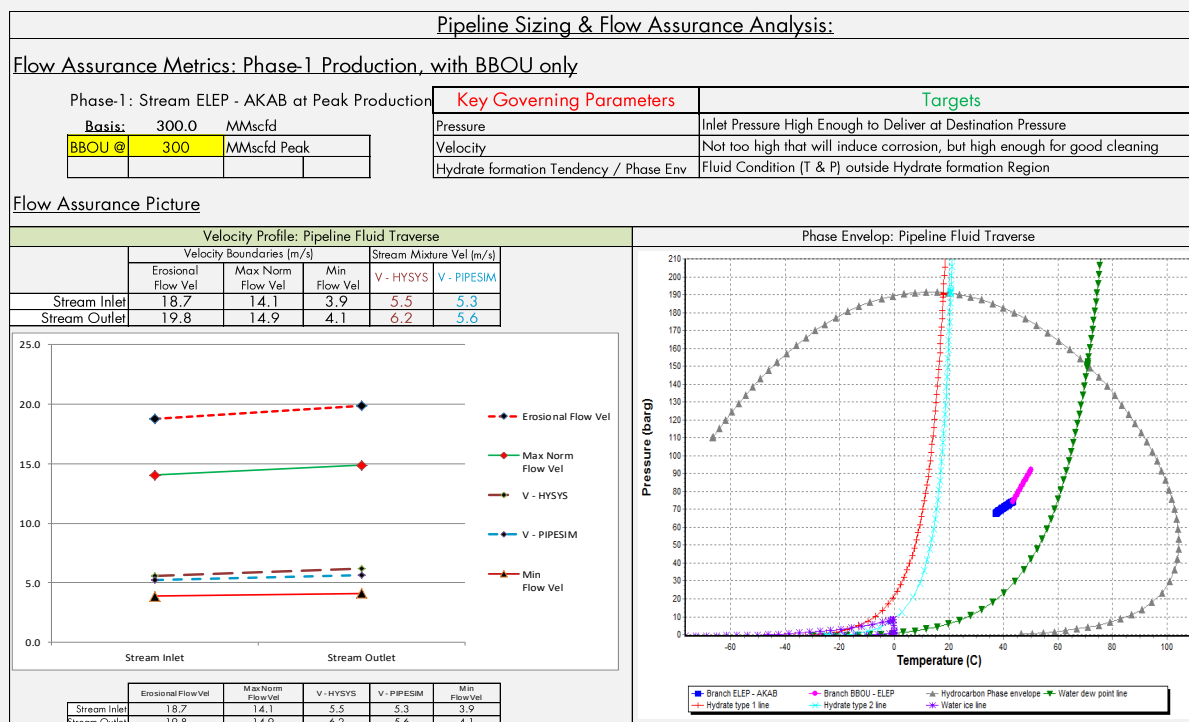


Figure 6.4: Flow Assurance Analysis – Scenario-1, Phase-1 Production

Scenario-2 (Phase-1 Production)

The flow parameters give positive indication of deliverability and flow assurance, giving the pressure, velocity, & fluid traverse profiles shown in figs 6.5 & 6.6 below.

Flow Parameters – Pressure & Velocity Profile: Scenario-2, Phase-1 Production

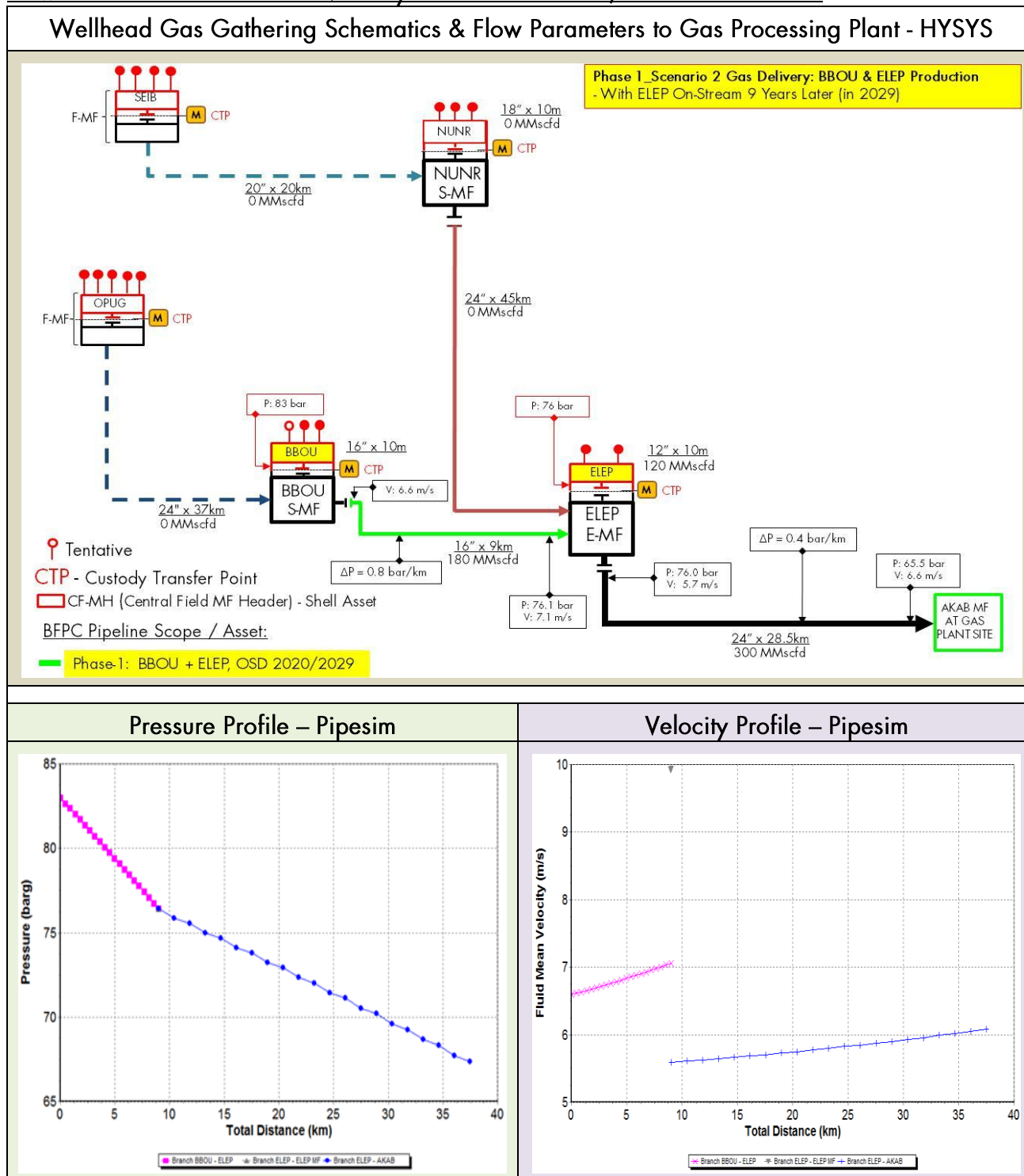
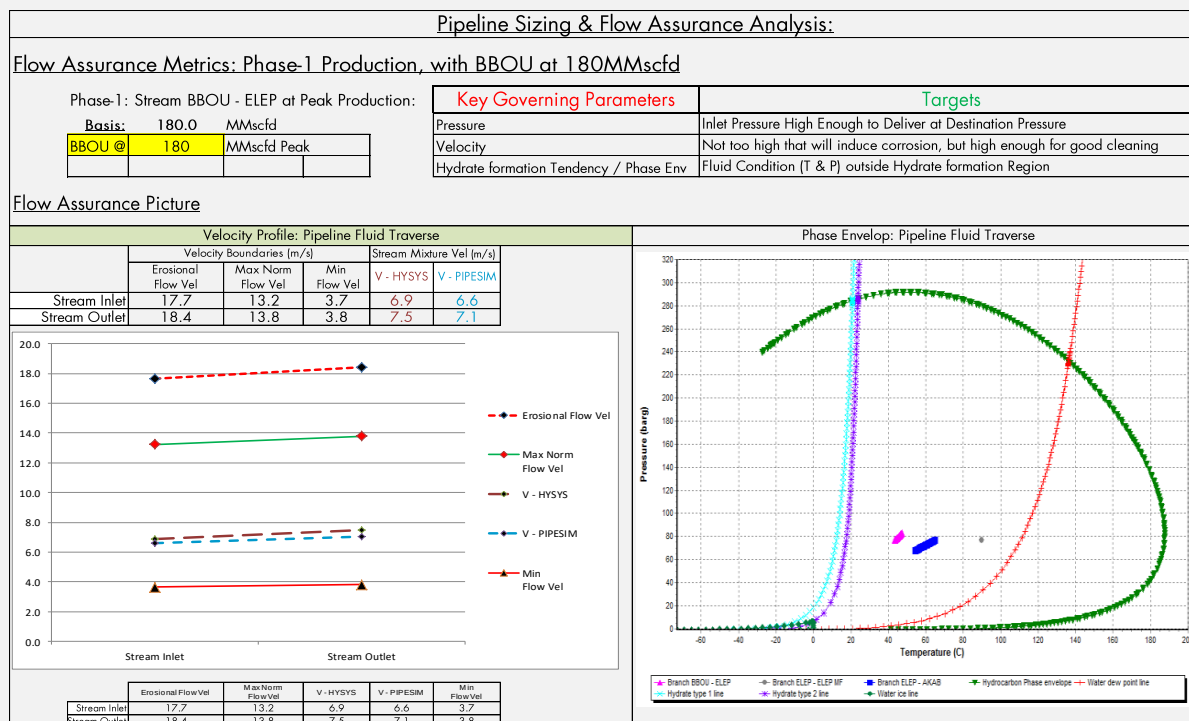


Figure 6.5: Flow Parameters and Pressure & Velocity Profile – Scenario-2, Phase-1 Production

Flow Parameters – Flow Assurance Analysis

BBOU – ELEP Flow Stream: Scenario-2, Phase-1 Production



ELEP – AKAB Flow Stream: Scenario-2, Phase-1 Production

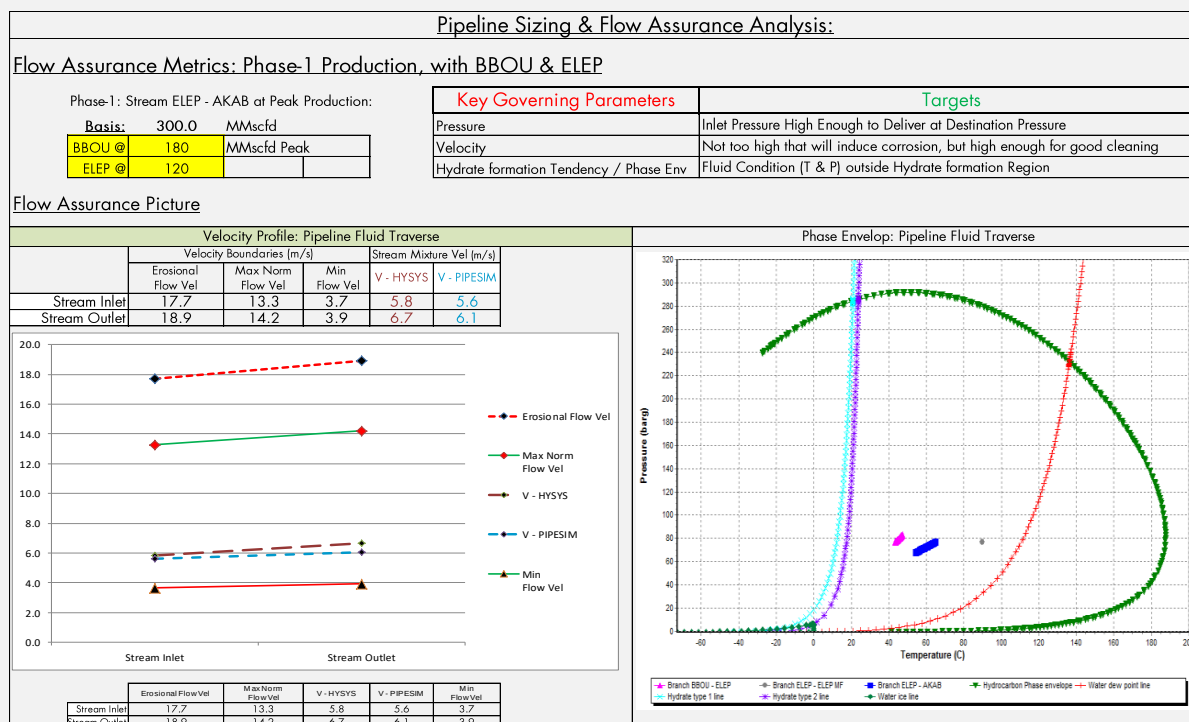


Figure 6.6: Flow Assurance Analysis – Scenario-2, Phase-1 Production

Scenario-3 (Phase-1 Production)

The flow parameters give positive indication of deliverability and flow assurance, giving the pressure, velocity, & fluid traverse profiles shown in figs 6.7 & 6.8 below.

Flow Parameters – Pressure & Velocity Profile: Scenario-3, Phase-1 Production

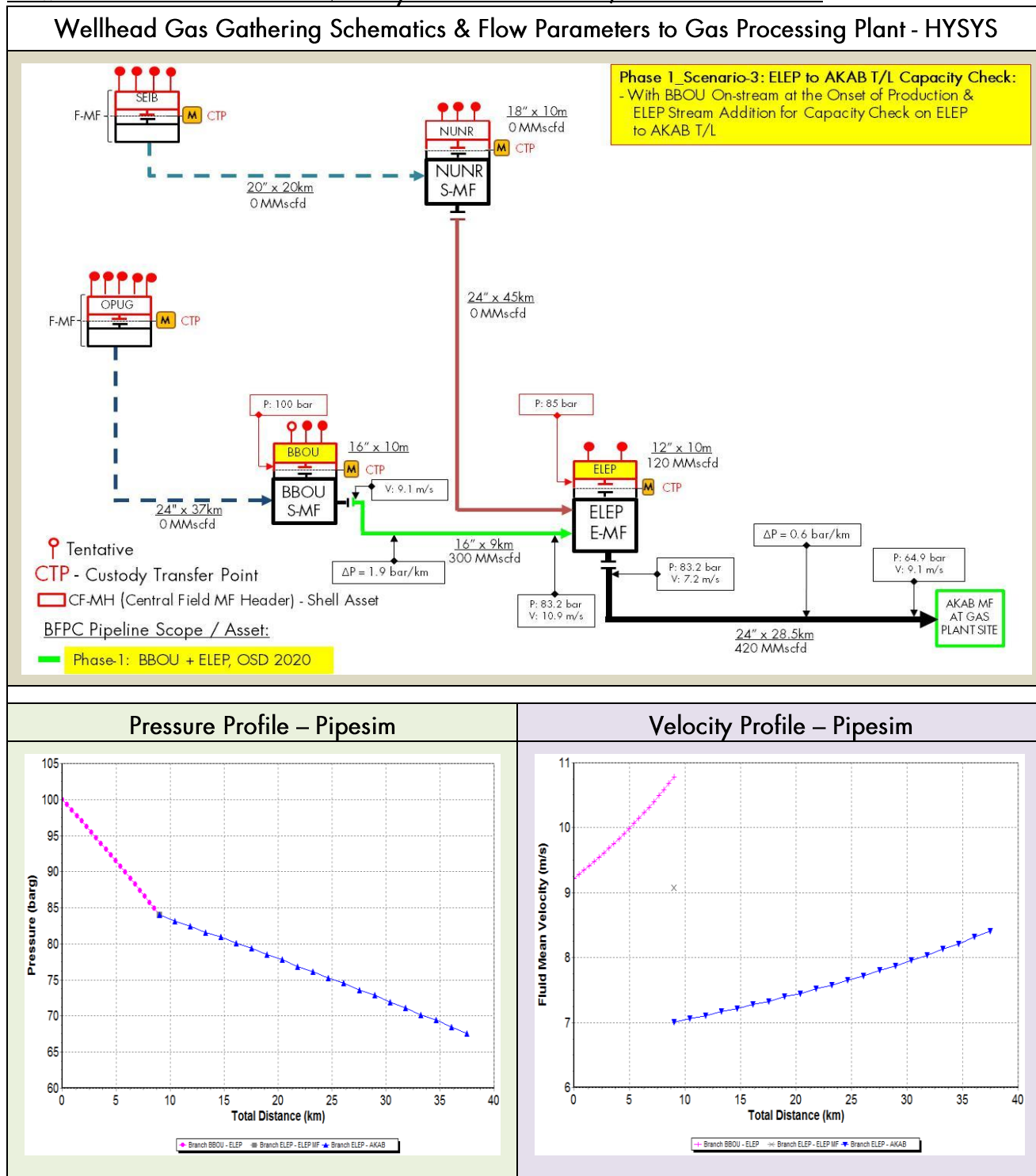
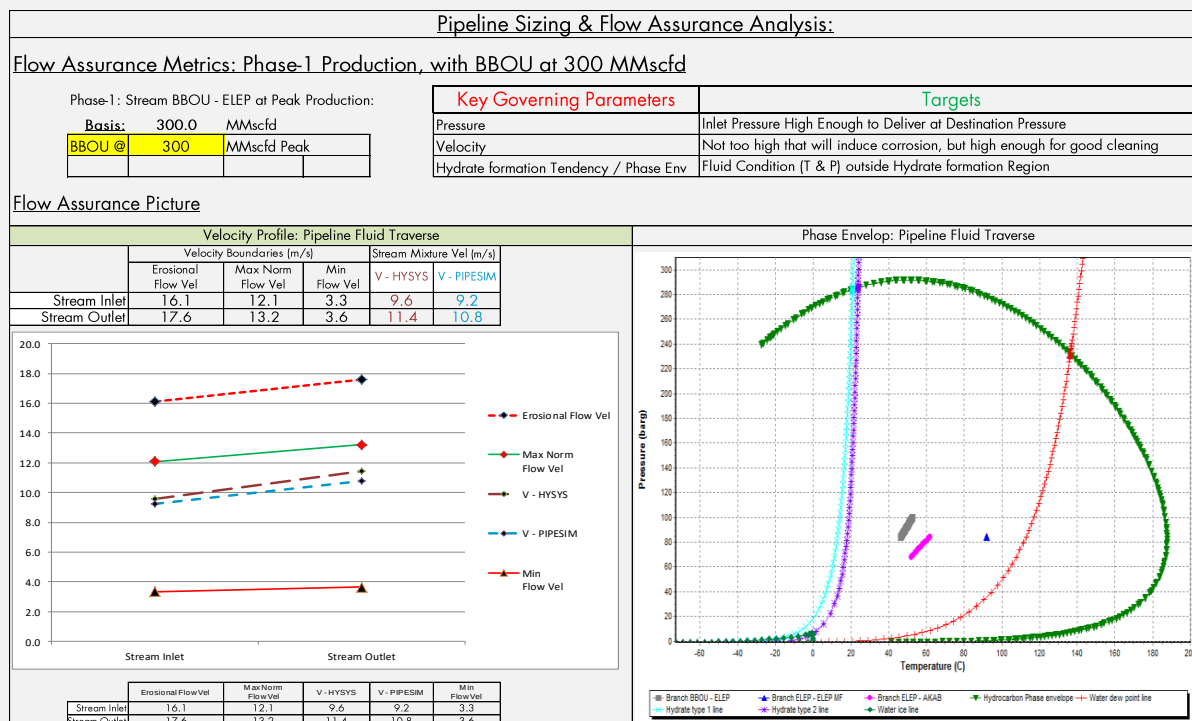


Figure 6.7: Flow Parameters and Pressure & Velocity Profile – Scenario-3, Phase-1 Production

Flow Parameters – Flow Assurance Analysis

BBOU – ELEP Flow Stream: Scenario-3, Phase-1 Production



ELEP – AKAB Flow Stream: Scenario-3, Phase-1 Production

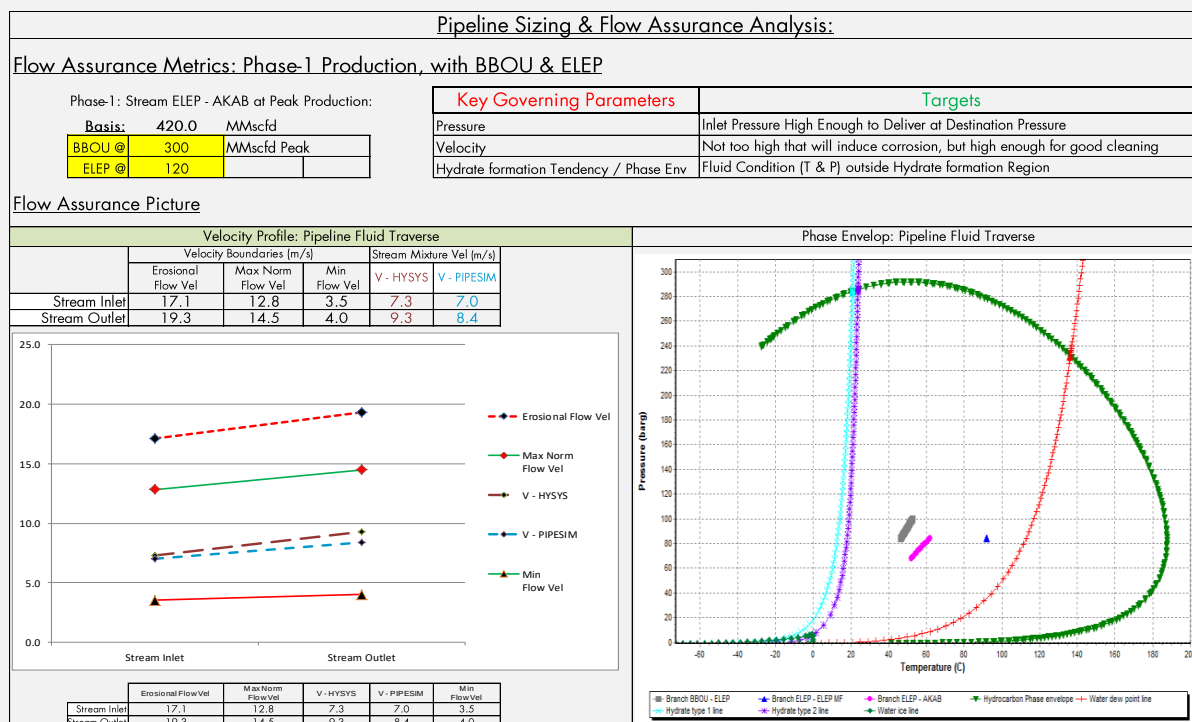


Figure 6.8: Flow Assurance Analysis – Scenario-3, Phase-1 Production

ELEP – AKAB Flow Parameters & Flow Assurance Analysis - Pipesim: Phase-1 Scenarios

The flow parameters give positive indication of deliverability and flow assurance, giving the pressure, velocity, & fluid traverse profiles shown in fig 6.9 below.

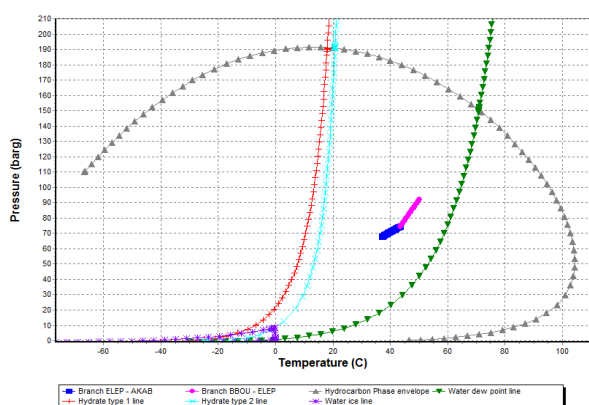
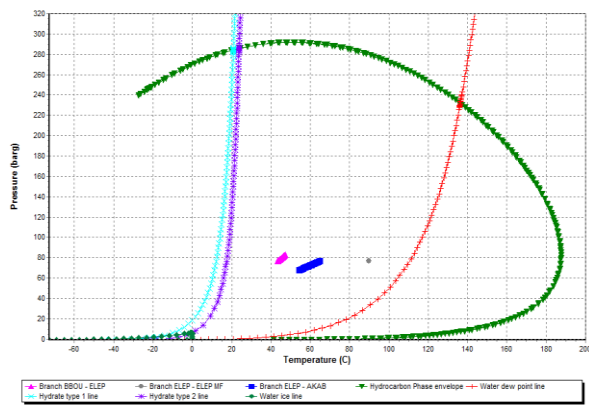
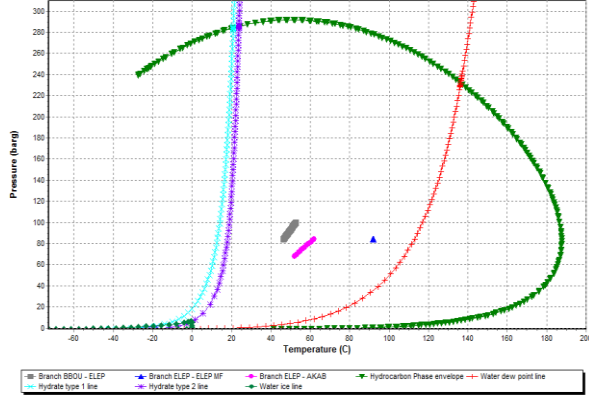
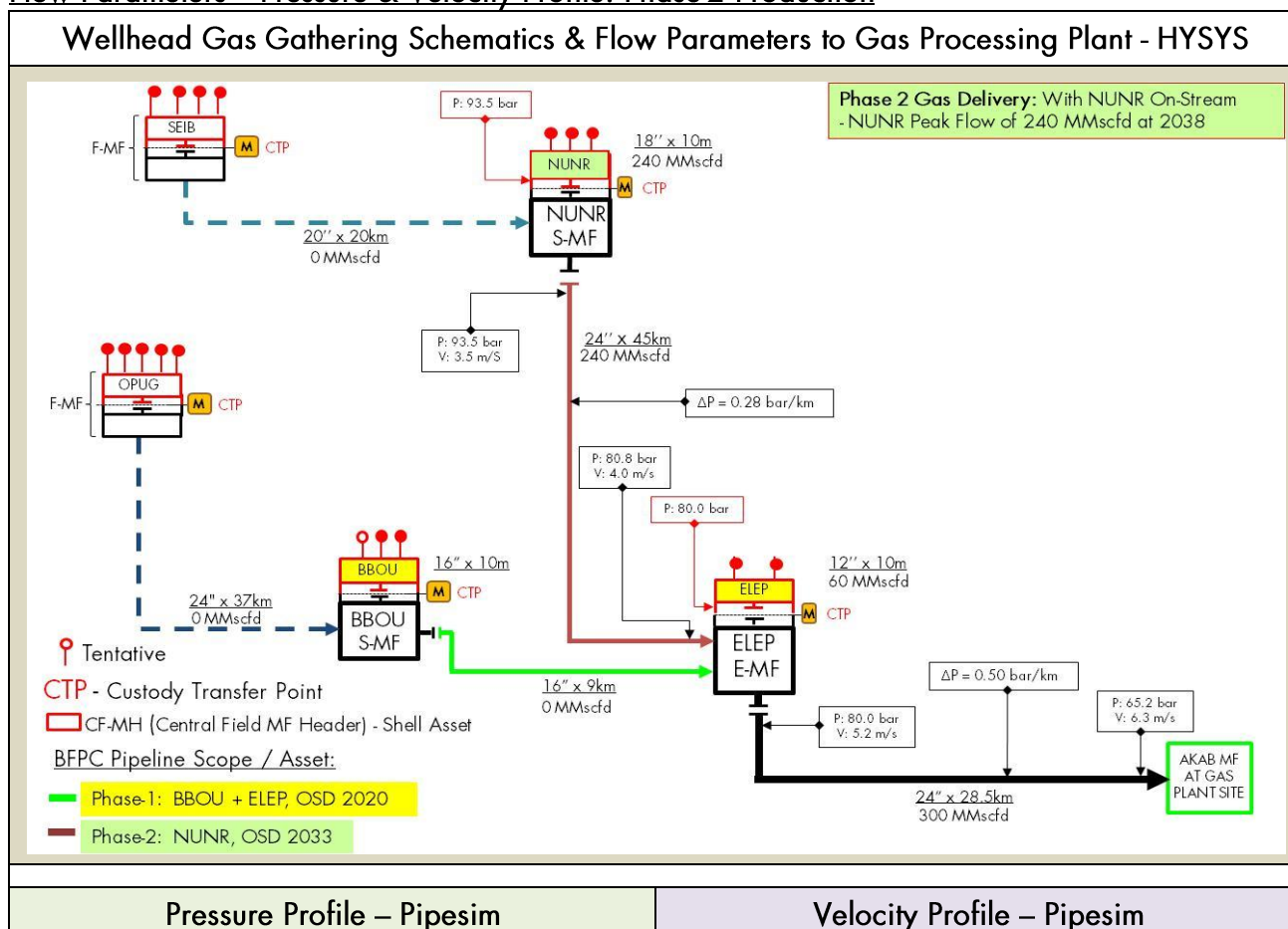
Scenario 1: BBOU – ELEP Bulkline Flow, with BBOU as the only production to AKAB:	Pressure (barg)		Velocity (m/s)		Phase Envelope
	Inlet	Outlet	Inlet	Outlet	
ELEP – AKAB Bulkline 300 MMscfd. <i>Resulting From:</i> <i>BBOU – ELEP Bulkline</i> <i>300 MMscfd</i> <i>ELEP – ELEP MF Bulkline</i> <i>0 MMscfd</i>	74.35	67.71	5.25	5.64	
Scenario 2: BBOU – ELEP Bulkline Flow, with production from both BBOU and ELEP to AKAB.	Pressure (barg)		Velocity (m/s)		Phase Envelope
	Inlet	Outlet	Inlet	Outlet	
ELEP – AKAB Bulkline 300 MMscfd. <i>Resulting From:</i> <i>BBOU – ELEP Bulkline</i> <i>180 MMscfd</i> <i>ELEP – ELEP MF Bulkline</i> <i>120 MMscfd</i>	76.41	67.37	5.59	6.08	
Scenario 3: Examines the robustness of the ELEP to AKAB Trunkline to handle up to 420 MMscfd,	Pressure (barg)		Velocity (m/s)		Phase Envelope
	Inlet	Outlet	Inlet	Outlet	
ELEP – AKAB Bulkline 420 MMscfd. <i>Resulting From:</i> <i>BBOU – ELEP Bulkline</i> <i>180 MMscfd</i> <i>ELEP – ELEP MF Bulkline</i> <i>120 MMscfd</i>	84.01	67.63	7.00	8.41	

Figure 6.9: Flow Assurance Analysis – ELEP - AKAB, Phase-1 Production

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The flow parameters give positive indication of deliverability and flow assurance, giving the pressure, velocity, & fluid traverse profiles shown in figs 6.10 & 6.11 below.

Flow Parameters – Pressure & Velocity Profile: Phase-2 Production



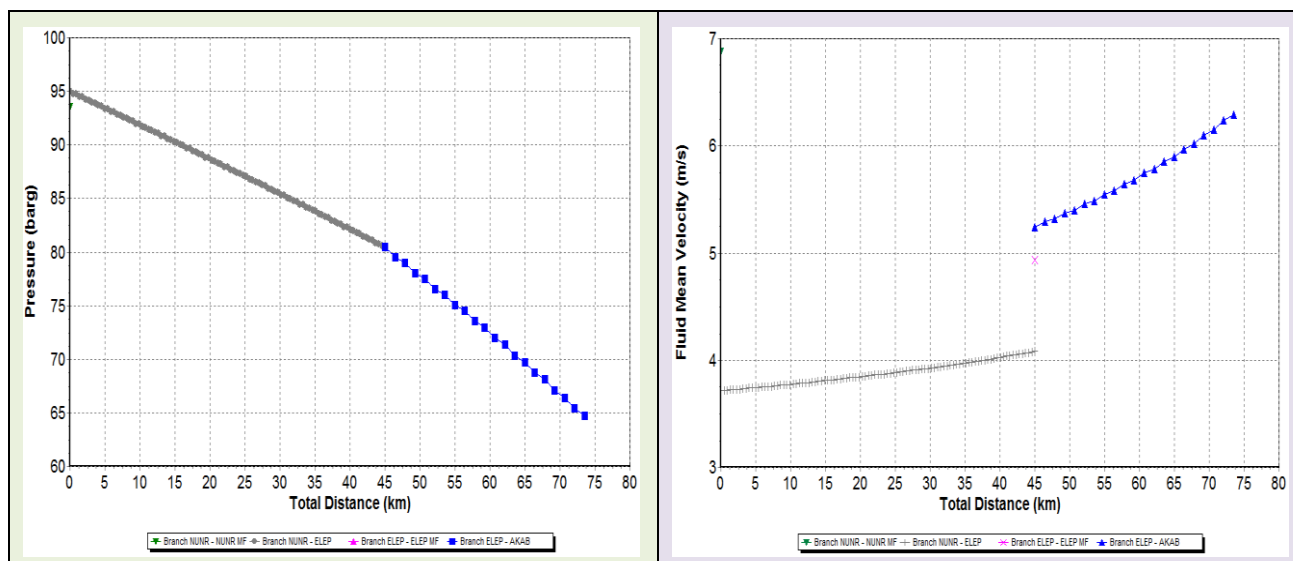
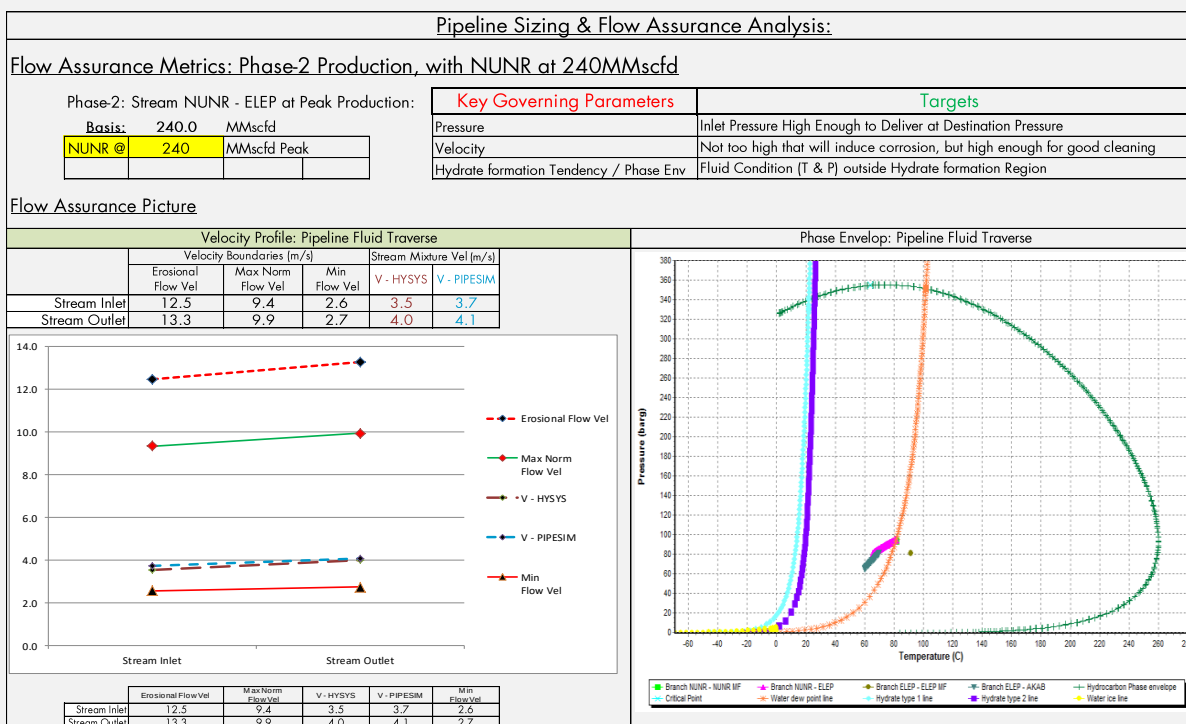


Figure 6.10: Flow Parameters and Pressure & Velocity Profile – Phase-2 Production
Flow Parameters – Flow Assurance Analysis

NUNR – ELP Flow Stream: Phase-2 Production



ELP – AKAB Flow Stream: Phase-2 Production

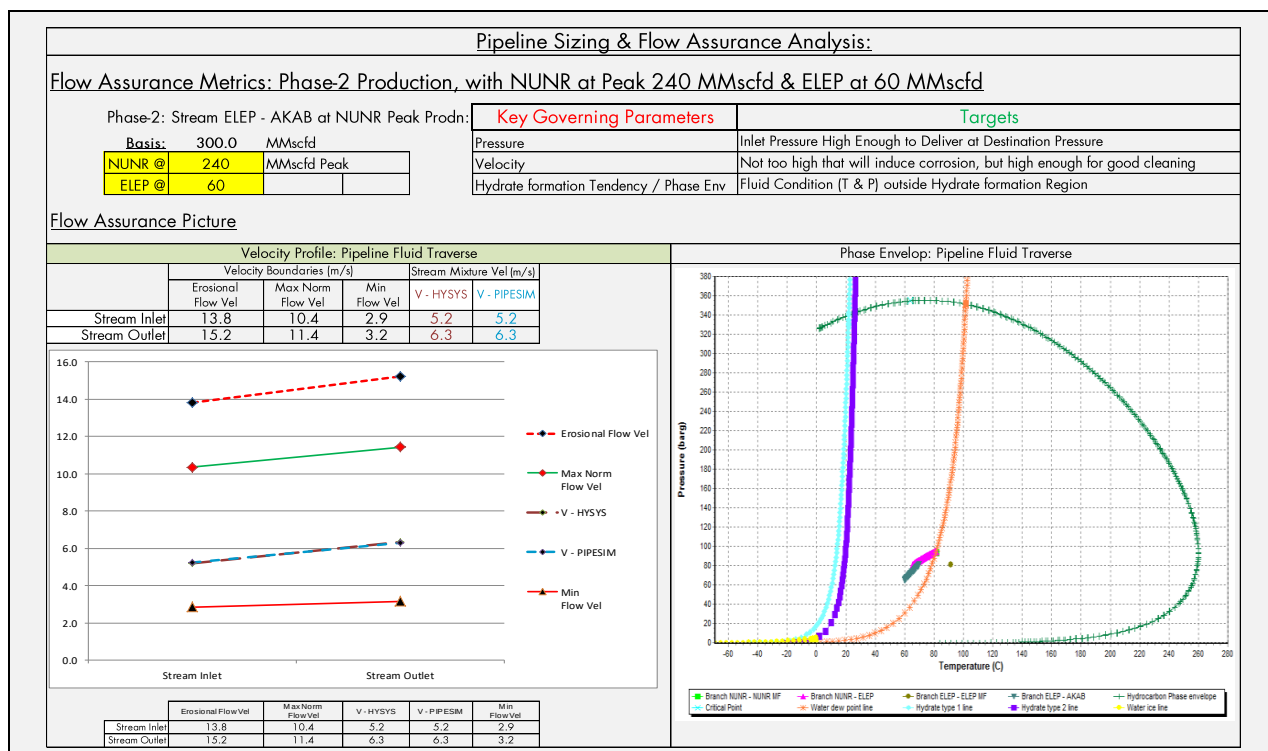


Figure 6.11: Flow Assurance Analysis – Phase-2 Production

1.1.2. Phase-3 Production (SEIB, NUNR & ELEP)

The flow parameters give positive indication of deliverability and flow assurance, giving the pressure, velocity, & fluid traverse profiles shown in figs 6.12 & 6.13 below

Flow Parameters – Pressure & Velocity Profile: Phase-3 Production

Wellhead Gas Gathering Schematics & Flow Parameters to Gas Processing Plant - HYSYS

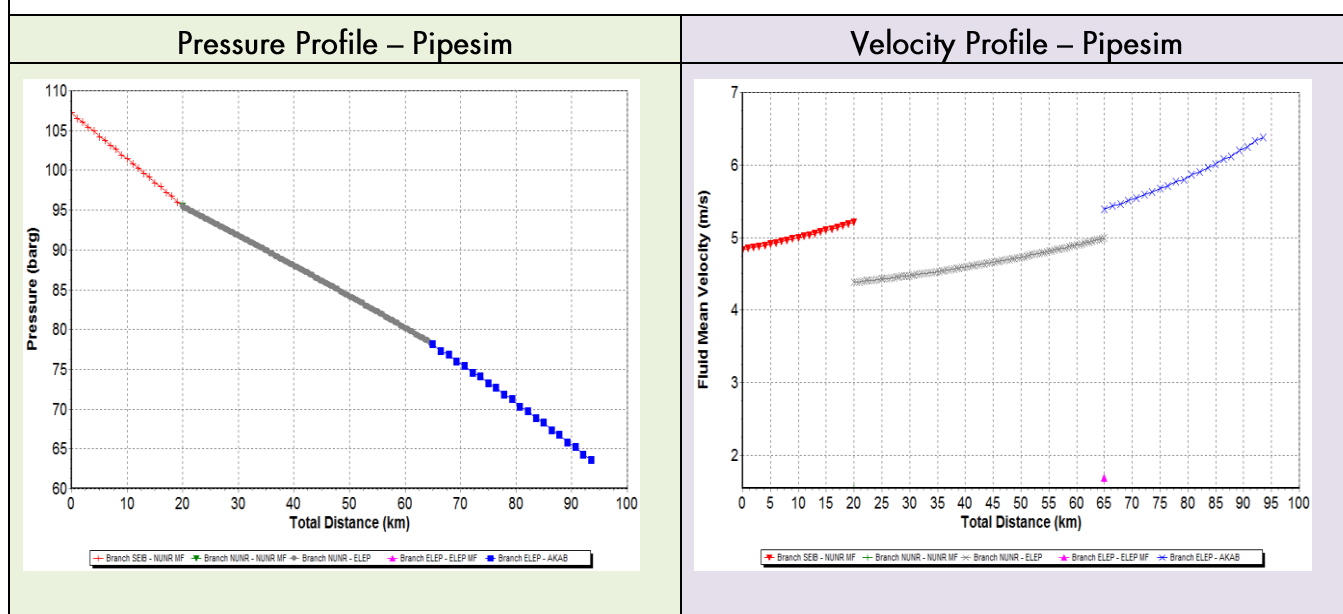
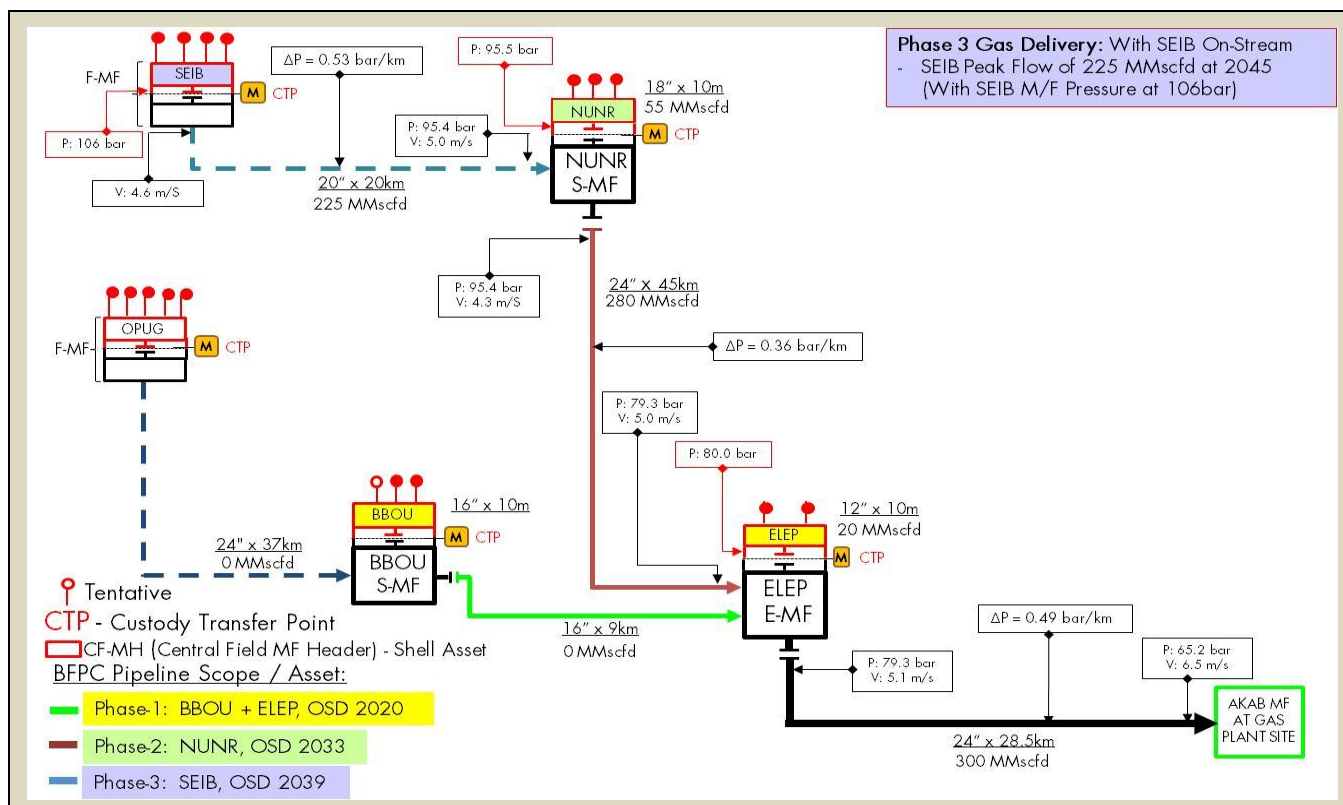


Figure 6.12: Flow Parameters and Pressure & Velocity Profile – Phase-3 Production
Flow Parameters – Flow Assurance Analysis

SEIB – NUNR Flow Stream: Phase-3 Production

Pipeline Sizing & Flow Assurance Analysis:

Flow Assurance Metrics: Phase-3 Production, with SEIB at 225 MMscfd

Phase-3: Stream SEIB - NUNR S-MF at Peak Production

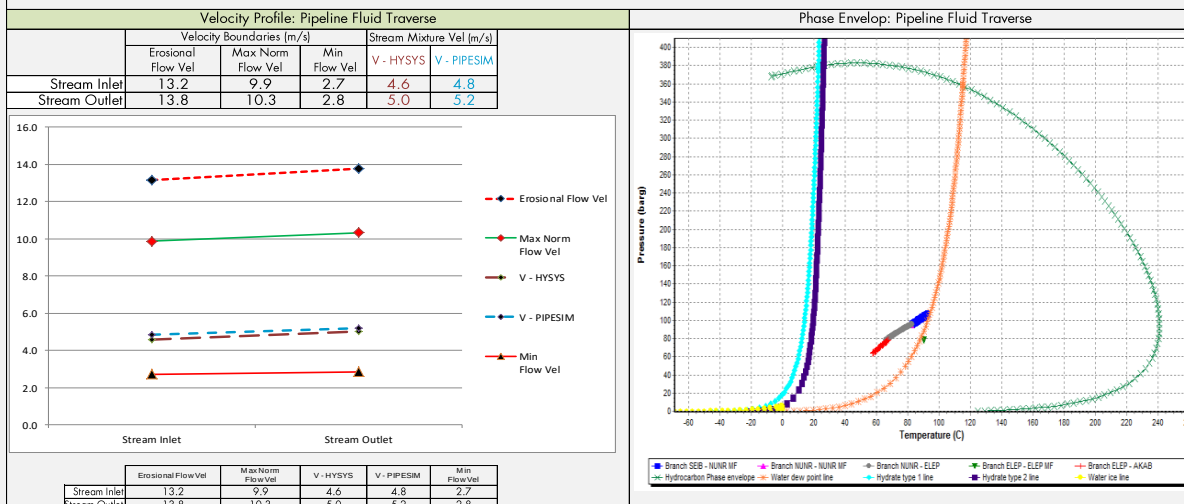
Basis:	225.0	MMscfd
SEIB @	225	MMscfd Peak

Key Governing Parameters

Pressure	Inlet Pressure High Enough to Deliver at Destination Pressure
Velocity	Not too high that will induce corrosion, but high enough for good cleaning
Hydrate formation Tendency / Phase Env	Fluid Condition (T & P) outside Hydrate formation Region

Targets

Flow Assurance Picture



NUNR - ELP Flow Stream: Phase-3 Production

Pipeline Sizing & Flow Assurance Analysis:

Flow Assurance Metrics: Phase-3 Production, with SEIB at Peak 225 MMscfd, & NUNR at 55MMscfd

Phase-3: Stream NUNR S-MF - ELP E-MF:

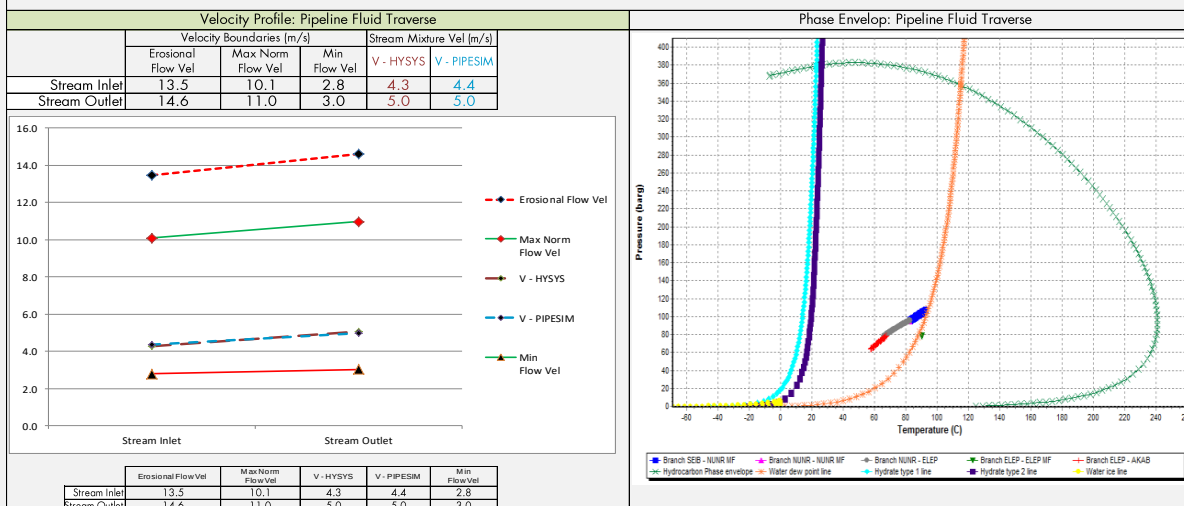
Basis:	280.0	MMscfd
SEIB @	225	MMscfd Peak
NUNR @	55	

Key Governing Parameters

Pressure	Inlet Pressure High Enough to Deliver at Destination Pressure
Velocity	Not too high that will induce corrosion, but high enough for good cleaning
Hydrate formation Tendency / Phase Env	Fluid Condition (T & P) outside Hydrate formation Region

Targets

Flow Assurance Picture



ELP - AKAB Flow Stream: Phase-3 Production

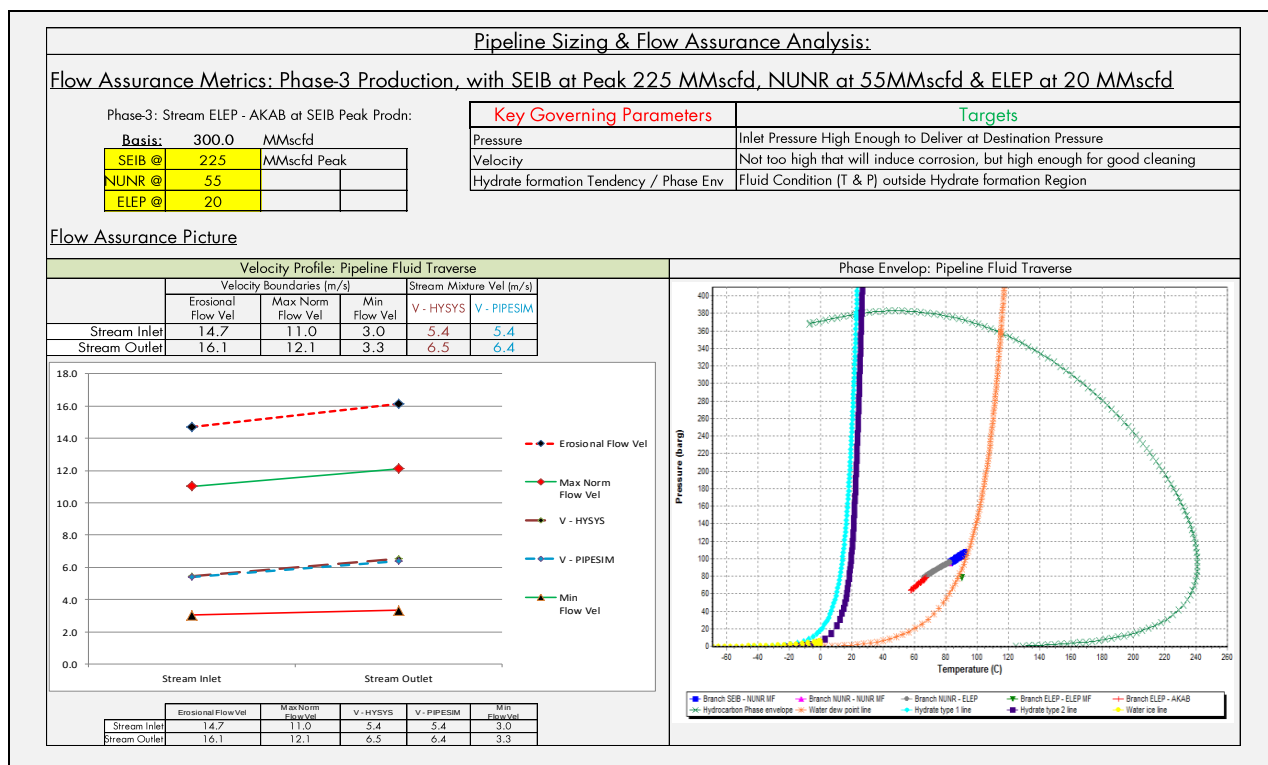


Figure 6.13: Flow Assurance Analysis – Phase-3 Production

1.1.3. Phase-4 Production (OPUG, SEIB, & NUNR)

The flow parameters give positive indication of deliverability and flow assurance, giving the pressure, velocity, & fluid traverse profiles shown in figs 6.14 & 6.15 below

Flow Parameters – Pressure & Velocity Profile: Phase-4 Production

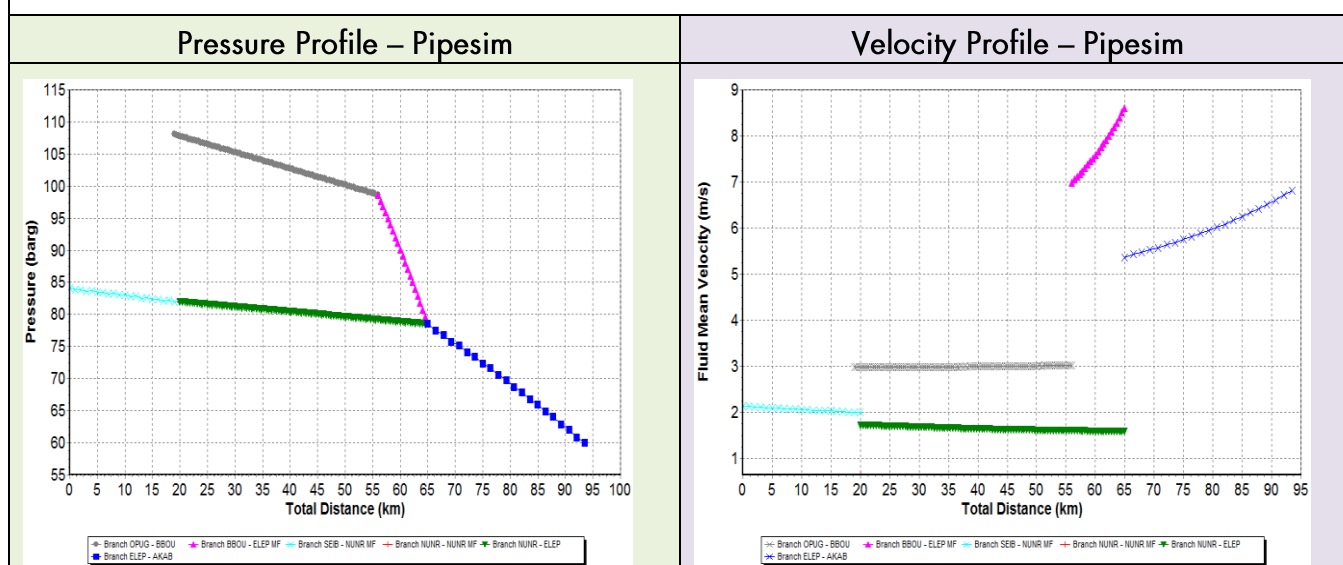
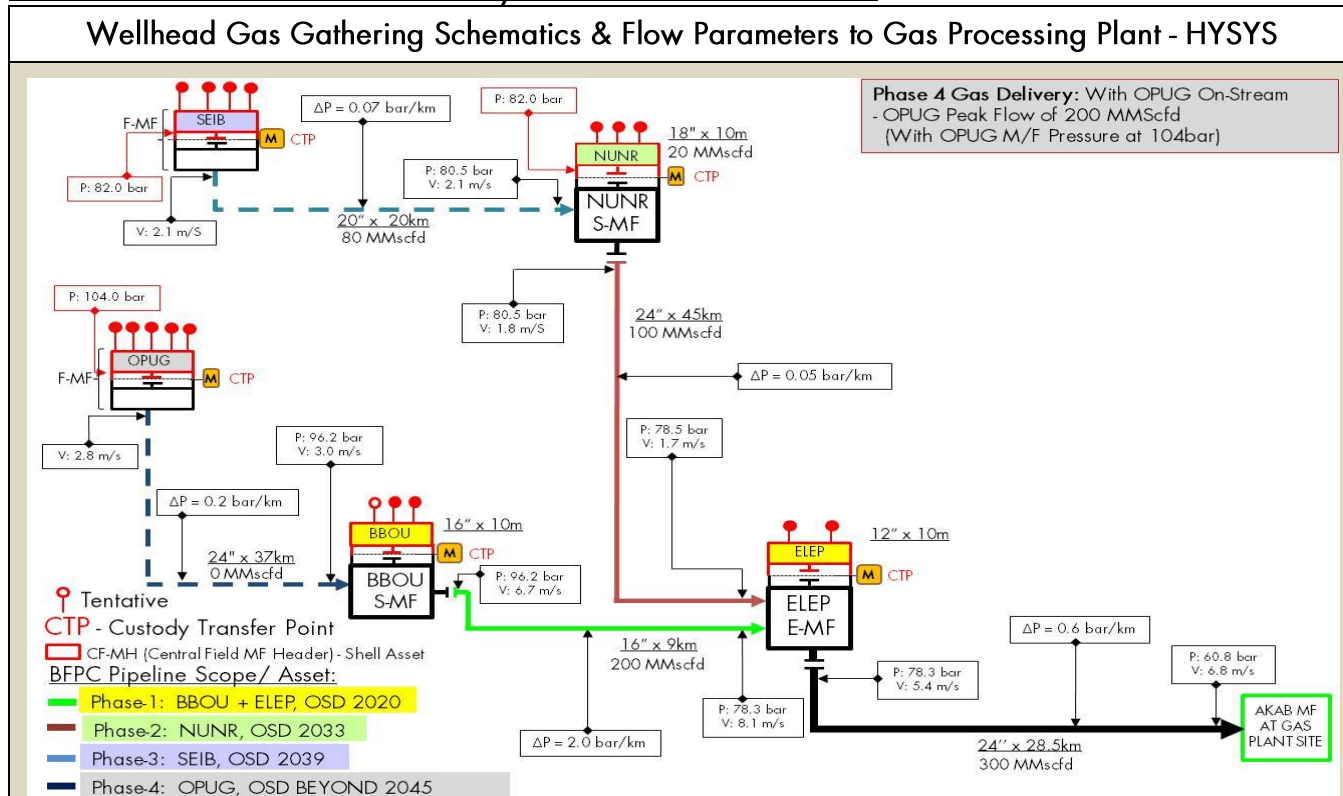


Figure 6.14: Flow Parameters and Pressure & Velocity Profile – Phase-4 Production

Flow Parameters – Flow Assurance Analysis

OPUG – BBOU Flow Stream: Phase-4 Production

Pipeline Sizing & Flow Assurance Analysis:

Flow Assurance Metrics: Phase-4 Production, with OPUG at 200MMscfd

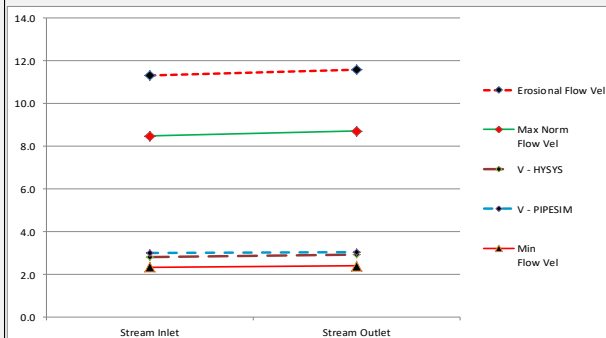
Phase-4: Stream OPUG - BBOU F-MF at Peak Production:

Basis:	200.0	MMscfd
OPUG @	200	MMscfd Peak

Key Governing Parameters	Targets
Pressure	Inlet Pressure High Enough to Deliver at Destination Pressure
Velocity	Not too high that will induce corrosion, but high enough for good cleaning
Hydrate formation Tendency / Phase Env	Fluid Condition (T & P) outside Hydrate formation Region

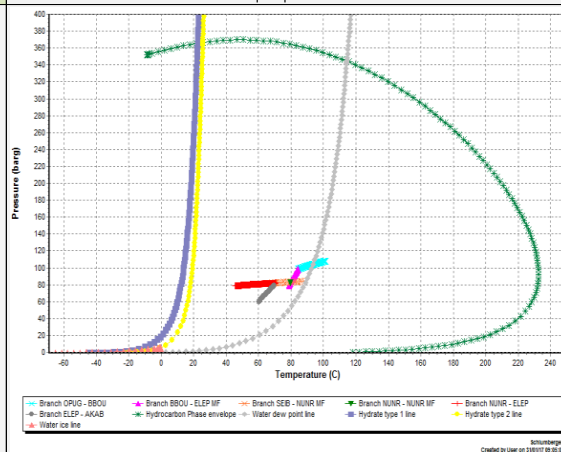
Flow Assurance Picture

	Velocity Profile: Pipeline Fluid Traverse			
	Velocity Boundaries (m/s)		Stream Mixture Vel (m/s)	
	Erosional Flow Vel	Max Norm Flow Vel	Min Flow Vel	V - HYSYS V - PIPESIM
Stream Inlet	11.3	8.5	2.3	2.8 3.0
Stream Outlet	11.6	8.7	2.4	2.9 3.0



	Erosional Flow Vel	Max Norm Flow Vel	V - HYSYS	V - PIPESIM	Min Flow Vel
Stream Inlet	11.3	8.5	2.8	3.0	2.3
Stream Outlet	11.6	8.7	2.9	3.0	2.4

Phase Envelop: Pipeline Fluid Traverse



BBOU - ELEG Flow Stream: Phase-4 Production

Pipeline Sizing & Flow Assurance Analysis:

Flow Assurance Metrics: Phase-4 Production, with OPUG at 200MMscfd

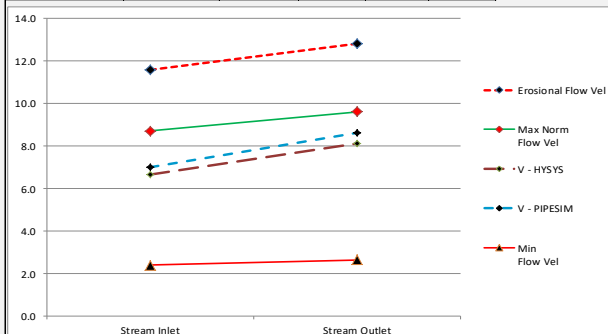
Phase-4: Stream BBOU F-MF to ELEG E-MF:

Basis:	200.0	MMscfd
OPUG @	200	MMscfd Peak

Key Governing Parameters	Targets
Pressure	Inlet Pressure High Enough to Deliver at Destination Pressure
Velocity	Not too high that will induce corrosion, but high enough for good cleaning
Hydrate formation Tendency / Phase Env	Fluid Condition (T & P) outside Hydrate formation Region

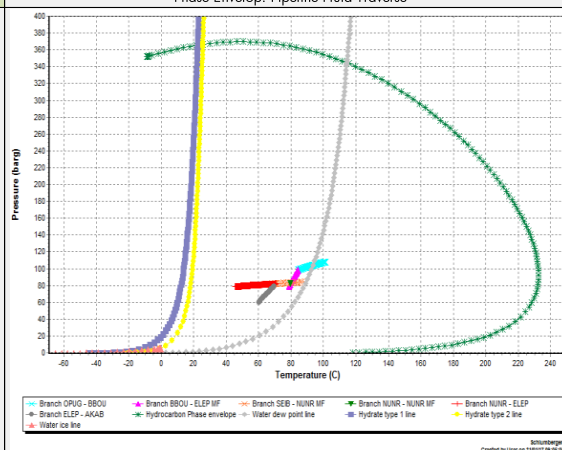
Flow Assurance Picture

	Velocity Profile: Pipeline Fluid Traverse			
	Velocity Boundaries (m/s)		Stream Mixture Vel (m/s)	
	Erosional Flow Vel	Max Norm Flow Vel	Min Flow Vel	V - HYSYS V - PIPESIM
Stream Inlet	11.6	8.7	2.4	6.7 7.0
Stream Outlet	12.8	9.6	2.7	8.1 8.6



	Erosional Flow Vel	Max Norm Flow Vel	V - HYSYS	V - PIPESIM	Min Flow Vel
Stream Inlet	11.6	8.7	6.7	7.0	2.4
Stream Outlet	12.8	9.6	8.1	8.6	2.7

Phase Envelop: Pipeline Fluid Traverse



SEIB – NUNR S-MF Flow Stream: Phase-4 Production

Pipeline Sizing & Flow Assurance Analysis:

Flow Assurance Metrics: Phase-4 Production, with SEIB at 80MMscfd

Phase-4: Stream SEIB to NUNR S-MF :

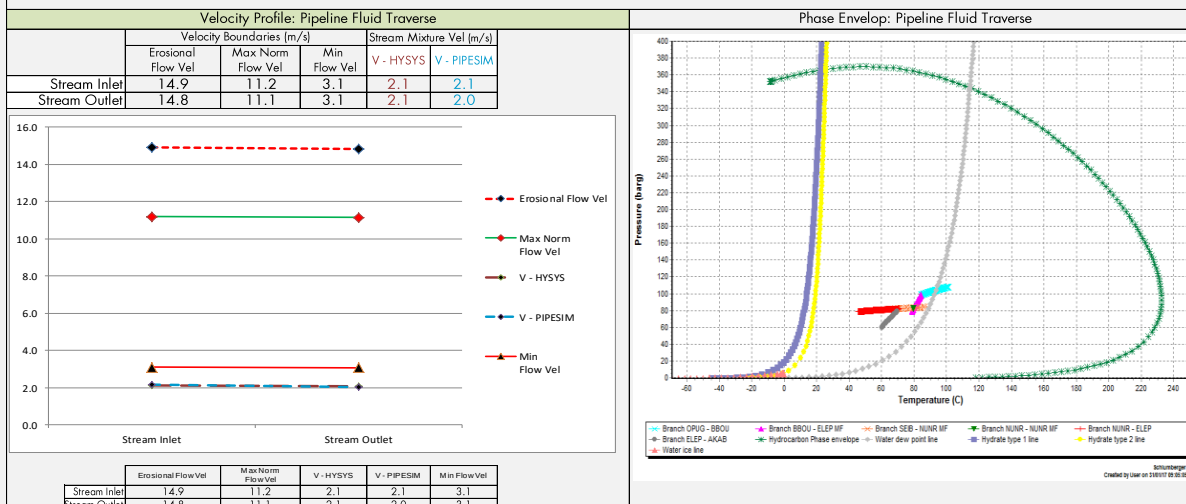
Basis:	80.0	MMscfd
OPUG @	80	MMscfd Peak

Key Governing Parameters

Pressure	Inlet Pressure High Enough to Deliver at Destination Pressure
Velocity	Not too high that will induce corrosion, but high enough for good cleaning
Hydrate formation Tendency / Phase Env	Fluid Condition (T & P) outside Hydrate formation Region

Targets

Flow Assurance Picture



NUNR S-MF – ELP Flow Stream: Phase-4 Production

Pipeline Sizing & Flow Assurance Analysis:

Flow Assurance Metrics: Phase-4 Production, with SEIB at 80MMscfd, & NUNR at 20MMscfd

Phase-4: Stream NUNR S-MF - ELP E-MF:

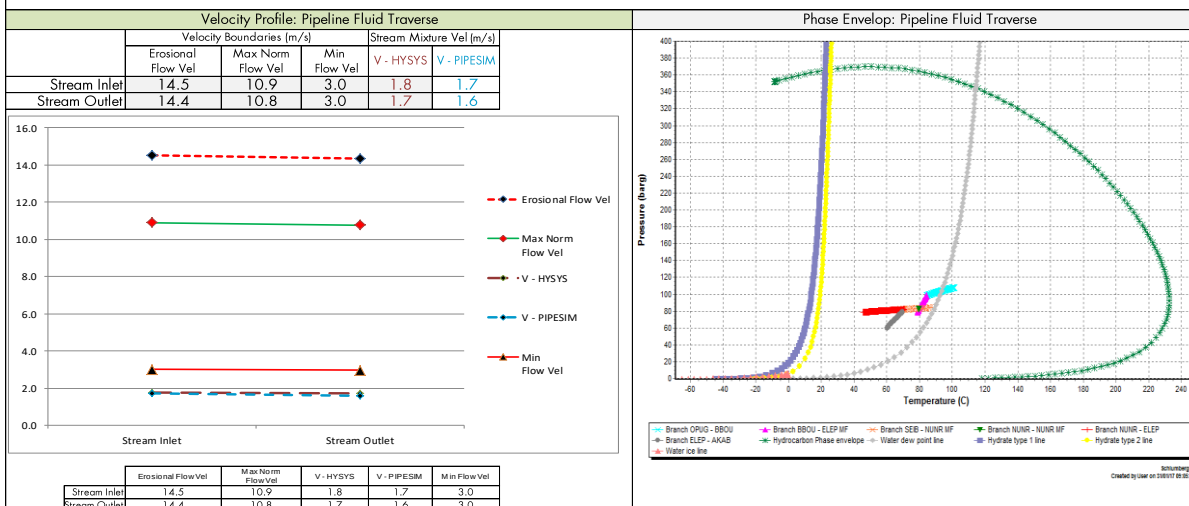
Basis:	100.0	MMscfd
SEIB @	80	MMscfd
NUNR @	20	

Key Governing Parameters

Pressure	Inlet Pressure High Enough to Deliver at Destination Pressure
Velocity	Not too high that will induce corrosion, but high enough for good cleaning
Hydrate formation Tendency / Phase Env	Fluid Condition (T & P) outside Hydrate formation Region

Targets

Flow Assurance Picture



ELP – AKAB Flow Stream: Phase-4 Production

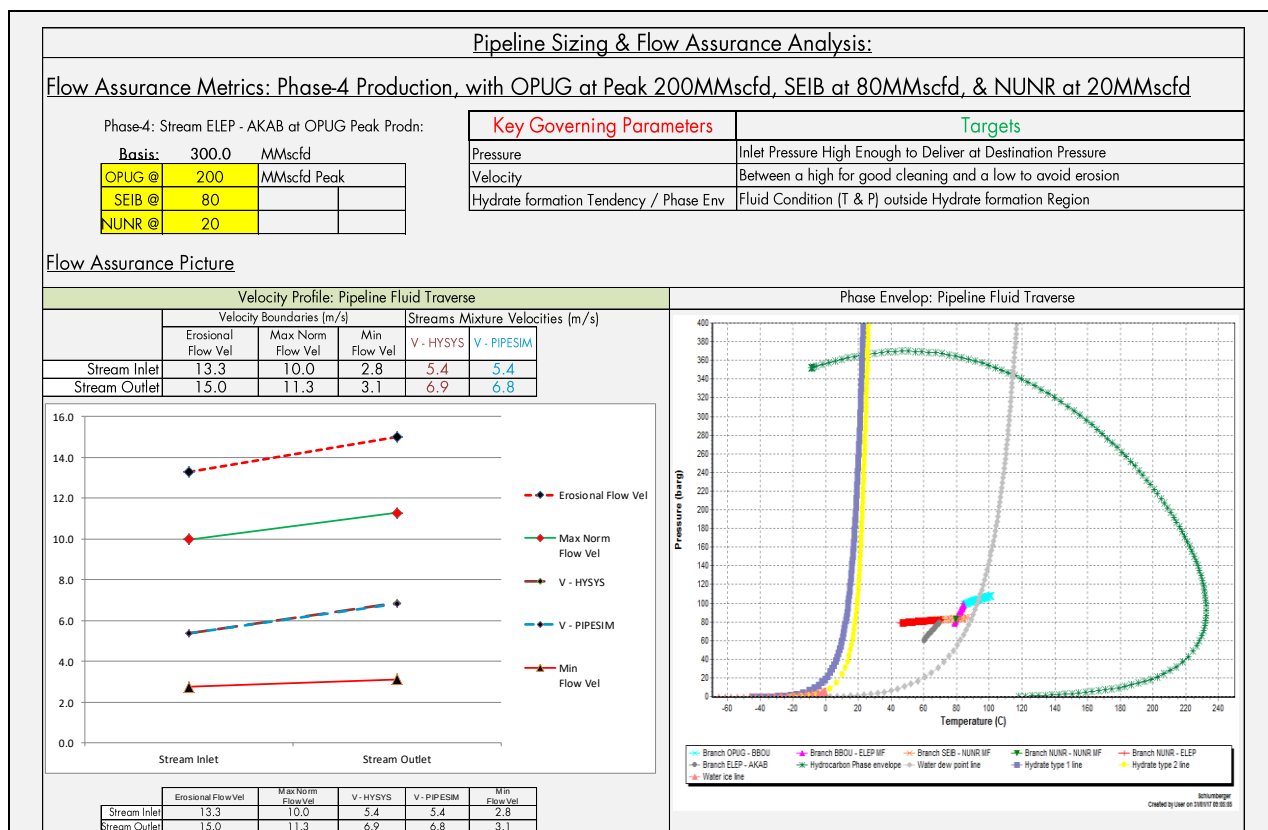


Figure 6.15: Flow Assurance Analysis – Phase-4 Production

2033 Production (Water Production Sensitivity)

The flow parameters give positive indication of deliverability and flow assurance, giving the pressure, velocity, & fluid traverse profiles shown in figs 6.16 & 6.17 below.

Flow Parameters – Pressure & Velocity Profile: 2033 Production

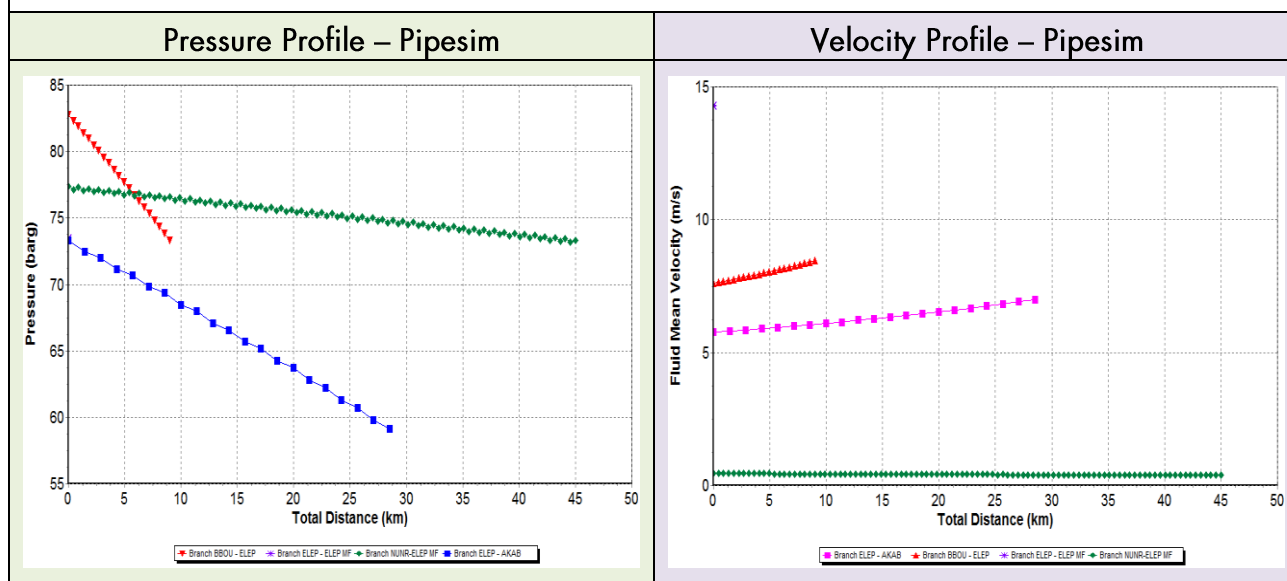
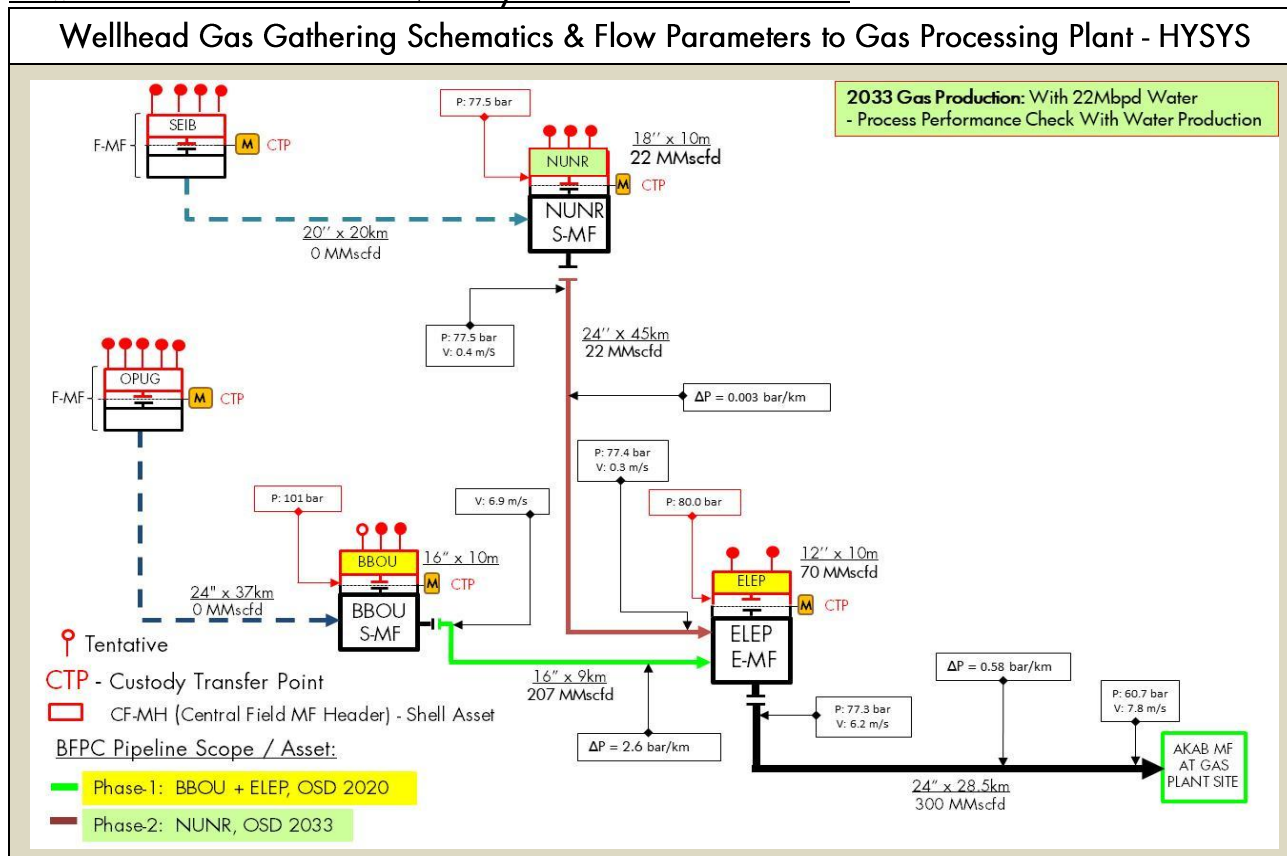


Figure 6.16: Flow Parameters and Pressure & Velocity Profile – 2033 Production

Flow Parameters – Flow Assurance Analysis

BBOU F-MF – ELEP Flow Stream: 2033 Production

Pipeline Sizing & Flow Assurance Analysis:

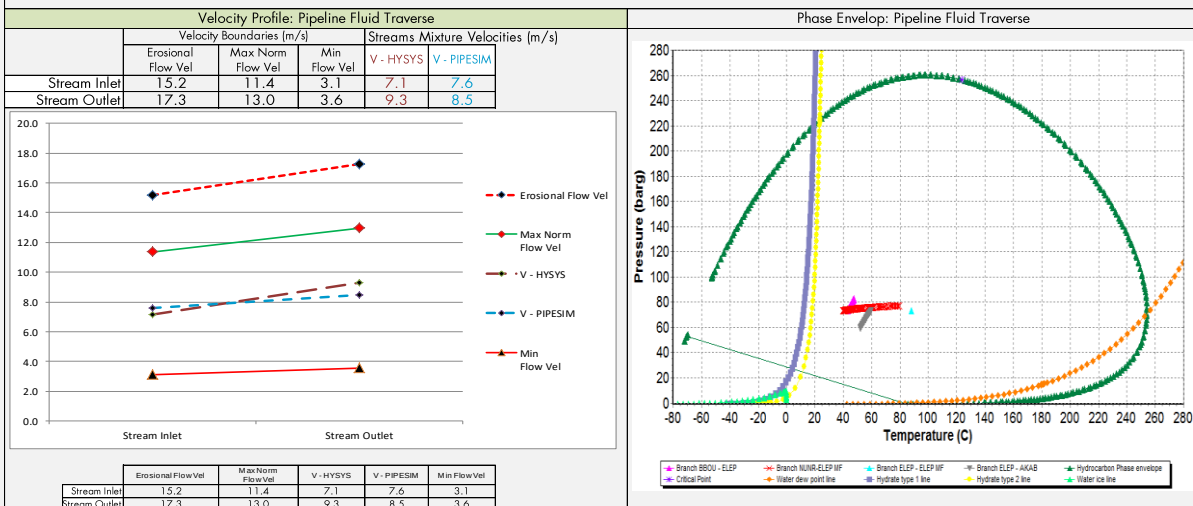
Flow Assurance Metrics: Water Sensitivity (2033) Production, with BBOU at 207MMscfd

Water Sensitivity (2033) Production: BBOU - ELEP:

Basis:	207.0	MMscfd
BBOU @	207	MMscfd Peak

Key Governing Parameters	Targets
Pressure	Inlet Pressure High Enough to Deliver at Destination Pressure
Velocity	Not too high that will induce corrosion, but high enough for good cleaning
Hydrate formation Tendency / Phase Env	Fluid Condition (T & P) outside Hydrate formation Region

Flow Assurance Picture



NUNR – ELEP Flow Stream: 2033 Production

Pipeline Sizing & Flow Assurance Analysis:

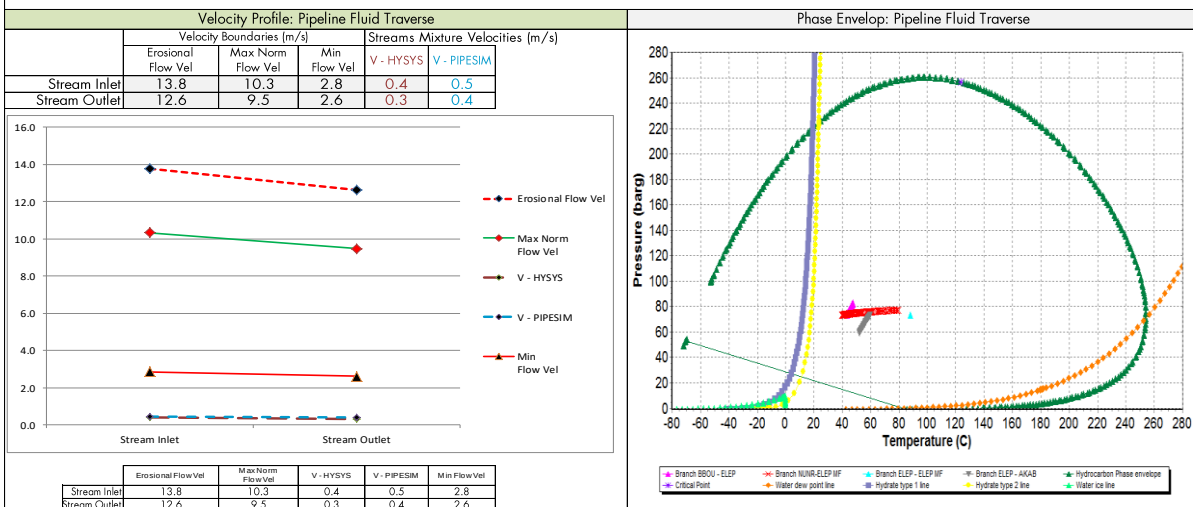
Flow Assurance Metrics: Water Sensitivity (2033) Production, with NUNRat 22MMScfd

Phase-2: Stream NUNR - ELEP at Peak Production:

Basis:	22.0	MMscfd
NUNR @	22	MMscfd Peak

Key Governing Parameters	Targets
Pressure	Inlet Pressure High Enough to Deliver at Destination Pressure
Velocity	Not too high that will induce corrosion, but high enough for good cleaning
Hydrate formation Tendency / Phase Env	Fluid Condition (T & P) outside Hydrate formation Region

Flow Assurance Picture



ELEP - AKAB Flow Stream: 2033 Production

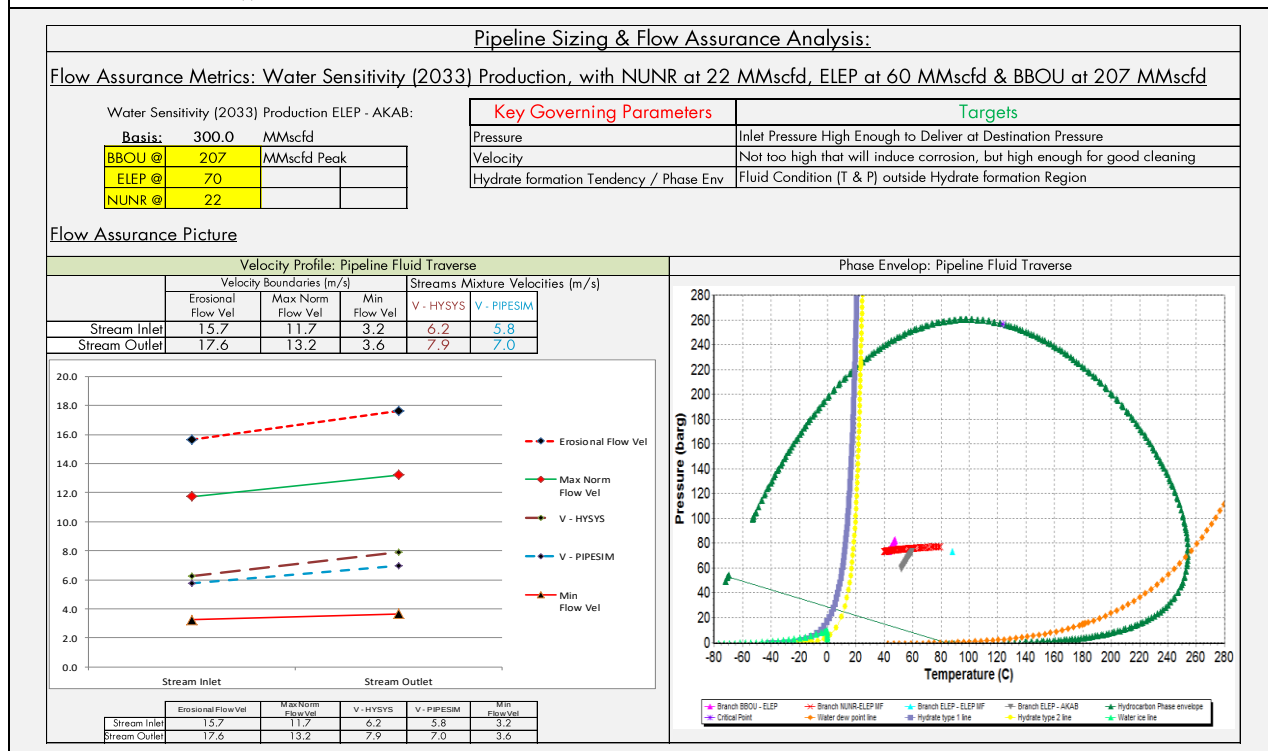


Figure 6.17: Flow Assurance Analysis – 2033 Production

ELEP – AKAB Flow Parameters & Flow Assurance Analysis - Pipesim: Phases 2 - 4

The flow parameters give positive indication of deliverability and flow assurance, giving the pressure, velocity, & fluid traverse profiles shown in fig 6.16 below.

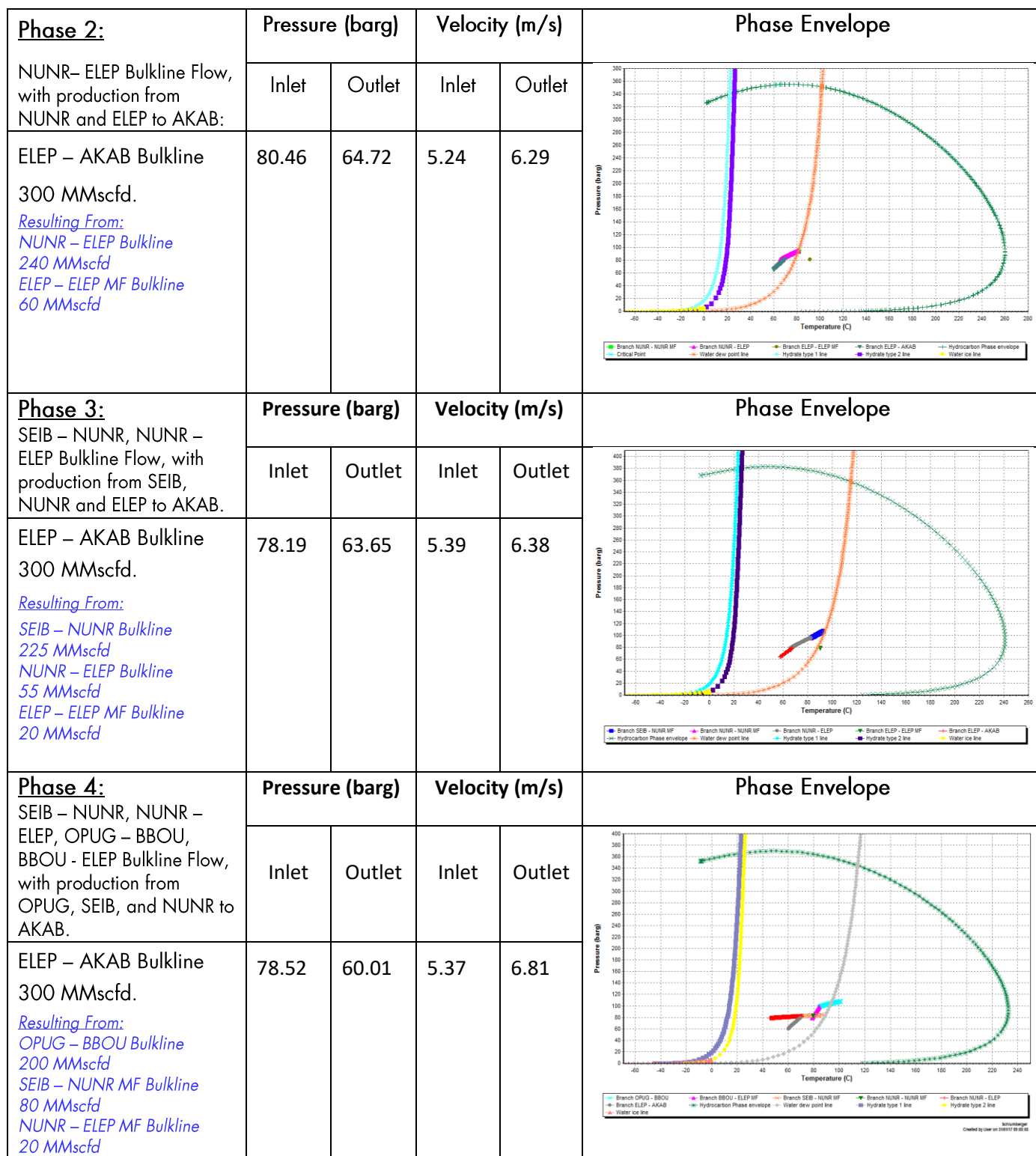


Figure 6.16: Flow Assurance Analysis – ELEP - AKAB, Phases 2 - 4 Production

7. Pipeline Materials Selection & Corrosion Analysis

Ref: EA SPDC-BFPC UGS-GEP MAT RPT-001-A01_Concept Select Study – Materials Selection & Corrosion Analysis

This chapter summarises the results of the above referenced Concept Select Study document in respect of the Materials Selection and Corrosion Analysis for the project.

The report covers the following subject areas:

-  The Pipelines' Materials Selection & Corrosion Analysis
-  The Coating Materials Choice
-  The Cathodic Protection System

7.1. Intent & Basis

Intent is Pipeline Integrity, and Materials Selection is the key to this intent.

The bases for the materials selection are the process & environmental conditions, and the operational requirements.

The process conditions constitute the physical conditions, - predominantly pressure and temperature, which the system must contend with in respect of stress; and the chemical conditions emanating from the impurities contained in the natural gas stream, - predominantly carbon dioxide, hydrogen sulfide (*though not in this case*), water vapor, oil, nitrogen, hydrates, which are capable of inducing internal corrosion and erosion of the system.

The environmental condition represents the ambient, which if corrosive will cause external corrosion of the system.

Both corrosions (internal & external) are undesirable and have to be mitigated.

The operational requirements constitute consideration of design, construction, operations, maintenance, threats, hazards, risks, safety and economics; and codes and standards that provide mandatory and optional requirements and guidelines.

Similarly, the bases for the corrosion analysis are the susceptibility of the material to corrosion, the corrosivity potential of the transported fluid that will cause internal corrosion to the pipeline system, and the environment which, if corrosive, will cause external corrosion.

These premises will be followed in the materials selection & corrosion analysis process to ensure proper selection of materials and the relational corrosion management system.

7.2. Purpose & Objective of the Study

The purpose is three-fold:

1. Identification of best pipeline materials for each of the line segments at optimised life cycle costs, utilising best industry practice.
2. Assessment of the risks associated with the specific selection.
3. Specification of the corrosion management system that will mitigate the risk.

The objective is the optimal specification of materials and the relational corrosion management systems that can contend with the process & environmental conditions, and operational requirements of the pipeline system.

7.3. Materials Options

The following materials were evaluated:

-  Mild Steel (MS) - Carbon Manganese Steels
-  MS plus Inhibition
-  Various CRA's (Austenitic SS, Ferritic SS, Martensitic SS, Duplex SS)
-  Precipitated Hardened Steel (with Al)
-  Pipeline with Internal CRA Lining or Cladding
-  Composite Materials
-  Non- Ferrous Materials

The technical feasibility followed a four stage evaluation for acceptance, defined on a **Green**, **Yellow**, and **Red** traffic light colouration system.

Red represents: rejection

Yellow represents: not totally acceptable because of some few limitations including costs, properties, maintainability etc.

Green represents total acceptance of the material

The technical feasibility for the materials was demonstrated through the following evaluations.

1. Feasibility of Mild Carbon Steel, - without Corrosion Inhibition
2. Feasibility of Mild Carbon Steel, - with Corrosion Inhibition
3. Feasibility of CRA Materials through the following steps:
4. Optional Process for Modification of the Fluid Content through the following steps:

7.4. The System & Fluid Under Consideration

The system under consideration is the pipeline network as shown in fig 7.1 below.

It comprises the following pipeline segments, coming into service in phases, and transporting raw wellhead gas to the Gas Processing Plant:

- 24" x 28.5 km ELEP – AKAB Trunkline
- 16" x 9 km BBOU – ELEP Bulkline
- 24" x 45 km NUNR – ELEP Bulkline
- 20" x 20 km SEIB – NUNR Bulkline, and
- 24" x 37 km OPUG – BBOU Bulkline
- Each of the five Field Manifold Headers of about 10 m length.

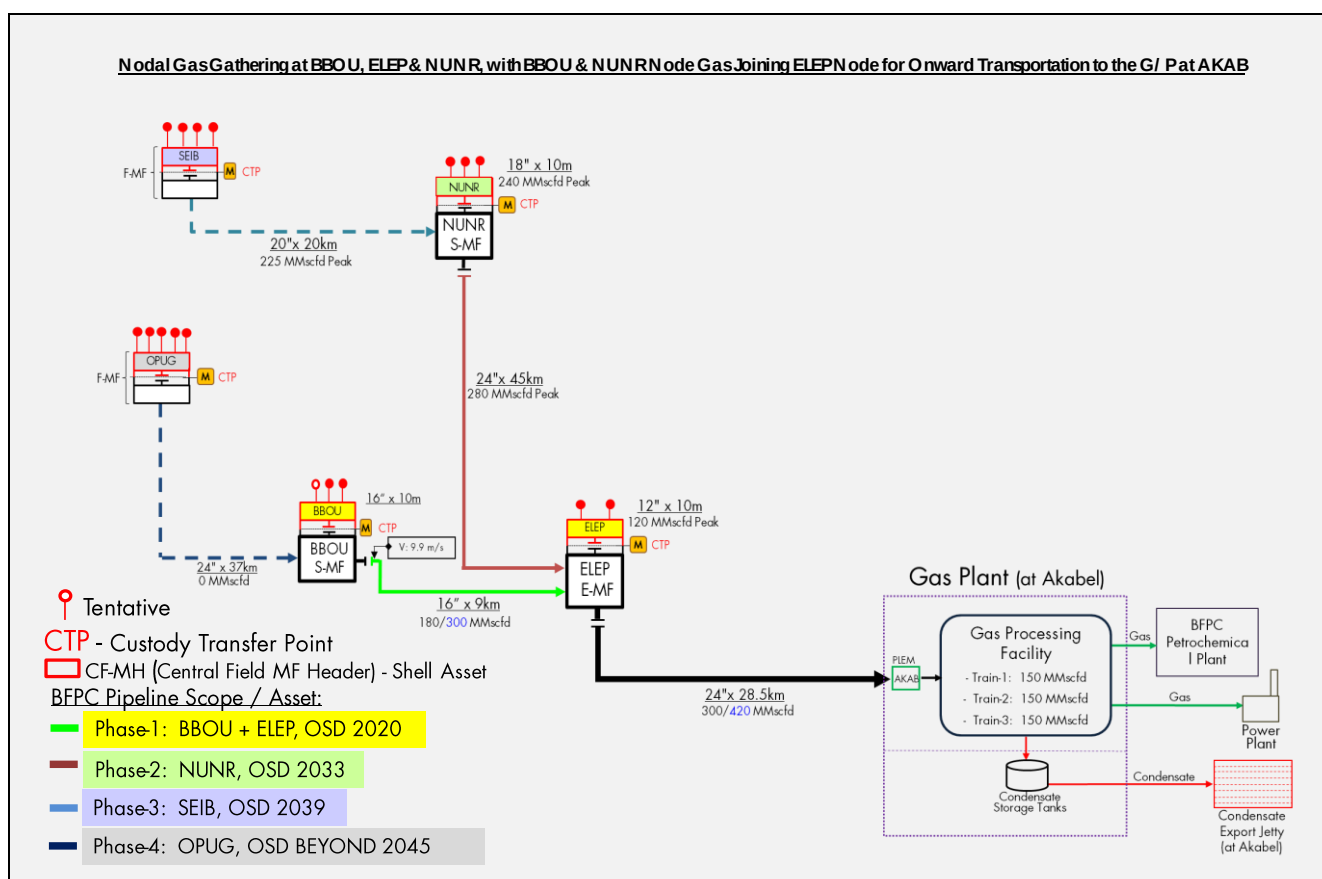


Figure 7.1: The Pipeline Network

The fluid transported is high pressure category D/E fluid (flammable fluid), in the pressure & temperature regions of 100 barg, and 60 °C, and containing CO₂ that causes corrosion.

However, it contains no H₂S that causes sour corrosion, or Hg that causes liquid metal embrittlement (LME) of alloys, or Naturally Occurring Radioactive Materials (NORM) which can accumulate to render the system radioactive over time.

7.5. The Fluid Impurities Composition

The impurities composition in the various reservoir fluids are shown in the table below:

Table 7.1: Reservoirs Conditions & Impurities Compositions (Water Saturated Compositions)

Component	Impurities Compositions - Mole %				
	BBOU	ELEP	NUNR	SIEB	OPUG
N ₂	0.07	0.00	0.00	0.00	0.10
CO ₂	0.59	2.17	0.77	1.53	2.21
H ₂ S	0.00	0.00	0.00	0.00	0.00
H ₂ O	0.34	2.19	0.63	0.95	1.08
Total	1.00	4.36	1.40	2.48	4.11
Reservoir Conditions					
Reservoir Pressure (bar)	223.8	476.6	500.6	450.6	356.9
Reservoir Temp (°C)	78	123	102	117	113.8
pCO ₂ (bar)	1.32	10.34	3.85	6.89	7.89

At the deducible partial pressures of CO₂ (pCO₂) above, the gas streams will result in significant corrosion of the pipeline system, judging from both the corrosion prediction calculation results from DeWaard-Milliams Model, and its CO₂ Corrosion Nomogram shown below.

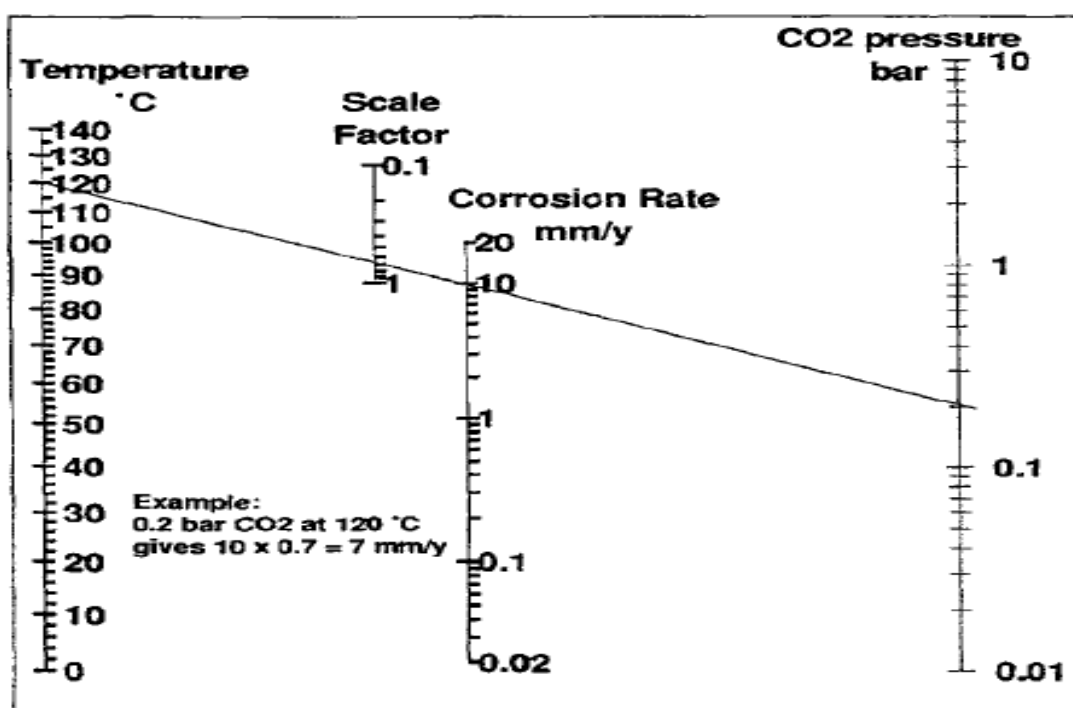


Fig 7.2: DeWaard-Milliams CO₂ Corrosion Nomogram

7.6. Issues & Considerations

Of the essence will be pressure containment by the system as well as corrosion prevention / mitigation in the system; and in that respect the following issues were given due considerations:

1. Materials of good strength & properties to withstand the high pressure under consideration and corrosive nature of the fluid to be transported.
2. High Pressure Transportation of Category D/E Fluid (flammable fluid), in the pressure region of 100 barg, and temperature region of 60 °C.
3. The Gas is potentially corrosive on account of its CO₂ content; and water, NGL, & particulates that are inherently present
4. CO₂ content is up to 2%, which for the empirically established 0.1 bar partial pressure threshold for significant CO₂ corrosion, implies corrosion of unprotected mild CS at pressures above 5 barg
5. The CO₂ content of up to 2%, renders the gas corrosive at pressures higher than 5 barg (corresponding to the empirically established 0.1 bar partial pressure threshold for significant CO₂ corrosion), - prompting the need for CRA materials or matching corrosion management design.
6. The Carbon Steel option would mean more internal wall thickness plus corrosion allowance
7. Internal corrosion will be substantial for which the traditional (normal) corrosion allowance of 3 mm will be insufficient provision.
8. Higher corrosion allowance, up to the traditional maximum of up to 10 mm will therefore be considered, and in that respect, high yield strength material with smaller wall thickness demand will be used, so as to keep the overall design wall thickness as small as possible.
9. Ambient environment is corrosive owing to joint contribution of humidity, acidity, and oxygen in air, and known potentially corrosive nature of the soil in the area.
10. Temporal Protection of the linepipes / pipelines, - that will give progressive protection during the period of construction, until the ICCP system is put in place

7.7. Recommended Materials

The recommended line pipe materials are as follows:

- API 5L X65 Carbon Steel for the main (long distance) Pipelines and Pig Barrels, and
- 25Cr UNS Code S32750 Stainless Steel for the Manifold's in-plot pipelines & piping.

The pictorial concept is presented in the fig 7.2 below using Elepa Manifold Platform as example that applies to all the Platforms.

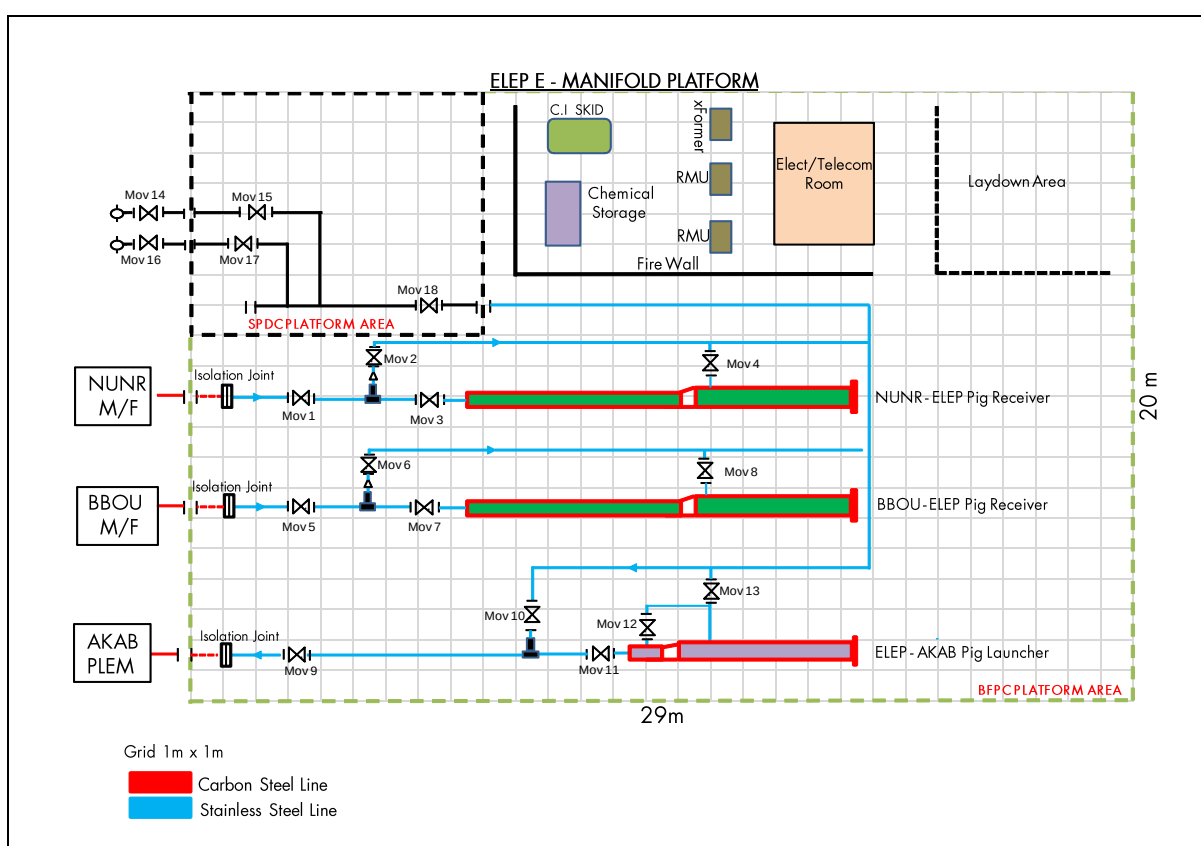


Figure 7.3: The Pipeline Materials Selection – Elepa Manifold

The recommended line materials are given on table 7.2, on the basis of traffic lights assessment in the concert of cost, corrosion & corrosion management analysis:

Red representing: total unacceptability.

Amber representing: not totally acceptable because of some few limitations, including cost & maintainability, but could be managed with very strict CMS.

Green representing: total acceptance of the material.

The hydraulic and wall thickness calculations for the carbon steel were based on the API 5L X65 line pipe.

Table 7.2: Traffic Light Assessment

TRAFFIC LIGHTS COMPARISON – based on Cost & Corrosion						
s/n	Line Segment	MS	MS with 3 mm ca plus CI Injection	Duplex SS	MS with 10 mm ca plus CI Injection	Selected Material
1	24" x 28.5 km ELEP – AKAB T/L					MS + 10 mm ca + Corrosion Inhibitor under very strict CMS
2	16" x 9 km BBOU – ELEP B/L					MS + 10 mm ca + Corrosion Inhibitor
3	16" x 10 m BBOU B/L					DUPLEX SS
4	12" x 10 m ELEP B/L					DUPLEX SS
5	24" x 45 km NUNR – ELEP B/L					MS + 10 mm ca + Corrosion Inhibitor under very strict CMS
6	18" x 10 m NUNR B/L					DUPLEX SS
7	20" x 20 km SEIB - NUNR B/L					MS + 10 mm ca + Corrosion Inhibitor under very strict CMS
8	20" x 10 m SIEB B/L					DUPLEX SS
9	24" x 37 km OPUG - BBOU B/L					MS + 10 mm ca + Corrosion Inhibitor under very strict CMS
10	24" x 10 m OPUG B/L					DUPLEX SS

Note:

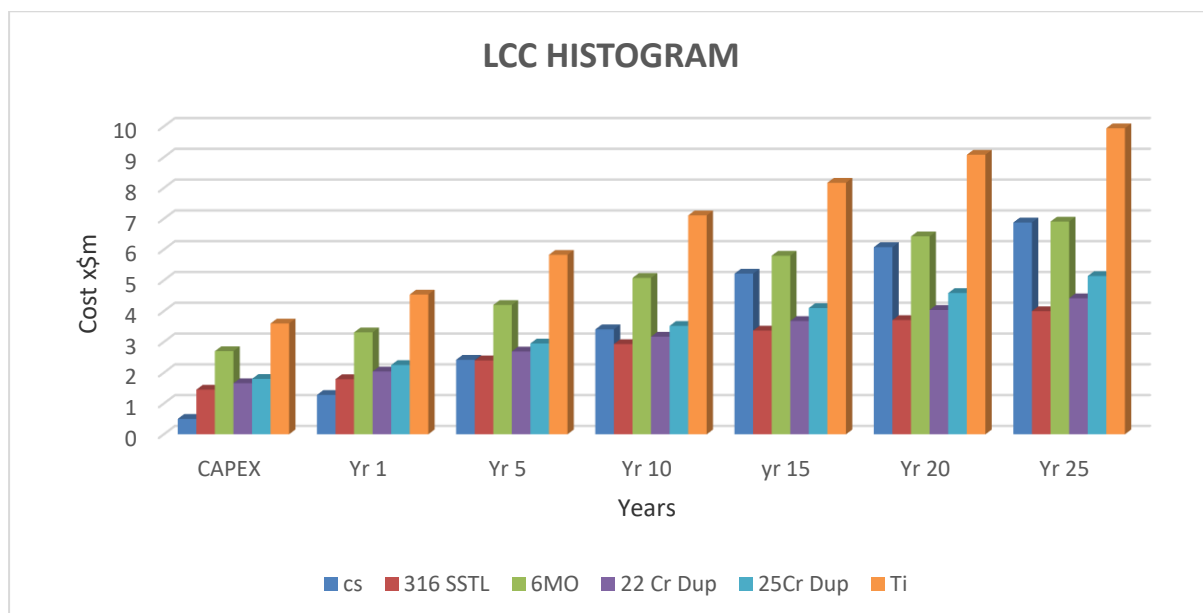
1. The Duplex SS are amber for the long distance pipelines owing to prohibitively high cost of installation, otherwise would be green on purely technical merit.

The cost, properties and, materials selection justifications were built up from bare MS through, MS with corrosion inhibition at 3mm & 10 mm ca (corrosion allowance) to CRA's.

The conclusion that was drawn favoured Duplex SS for the Manifold in-plot short distance lines, and API 5L X65 MS for the long distance pipelines. The API 5L X65 is more acceptable than CRA or Duplex SS, considering cost.

From the corrosivity potential of the transported fluid the MS with 10 mm ca and corrosion inhibitor injection is quite acceptable for the BBOU – ELEP B/L, while the

other long distance P/L are only manageably acceptable requiring strict / stringent measures to be taken in terms CMS (Corrosion Management System) in order to achieve the desired service life.



7.8. Materials Specifications

The materials specifications for the various pipeline segments are follows:

- ELEPA – AKABEL 24" x 28.5 km TRUNKLINE / AKABEL MANIFOLD
 - ✓ 24" API 5L X65, wt 31.75 mm for the Trunkline; with the use of Corrosion Inhibitor in the operational phase.
 - ✓ 24" API 5L X65, wt 31.75 mm for the Trunkline; with the use of Corrosion Inhibitor in the operational phase.
 - ✓ API 5L X65 for the pig barrels, and associated kicker line & balance line.
 - ✓ Duplex SS – 25Cr UNS Code S32750 for Manifold In-plot Piping
- ELEPA F-MF – ELEP E-MF 12" x 10 m BULKLINE
 - ✓ Duplex SS – 25Cr UNS Code S32750

- BUBOUWE – ELEPA 16" x 9 km BULKLINE / BUBOUWE MANIFOLD
- ✓ 16" API 5L X65, wt 31.75 mm for the Bulkline; with the use of Corrosion Inhibitor in the operational phase.
 - ✓ 25Cr UNS Code S32750 Stainless Steel for the short piece 16" x 10 m In-plot Manifold Bulkline that offtakes BBOU production from SPDC BBOU Field Manifold Header for delivery into the BBOU – ELEB Bulkline, either directly or through the Pig Barrel.
 - ✓ Duplex SS – 25Cr UNS Code S32750 for Manifold In-plot Piping

8. Manifold Platforms Power Supply Options

Ref: EA SPDC-BFPC UGS-GEP ELE RPT-001-A01_Concept Select Study – Manifold Platforms Power Supply Options

This chapter summarises the results of the above referenced Concept Select Study document in respect of the Materials Selection and Corrosion Analysis for the project.

The report covers the following subject areas:

-  The Pipelines' Materials Selection & Corrosion Analysis
-  The Coating Materials Choice
-  The Cathodic Protection System

8.1. Intent & Basis

Intent is Pipeline Integrity, and Materials Selection is the key to this intent.

The bases for the materials selection are the process & environmental conditions, and the operational requirements.

The process conditions constitute the physical conditions, - predominantly pressure and temperature, which the system must contend with in respect of stress; and the chemical conditions emanating from the impurities contained in the natural gas stream, - predominantly carbon dioxide, hydrogen sulfide (*though not in this case*), water vapor, oil, nitrogen, hydrates, which are capable of inducing internal corrosion and erosion of the system.